STRUCTURAL SYSTEMS 1 *ARC 2713*

STEEL BUILDING: ROOF STRUCTURE

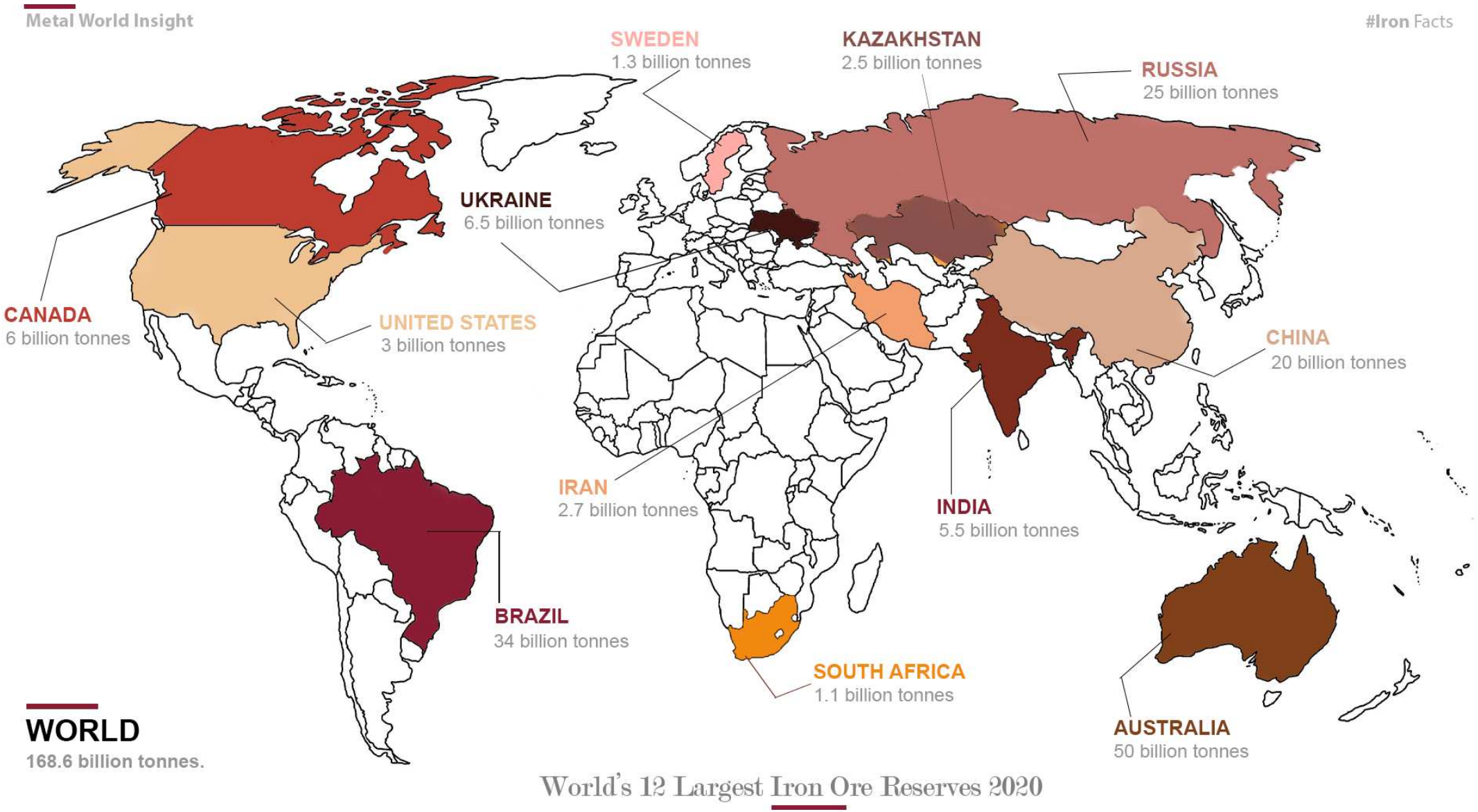
Lecture 9

1. **STEEL material properties and strengths**
2. **STEEL elements**
3. **STEEL BUILDING project parameters**
4. **STEEL BUILDING frame options**
5. **STEEL BUILDING step-by-step process for roof deck, joists, and joist girders**
6. **FIELD TRIP to the low-bay structural engineering lab**

STEEL

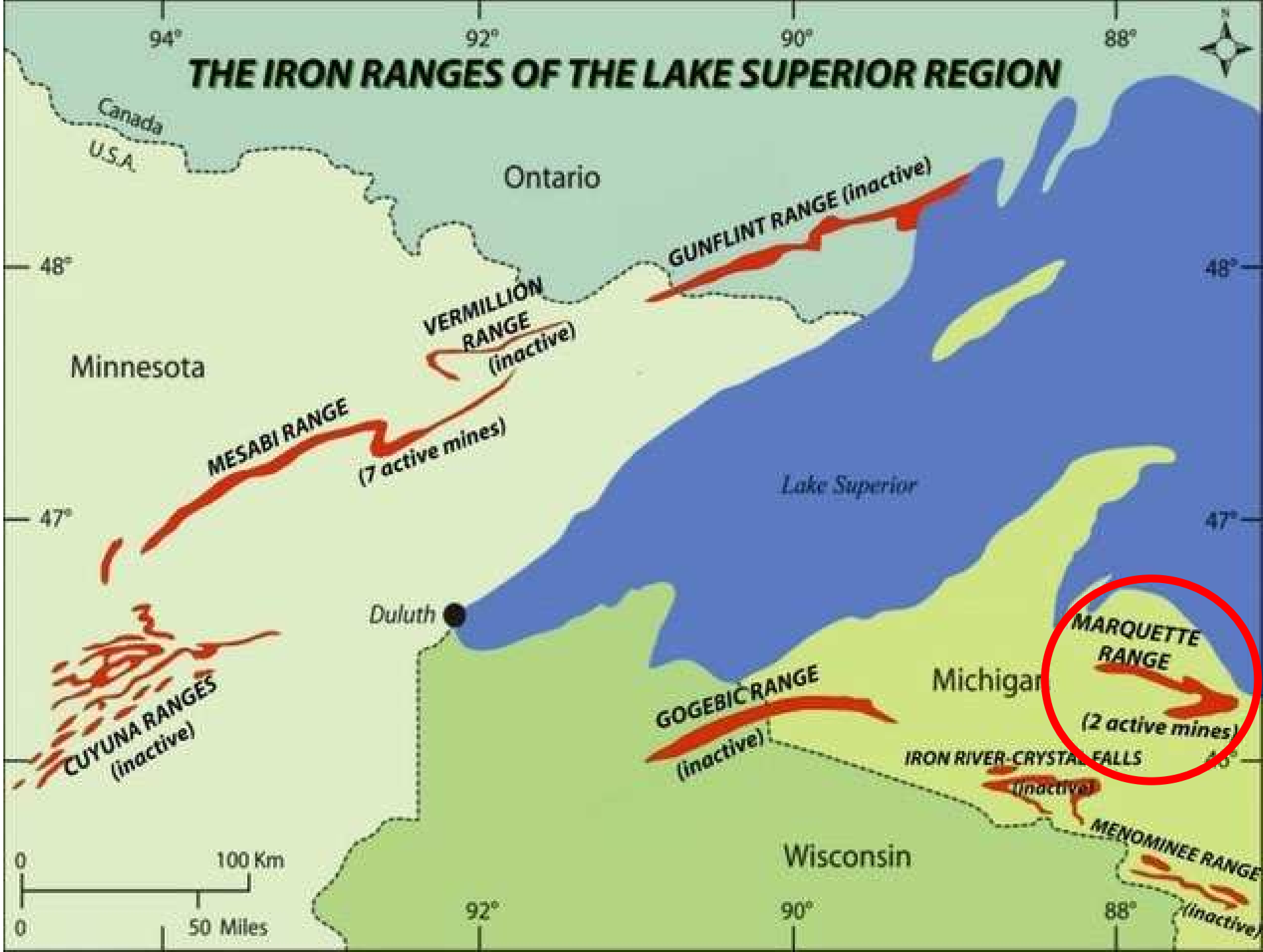
…what do blood and steel have in common?

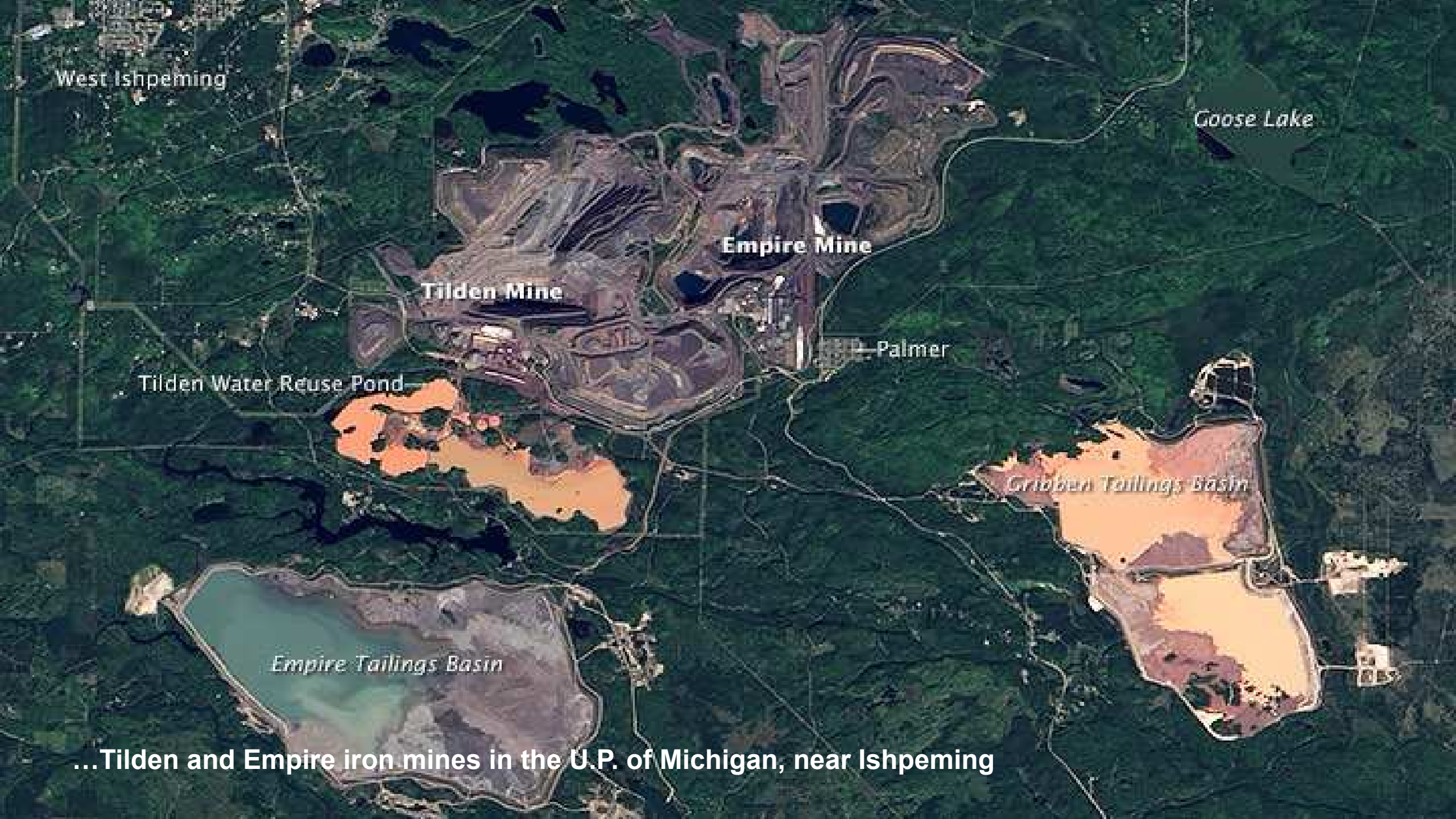

FRESH
BLOOD
SET-IN
BLOOD
STAIN



…iron reserves in the World

…Iron Range


THE IRON RANGES OF THE LAKE SUPERIOR REGION
94°
92°
90°
88°
Canada
U.S.A.
Ontario
48°
GUNFLINT RANGE (inactive)
VERMILLION RANGE (inactive)
Minnesota
MESABI RANGE
(7 active mines)
47°
Lake Superior
Duluth
CUYUNA RANGES (inactive)
GOGEBIC RANGE (inactive)
Michigan
MARQUETTE RANGE
(2 active mines)
IRON RIVER-CRYSTAL FALLS
(inactive)
MENOMINEE RANGE
(inactive)
Wisconsin
0
100 Km
0
50 Miles

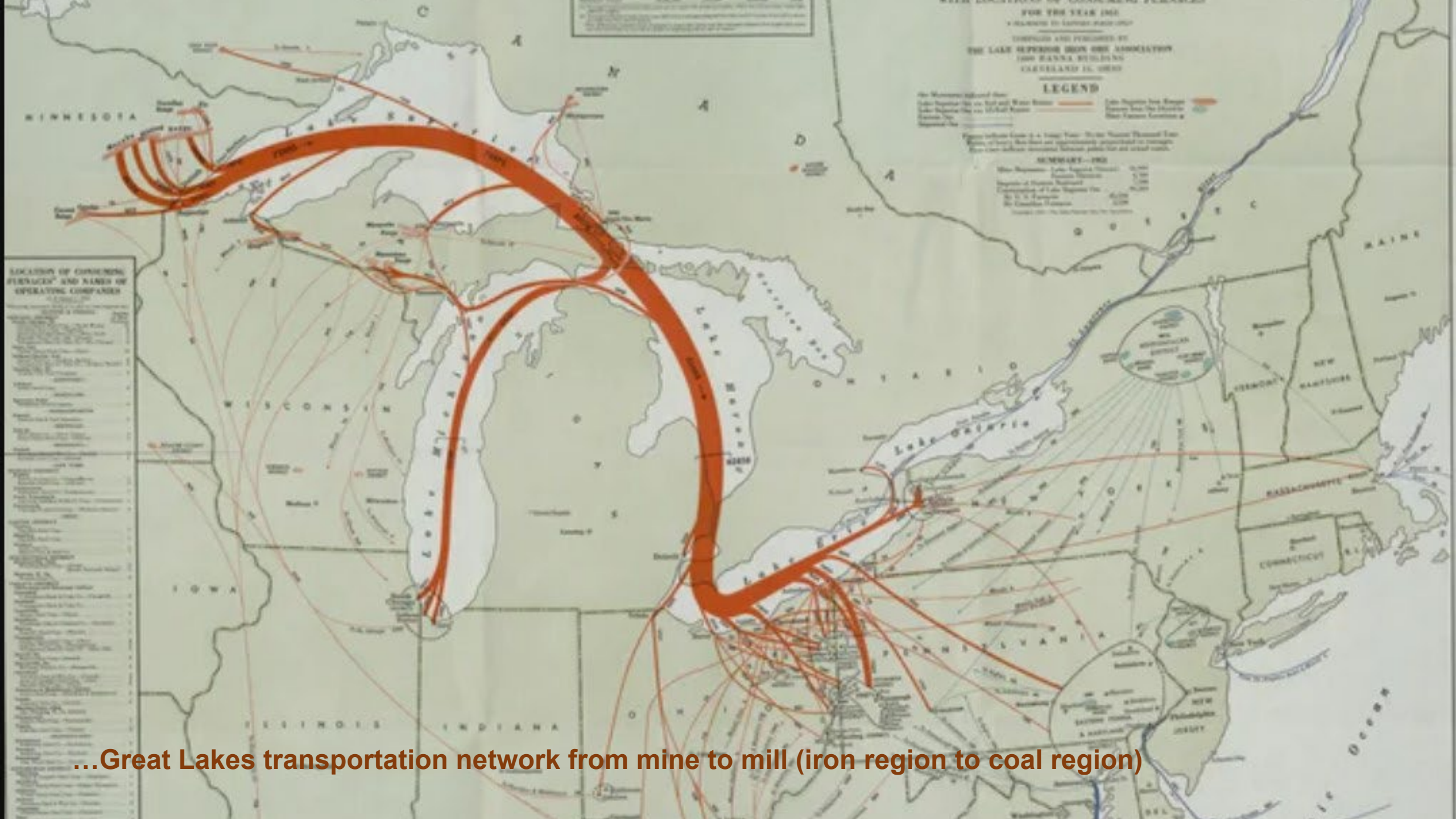


…Tilden and Empire iron mines in the U.P. of Michigan, near Ishpeming


…open pit and tailing pond



…taconite pellets



…Great Lakes transportation network from mine to mill (iron region to coal region)



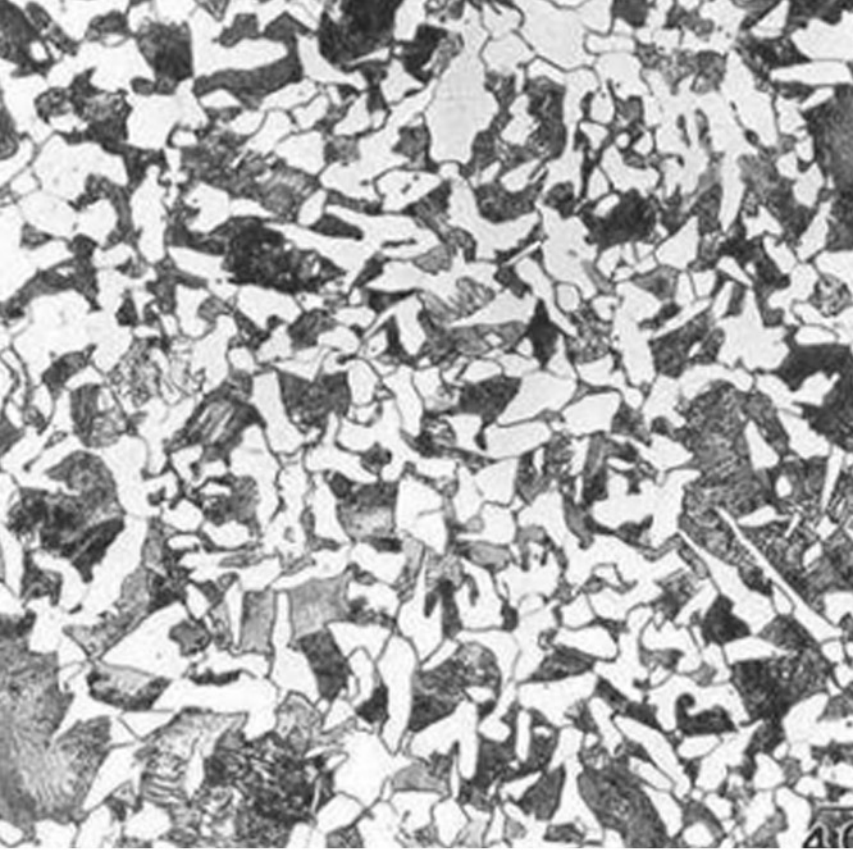
…ore shipped to steel mill



…solid heated to liquid



…liquid formed to solid



STEEL (200x)
Extremely tight particle bonds
Atoms arranged in lattices
Microscopic crystalline structure
Interlocking grain geometry
Complex mix of phases
Minor variations (tested)

…material characteristics and properties define their ability to handle Stress & Strain

STEEL *virtues as a structural material*

…extraordinary strength, durability, ductility, predictability and resilience

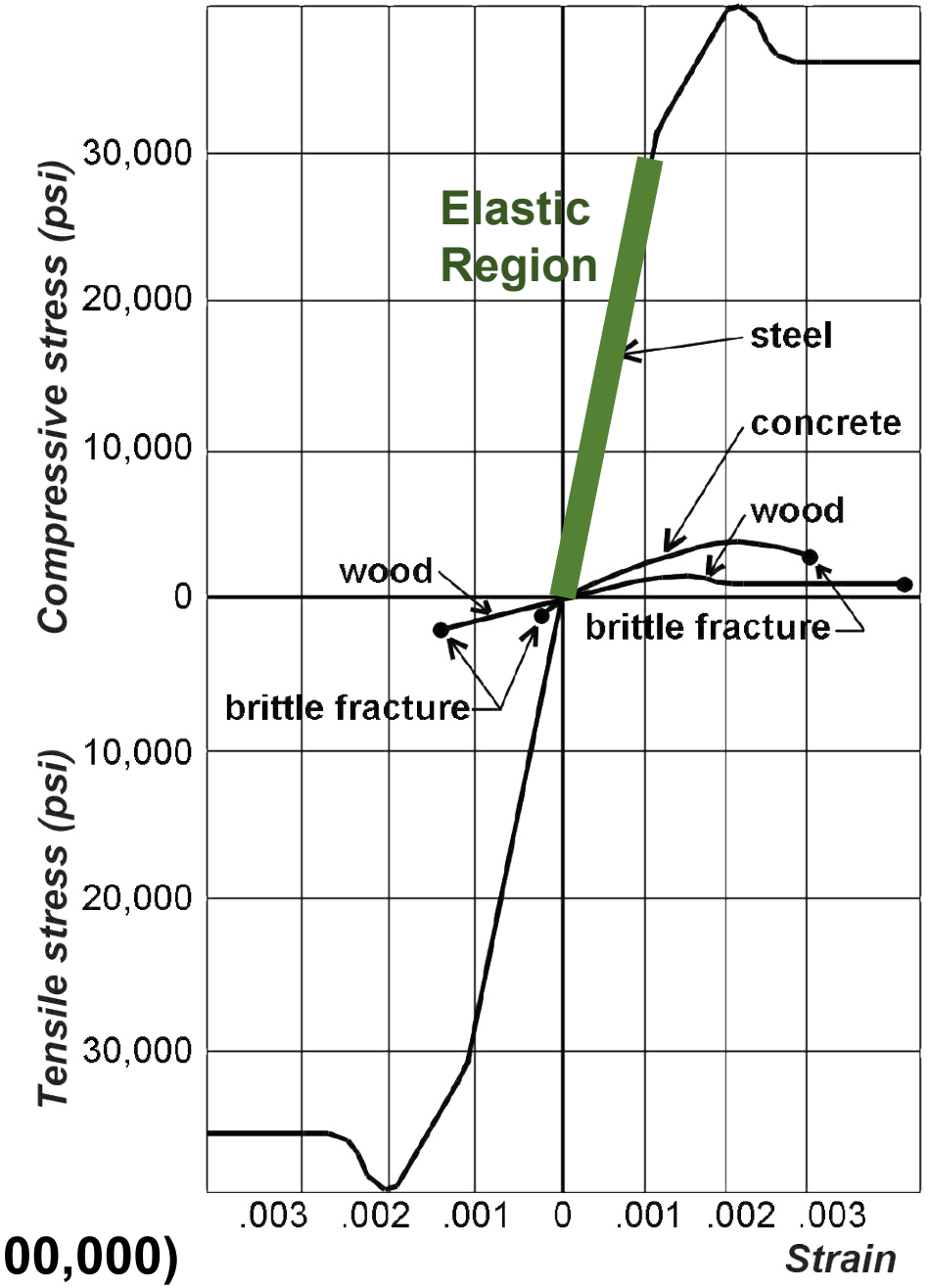
…workable and adaptable into numerous structural elements and components

…unparalleled in compression, tension & bending to resist structural forces

…highly recyclable (U.S. recycles 70% of steel, 30% from new production)

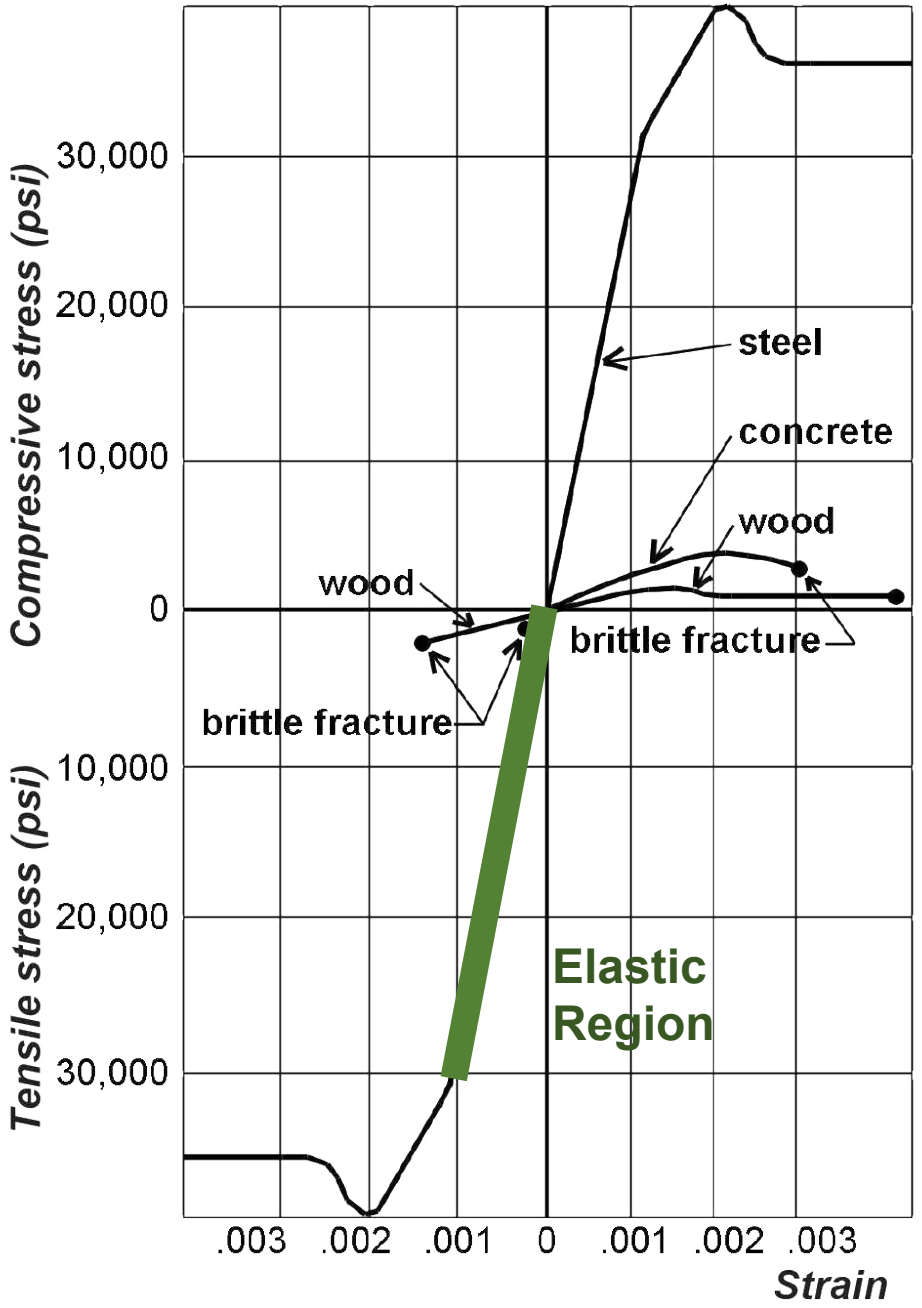
…steel is beautiful

MATERIAL strengths



STEEL compressive stress 30,000 - 50,000 psi (up to 100,000)

MATERIAL strengths



STEEL tensile stress 30,000 - 50,000 psi (up to 100,000)

STEEL MATERIAL measures of deformation or failure

ELASTICITY (temporary deformation)
…the ability of a material structural element to return to its original shape when a load is removed (steel is highly workable (hot or cold) with outstanding elasticity)

PLASTICITY (residual deformation) **SAFE WARNING BEFORE FAILURE**
… the ability of a material structural element to resist deformations which do not disappear when a load is removed (steel has high plasticity in compression and tension)

STEEL MATERIAL measures of strength

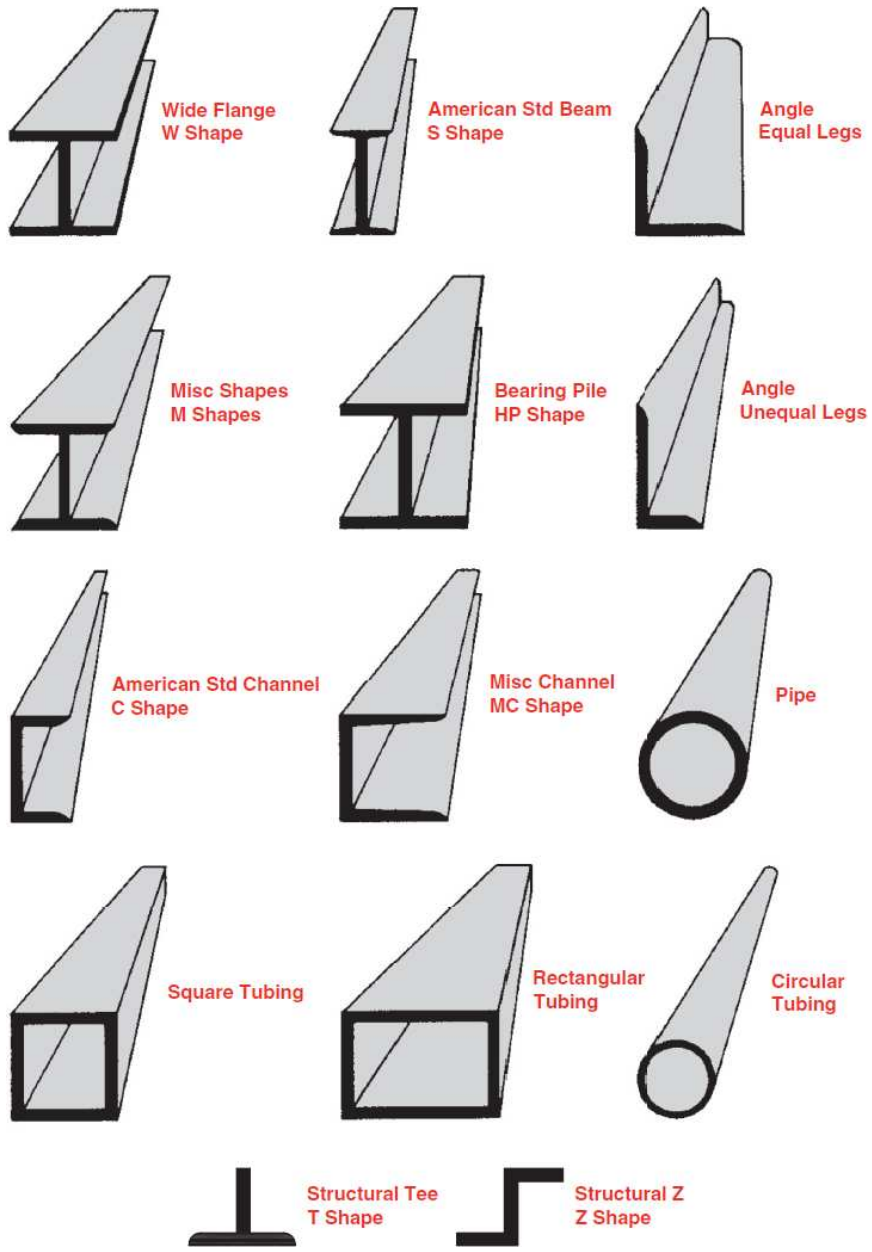
STRESS (unit of measure: psi)
…the push or pull on a material structural element, divided by the area it is applied to
(steel is highly resistant to stresses in both tension and compression and has a great strength-to-weight ratio)

STRAIN (unit of measure: inches)
…the tiny changes in length due to tension or compression, when divided by the original length of a material structural element
(steel has high ability and predictability to resist strain)

STEEL MATERIAL measures of strength

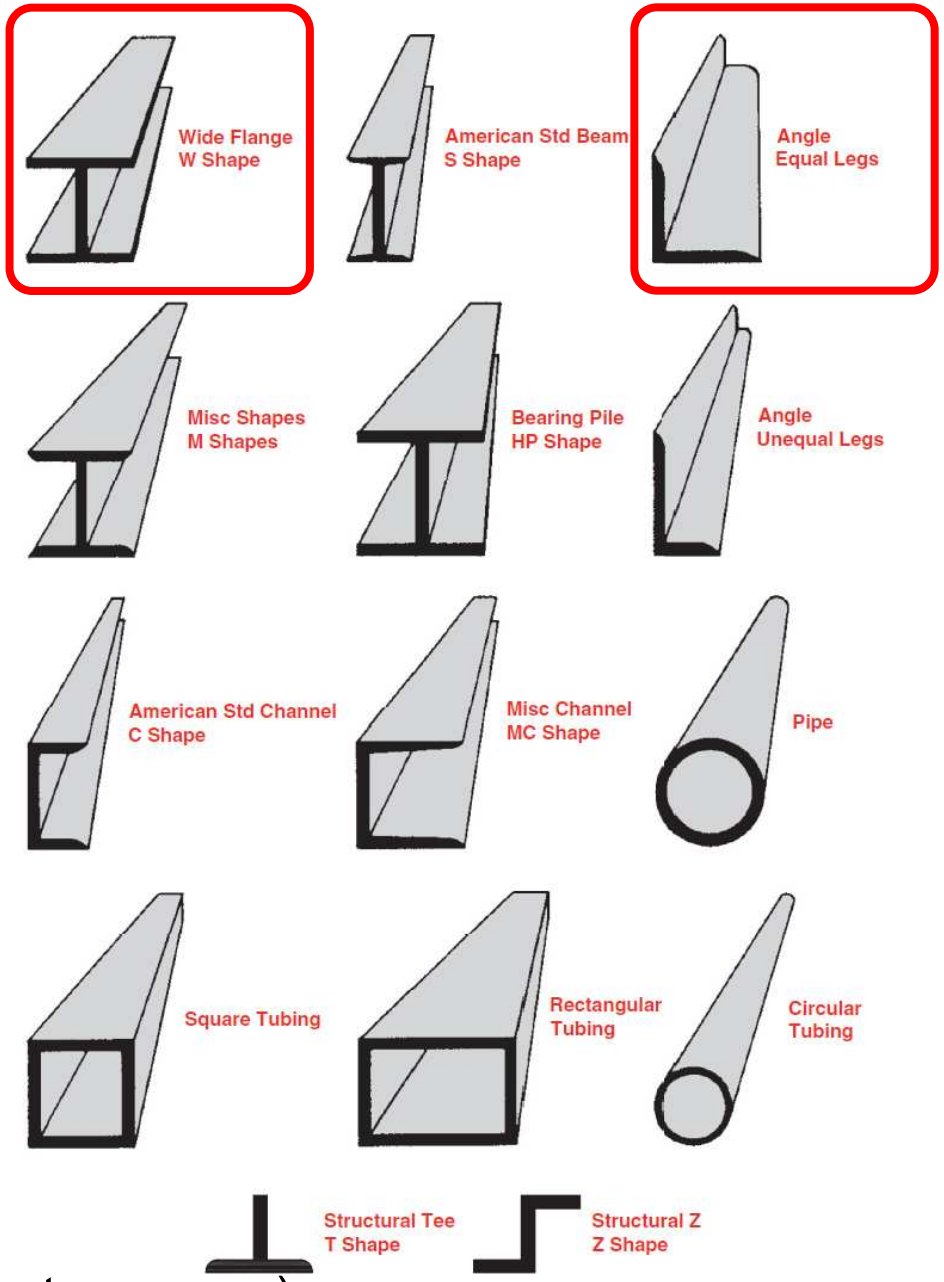
…how is steel optimized as structural shapes or elements?





Structural Steel sections



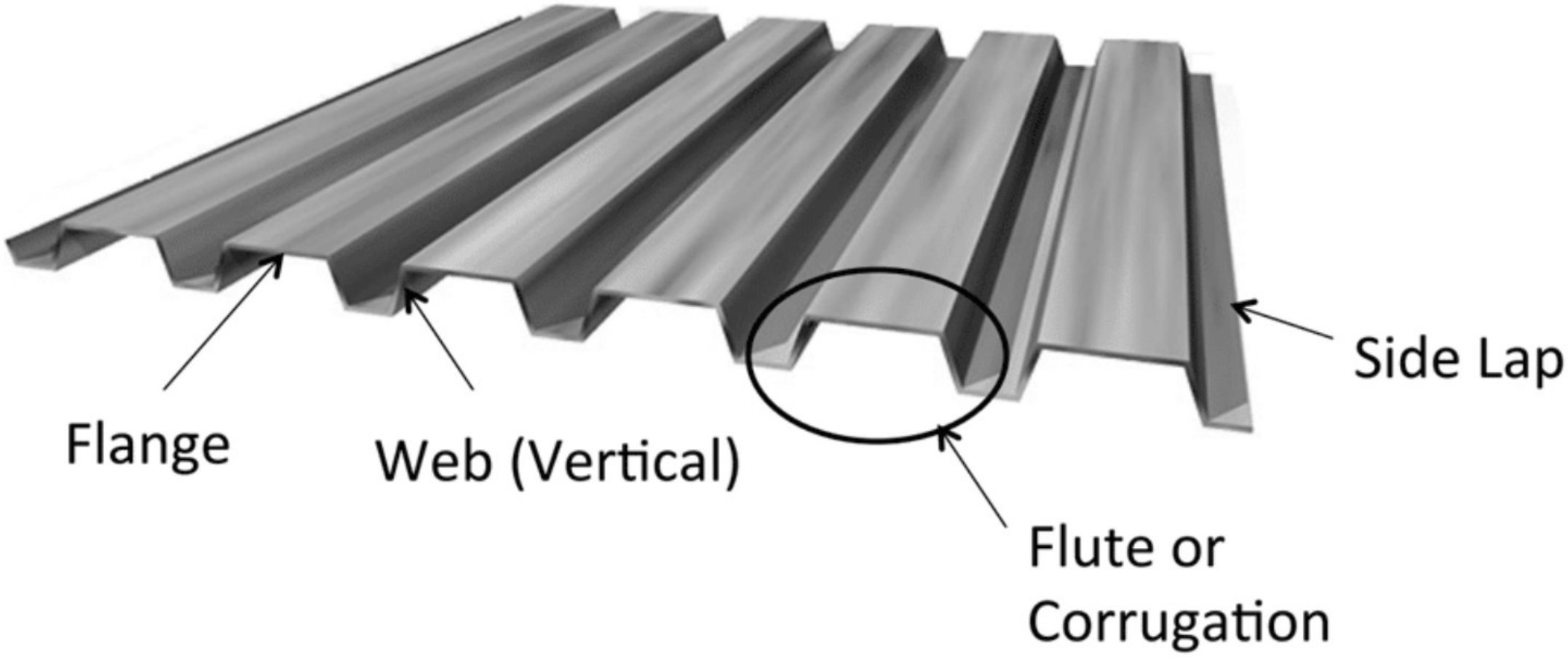


Structural Steel sections (wide-flange, angle, tube are most common)

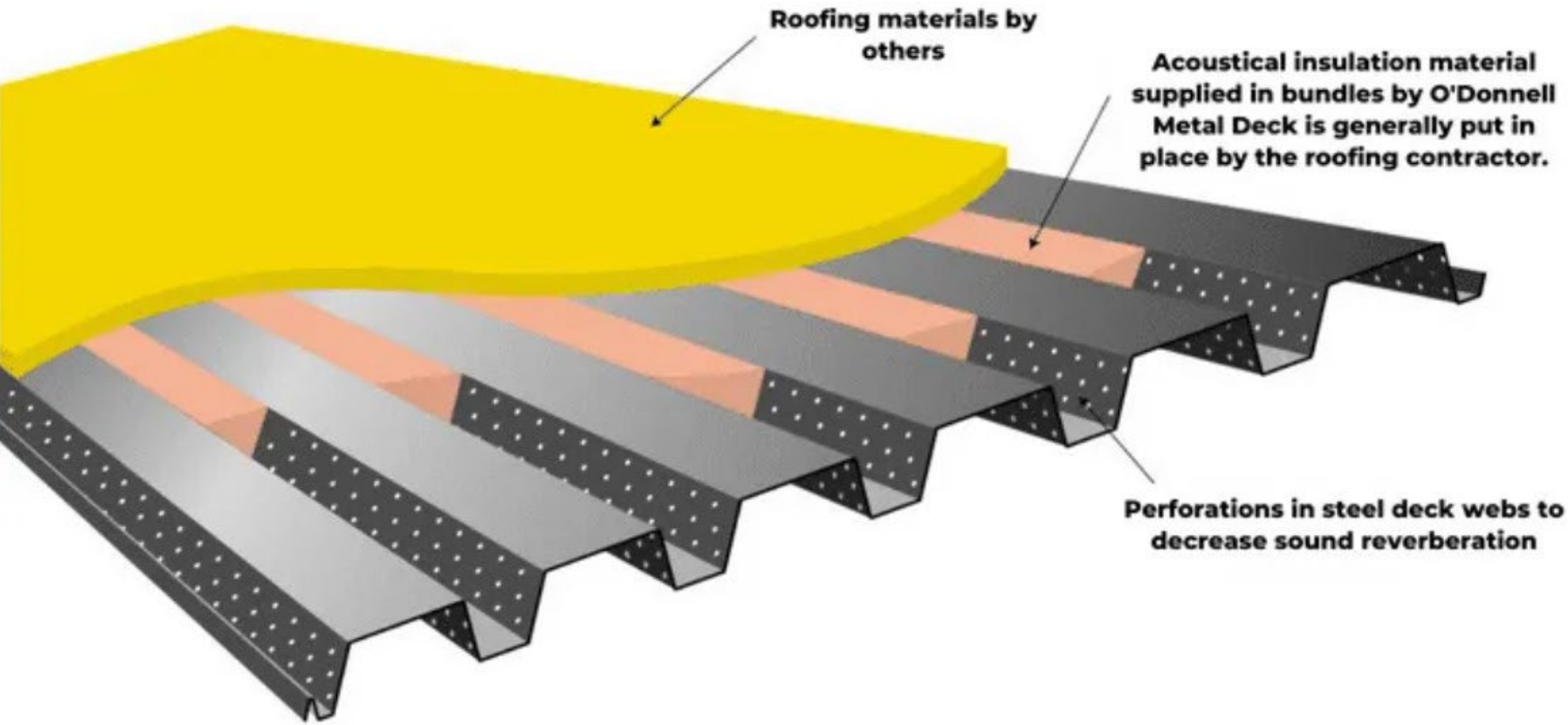




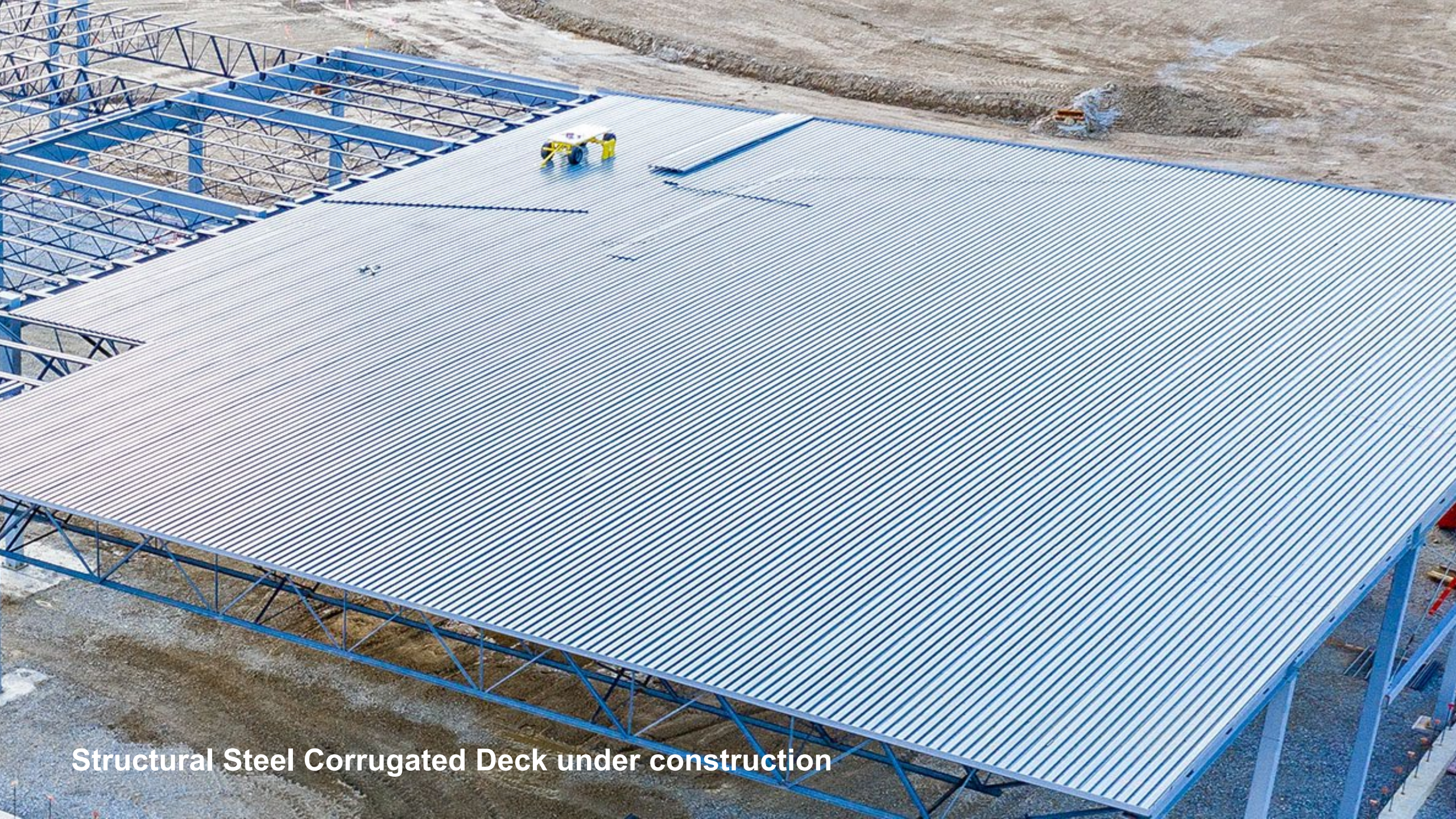
Structural Steel Corrugated Deck (B-Type 1.5” corrugated most common for roofs)



Components of a Steel Deck



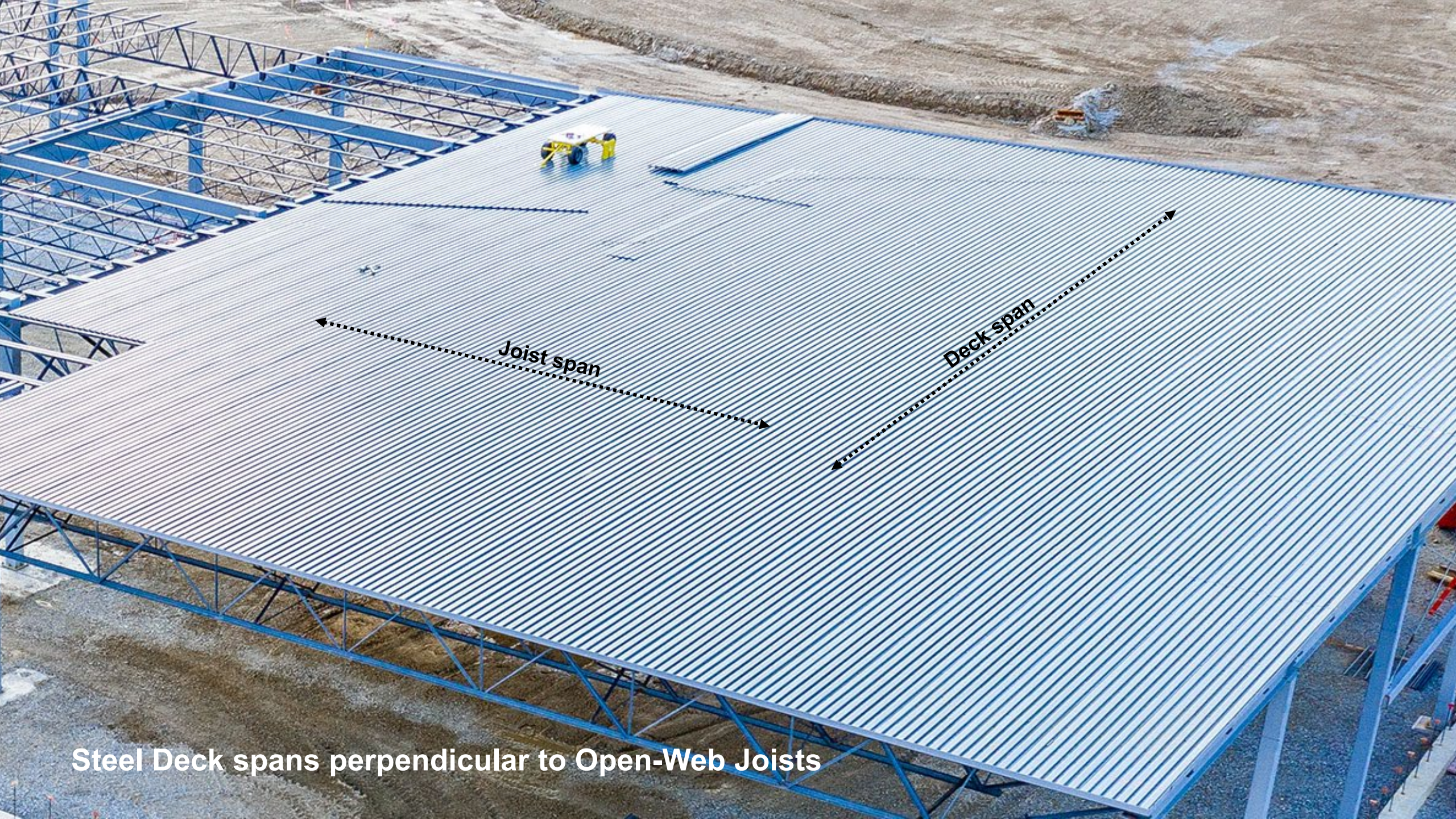
Acoustic Deck is common when structure is exposed as finished ceiling



Structural Steel Corrugated Deck under construction

STEEL SPANS

…which direction does the Deck span? …direction the joists span?



Steel Deck spans perpendicular to Open-Web Joists

…what element supports the tertiary structural deck?



Structural Steel Corrugated Deck (B-Type 1.5” corrugated most common)



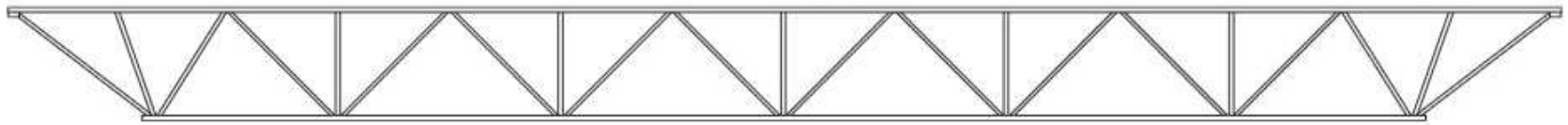
Structural Steel Corrugated Deck, prefabrication and construction technology



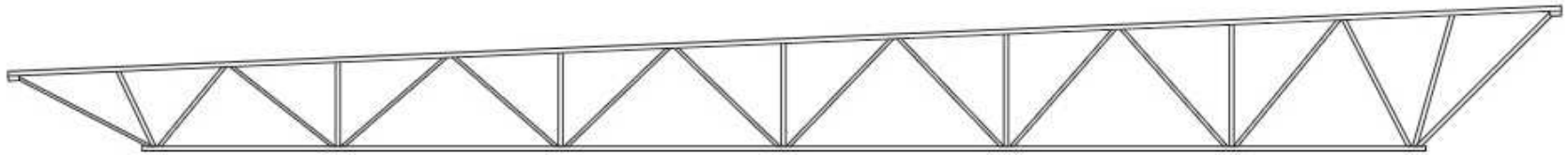
Structural Steel Corrugated Deck attached to Open-Web Steel Joists (double-span Deck)



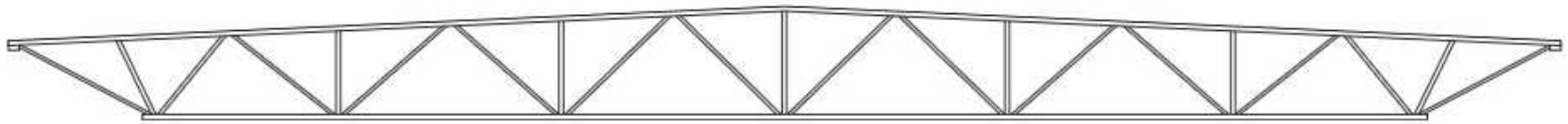
Structural Steel Corrugated Deck with Cross-Bracing of Open-Web Steel Joists



Parallel Chord Joist

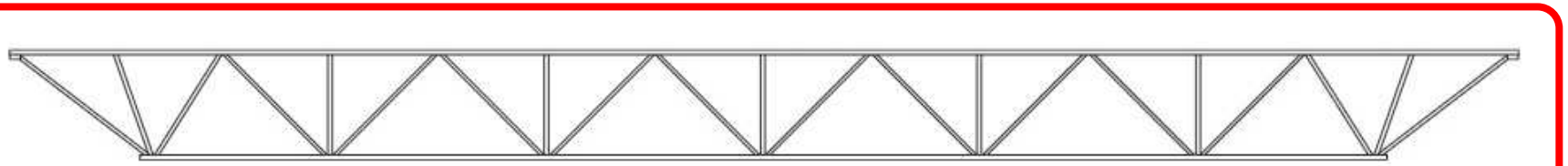


Single-Pitched Top Chord Joist

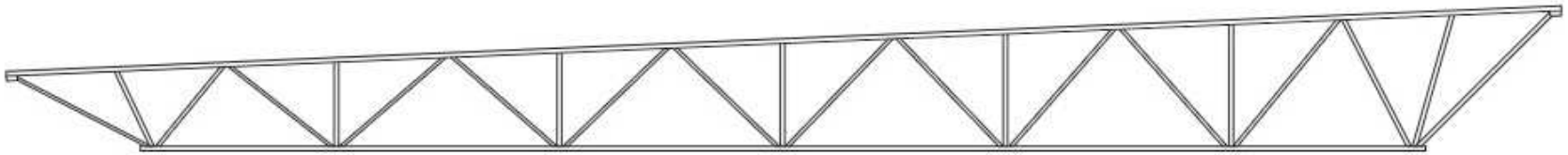


Double-Pitched Top Chord Joist

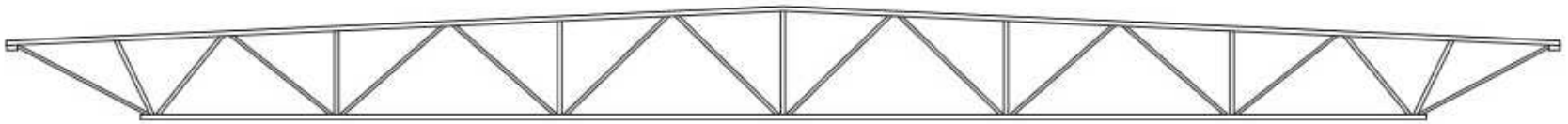
Open-Web Steel Joist Types (perhaps the most common roof structure in the U.S.)



Parallel Chord Joist

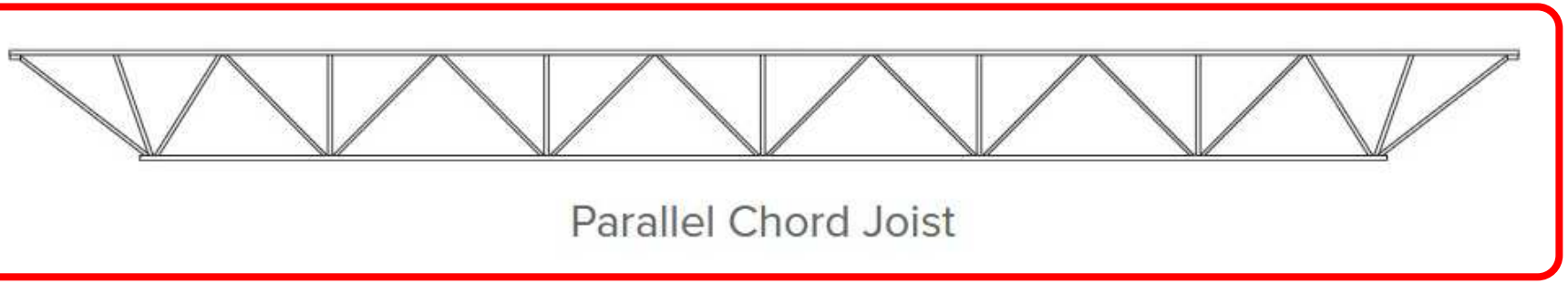


Single-Pitched Top Chord Joist



Double-Pitched Top Chord Joist

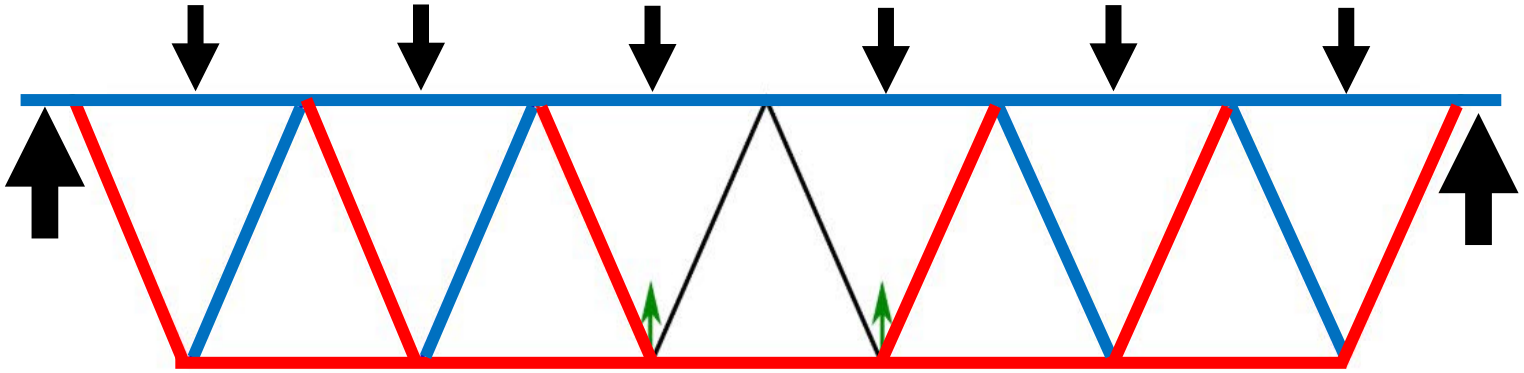
Parallel Chord Joist (the most common of the type)



…what truss type is this? Howe, Pratt, or Warren?

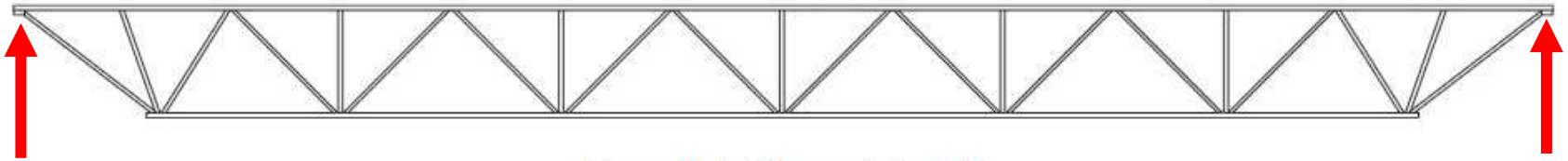
Parallel Chord Joist*…is a structural "truss"*

WARREN Truss



WARREN…diagonals alternate between compression or tension (can change the number of panels based on purlin spacing)

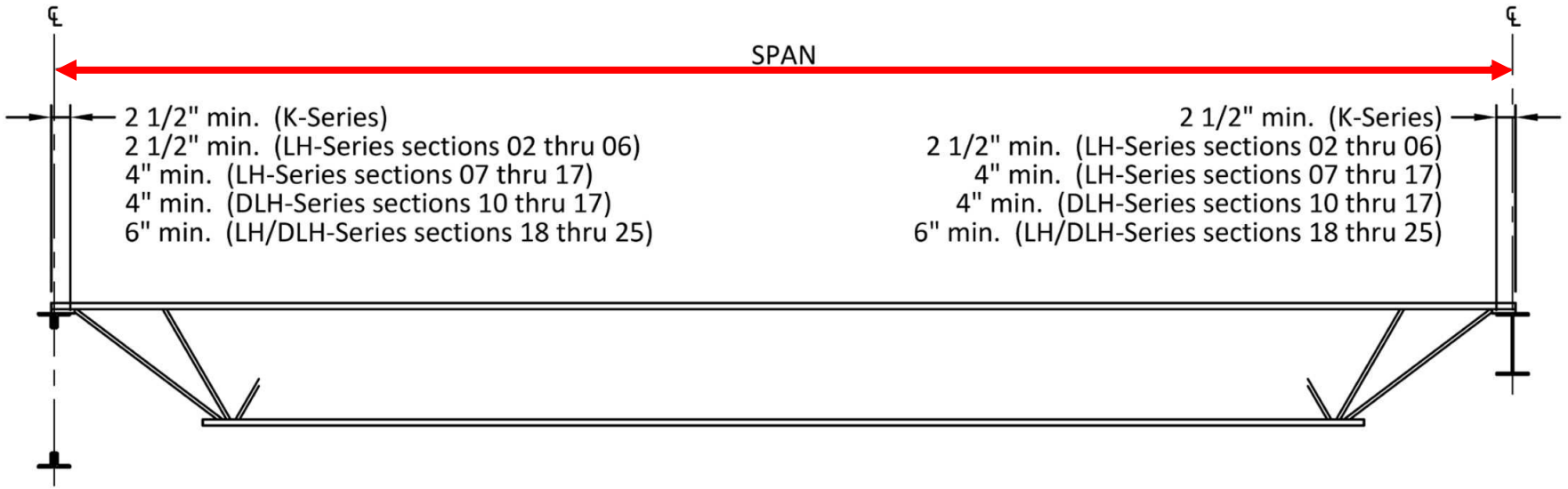
WARREN TRUSS magnitude of compression forces (blue) and tension forces (red)



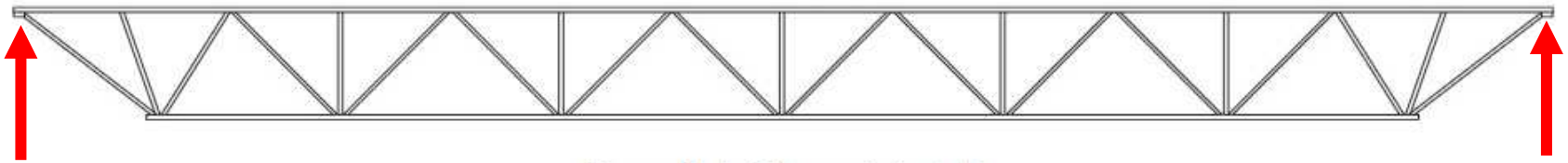
Parallel Chord Joist

Joists are top-chord bearing

Definition of Span



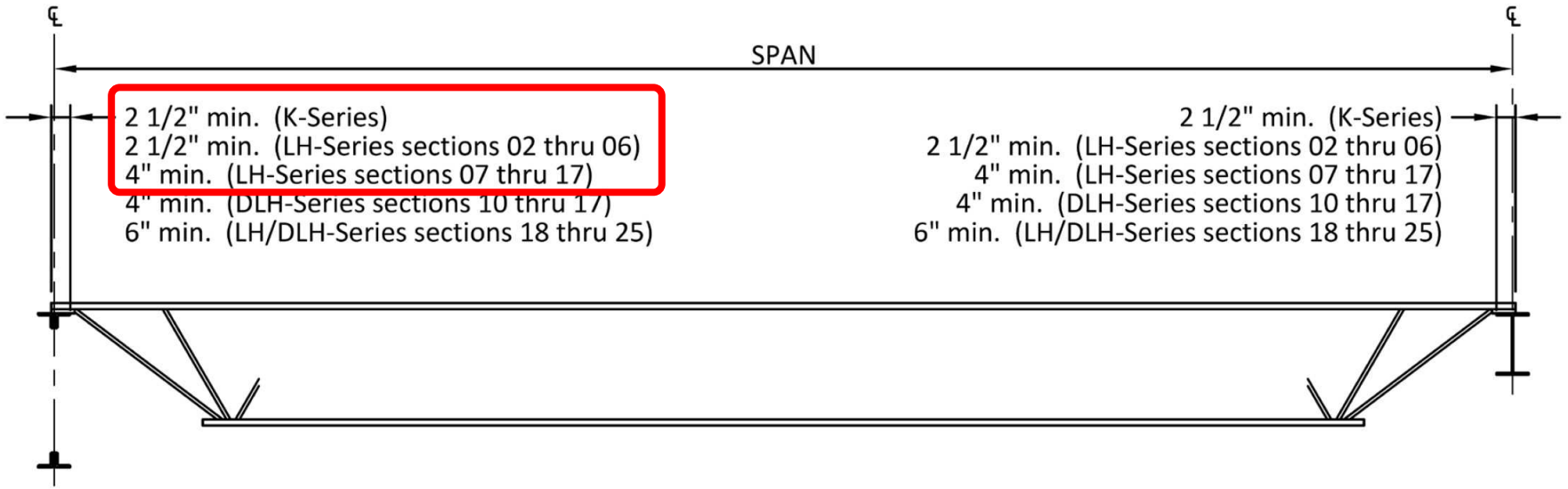
Span defined by centerline of support element (Joist Girder or Steel Beam or ?)



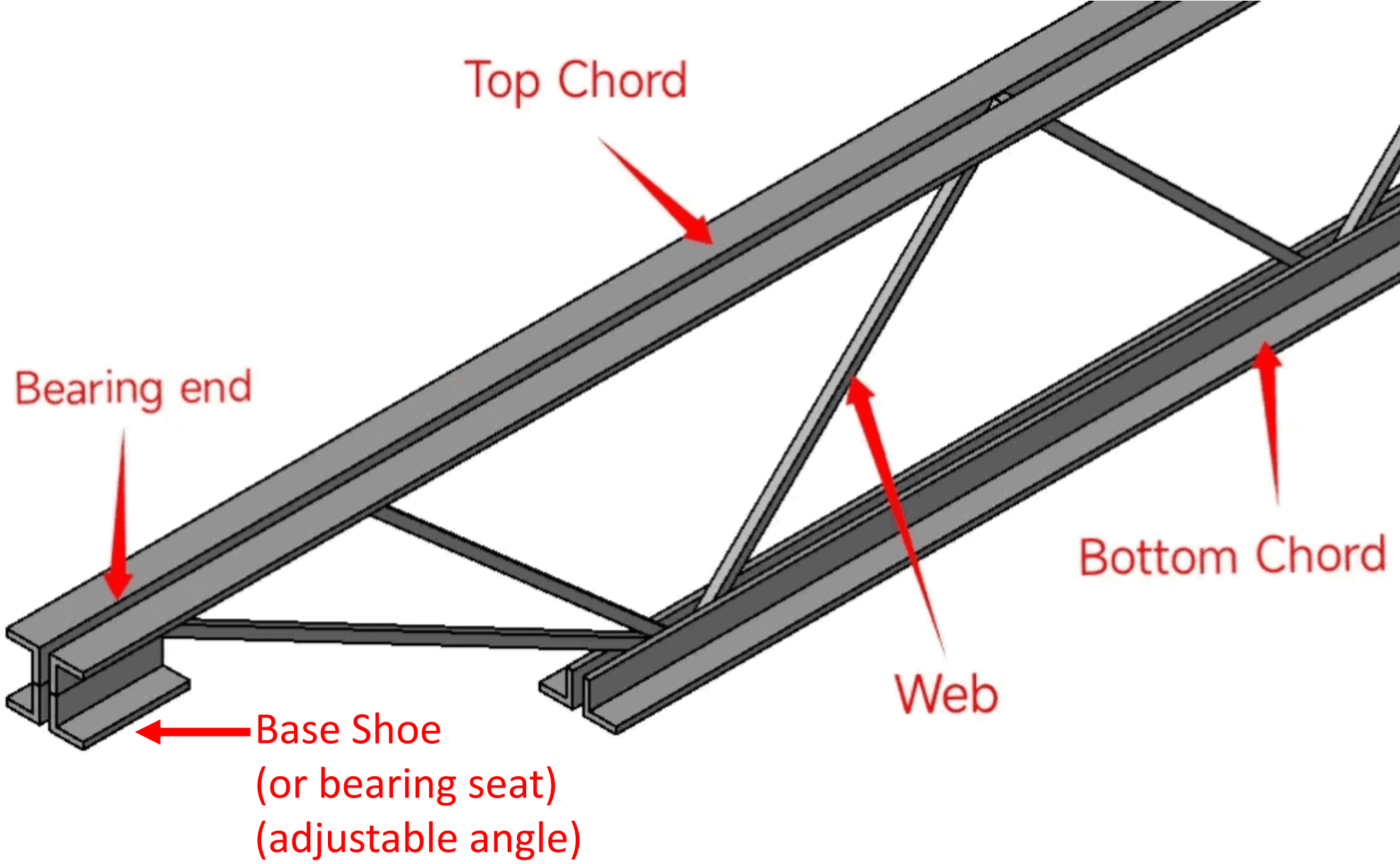
Parallel Chord Joist

Joists are top-chord bearing

Definition of Span



K-Series or LH-Series Joists (most common types, to be used for our project)



Components of an Open-Web Steel Joist (steel angles at chords; smaller angles at webs)

…what element supports the open-web steel joists?



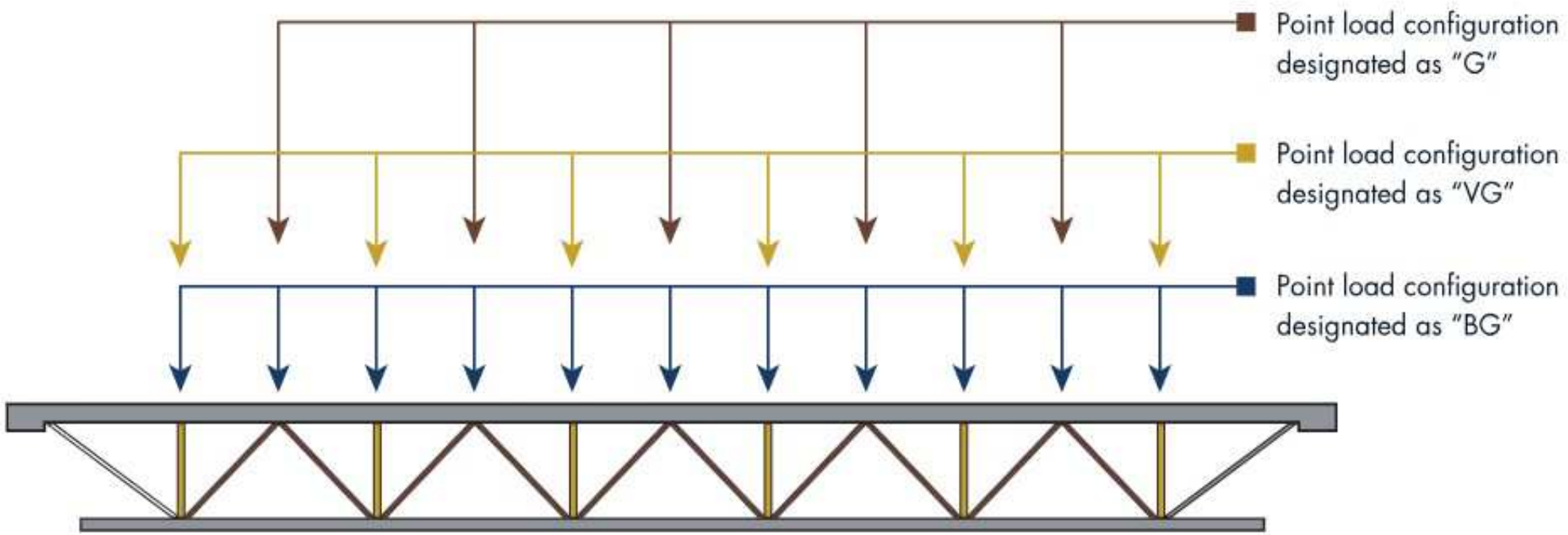
Structural Steel Corrugated Deck (1.5” corrugated most common)



Open-Web Steel Joists supported by Joist Girders


Joist Girders supported by Columns (HSS Tube or Wide-Flange) (top-chord bearing)

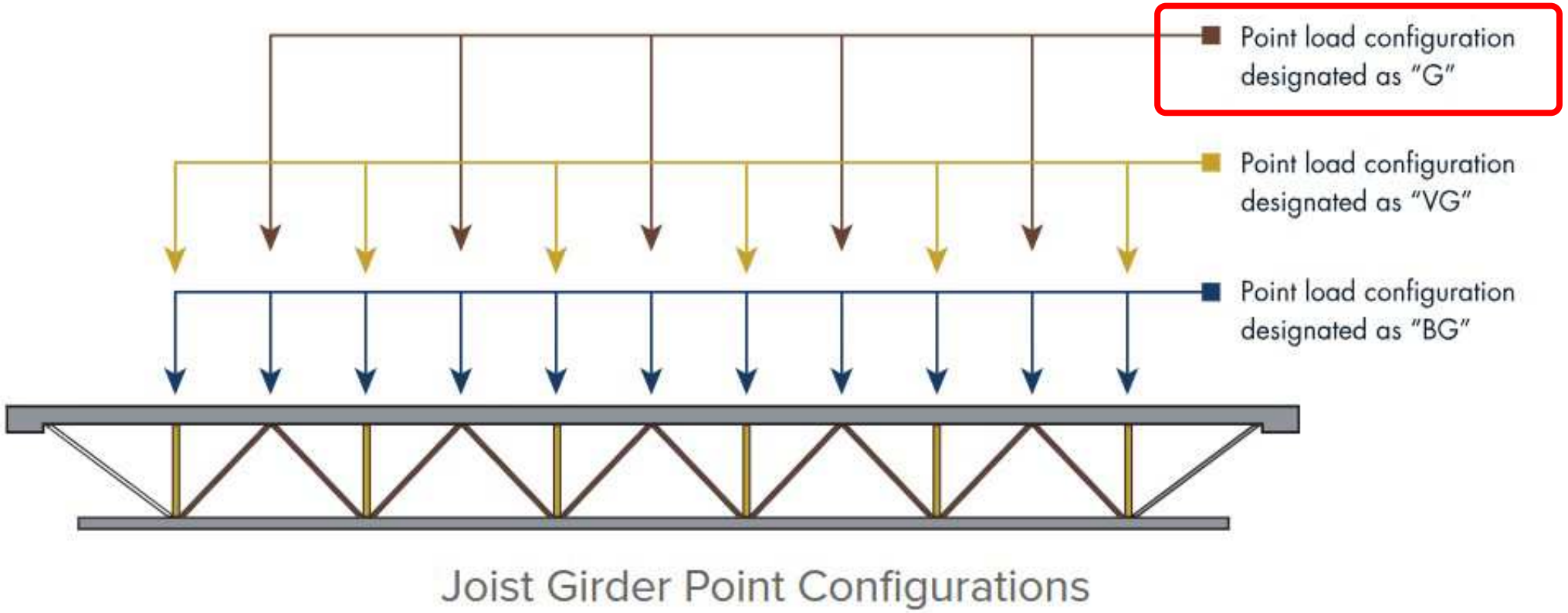
Joists Girders and Open-Web Joists are fully integrated with each other



Joist Girder Point Configurations

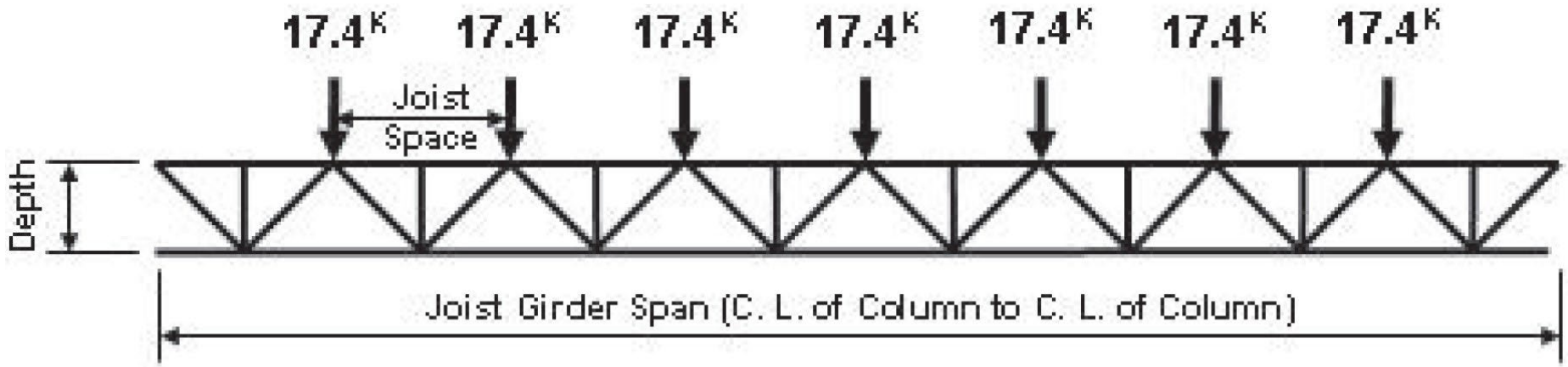
Joist Girder (parallel chord is the most common of the type)

Joists Girders and Open-Web Joists are fully integrated with each other



Joist Girder (we will use the most common G-type for point load configuration)

Definition of Span, and Nodes for Bearing of Open-Web Joists (G-configuration)

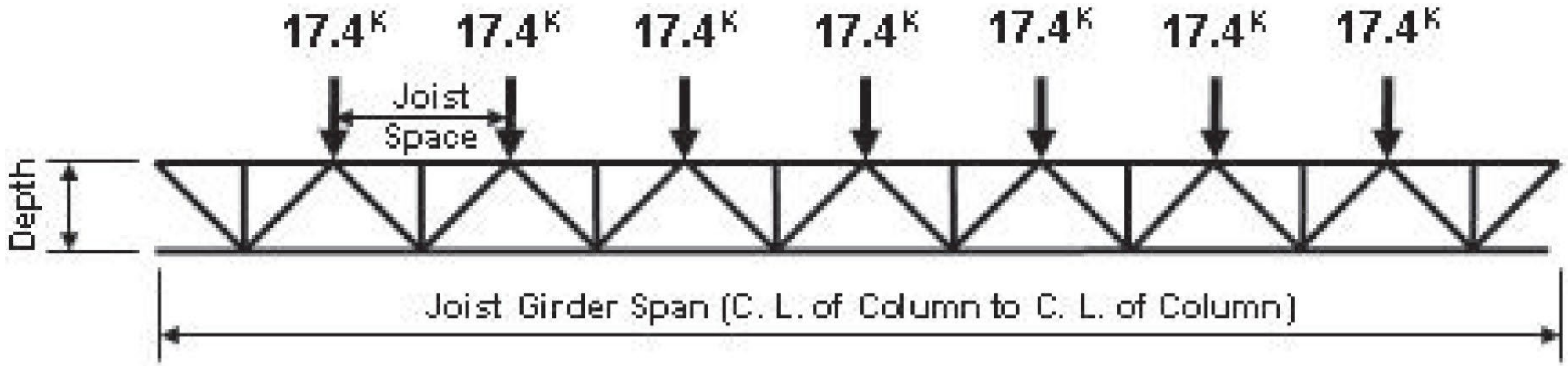


STANDARD DESIGNATION

44G	8N	17.4F
Depth in Inches	Number of Joist Spaces	Factored Load in Kips at Each Panel Point

Joist Girder (always designed in unison with Open-Web Joists and structural Deck)

Definition of Span, and Nodes for Bearing of Open-Web Joists (G-configuration)



STANDARD DESIGNATION

44G	8N	17.4F
Depth in Inches	Number of Joist Spaces	Factored Load in Kips at Each Panel Point

Joist Girder standard definitions in Span & Load Tables (used for roof structure)

STEEL BUILDINGS Corrugated Deck, Open-Web Joists & Joist Girders

…where do we see steel buildings with these elements?



…most big box stores (spans from 30’ to 40’, and up to 60’)

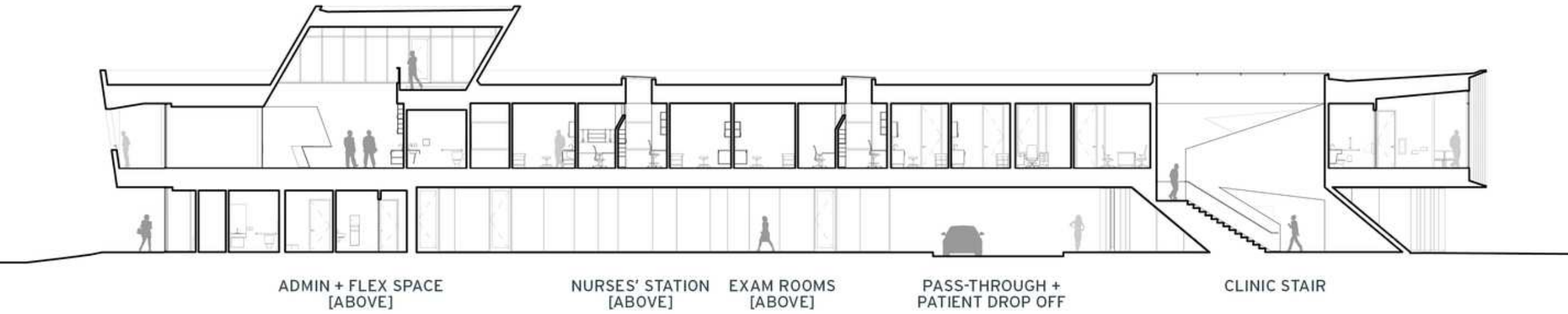

…look up at common buildings under construction

…they are also seen in works of architecture



HARVEY PEDIATRIC CLINIC Rogers, Arkansas **Marlon Blackwell, architect**

…open webs of structure allow integration of mechanical and electrical systems

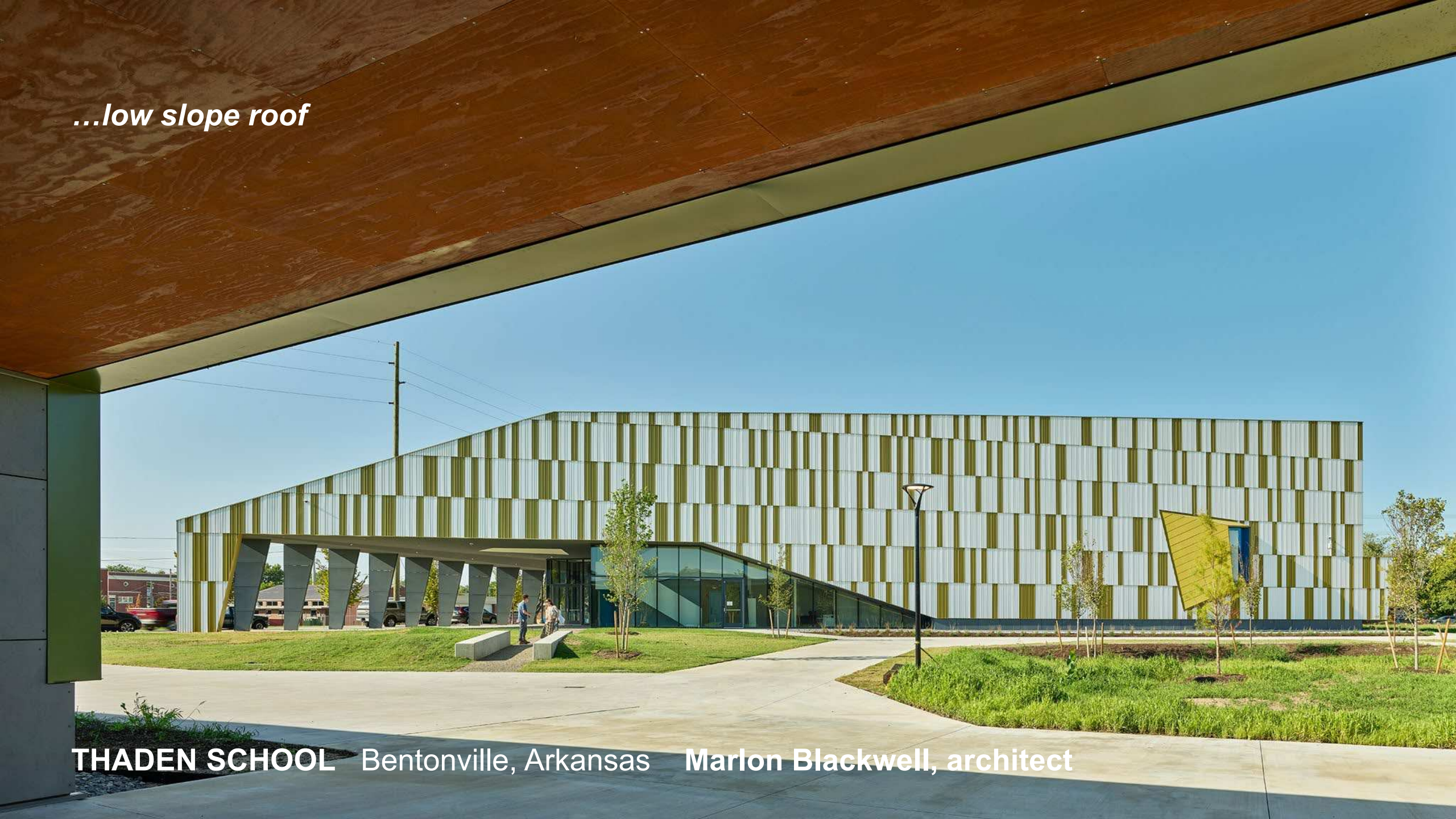


SECTION THROUGH CLINIC

0' 8' 16' 32'



…low slope roof



THADEN SCHOOL Bentonville, Arkansas **Marlon Blackwell, architect**

...low slope roof



MARYGROVE SCHOOL Detroit, Michigan **Marlon Blackwell, architect AIA National Award**

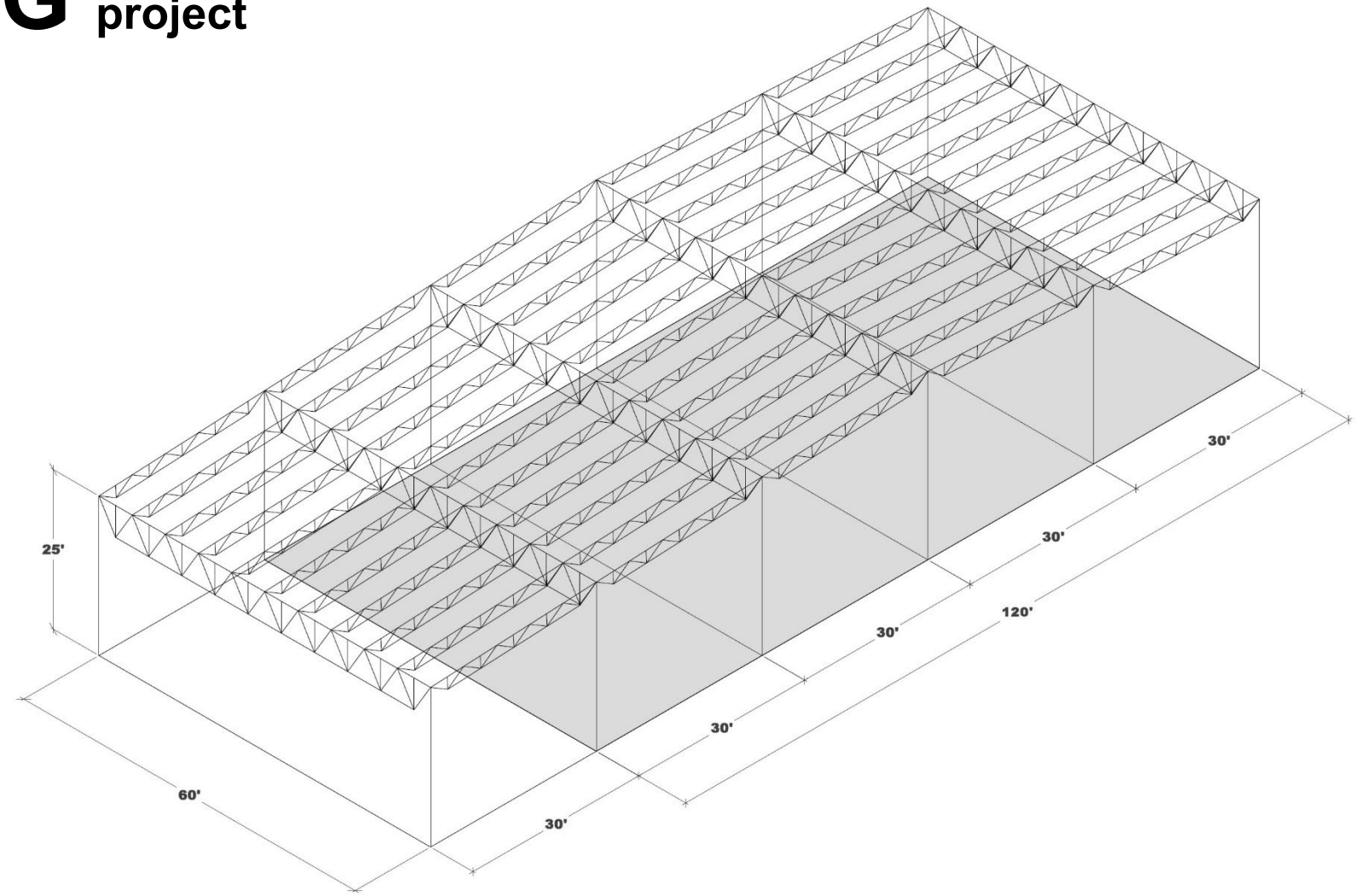
STEEL BUILDING project

…what roof can we make with steel corrugated Deck, Joists, and Joist Girders?

STEEL BUILDING project

…how can we optimize a structural system with these material elements?

STEEL BUILDING project



WEEK ONE

…configure roof structure

…define horizontal bracing

WEEK TWO

…configure floor structure

…configure lateral structure

WEEK THREE

…foundation/footing

PROJECT DUE: October 26

STEEL BUILDING project

PROJECT LEARNING OBJECTIVES

Demonstrate ability **to plan and configure the structure of a simple steel building.**
Demonstrate understanding **of basic modes of structural communication**
Demonstrate understanding **of basic structural systems in response to loads & forces**

…keep learning from the wood building project and theory lectures 1 - 5

STEEL BUILDING project

WEEK ONE (Oct 6-12)
…configure roof structure

Step 1: select a 3060 or 4060 layout and roof form, and decide on joist span direction
Step 2: define load on roof deck and roof deck span (related to joist girder panels)
Step 3: identify joist-space options for joist girder (based on joist girder tables)
Step 4: define open-web joists by load in plf from deck (based on joist tables)
Step 5: define primary joist girders (based on joist girder tables)
Step 6: 3d model overall configuration using Revit family (available on Canvas)
(Step 7: define horizontal bracing of open-web joists)

STEEL BUILDING project

PROJECT CONCEPTUAL PARAMETERS

***…Economy** of material and optimization of structural elements is the prime driver*

***…Functional** building size of 60 feet by 120 feet (nominal) conditioned space, and an open porch at one end*

***…Aesthetics** guided by Economy and Function*

…Economy as main conceptual parameter…"minimum material does maximum work"

STEEL BUILDING project

PROJECT PARAMETERS

ENGINEERING OFFICE: Ypsilanti, Michigan (60’ x 120’ nominal footprint, with one open-porch bay)
Snow Load: 20 psf Roofing DL: 15 psf

GROCERY STORE: Pinconning, Michigan (60’ x 120’ nominal footprint, with one open-porch bay)
Snow Load: 40 psf Roofing DL: 15 psf

PUBLIC LIBRARY: Houghton, Michigan (60’ x 120’ nominal footprint, with one open-porch bay)
Snow Load: 100 psf Roofing DL: 15 psf

Note: students rotate to the next building use & load criteria in line *(Grocery to Library etc…)*

STEEL BUILDING project

PROJECT PARAMETERS FOR ROOF

TERTIARY ELEMENTS: (span determined by specific snow load and ASD load calculations)

Type-B structural decking: 1.5” thick, 36” width (see span & load table)
Roof Deck must have a cantilever beyond Open-Web Steel Joists (only in deck span direction)

SECONDARY ELEMENTS: (spacing determined by corrugated steel deck span)

Open-Web Steel Joists: K or LH Series (size and self-weight by economy span & load tables)
Joists must have a top-chord extension for a cantilever beyond building footprint

PRIMARY ELEMENTS: (panel size determined by open-web joist spacing)

Joist Girders: Type-G configuration (size and self-weight by span & load tables for joist loads)
Joist Girders must have a top-chord extension for a cantilever beyond building footprint
Joist Girders supported on steel Columns (HSS tube or Wide-Flange)

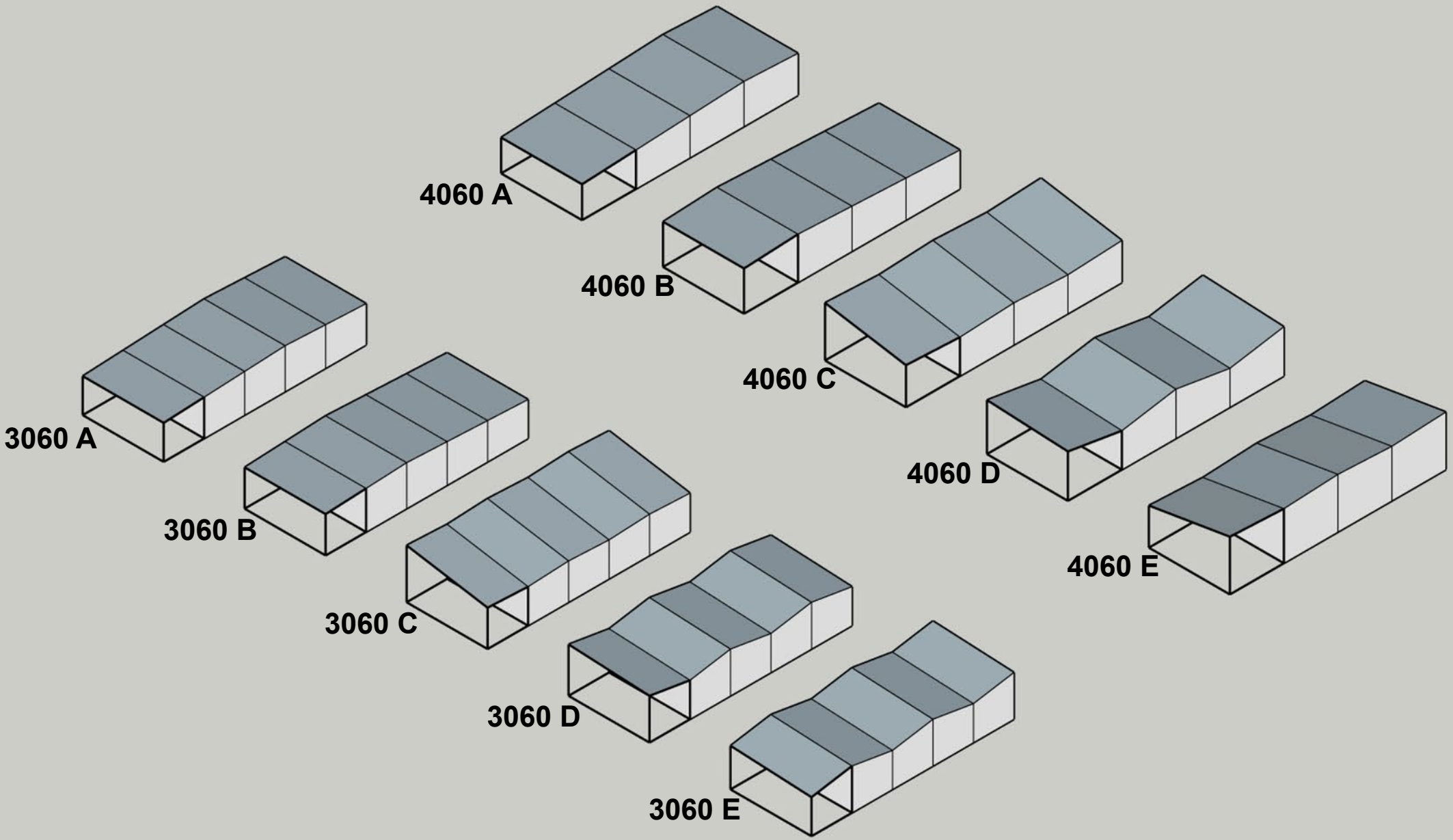
STEEL BUILDING project

BAY SIZE and ROOF CONFIGURATION OPTIONS with PARAMETERS

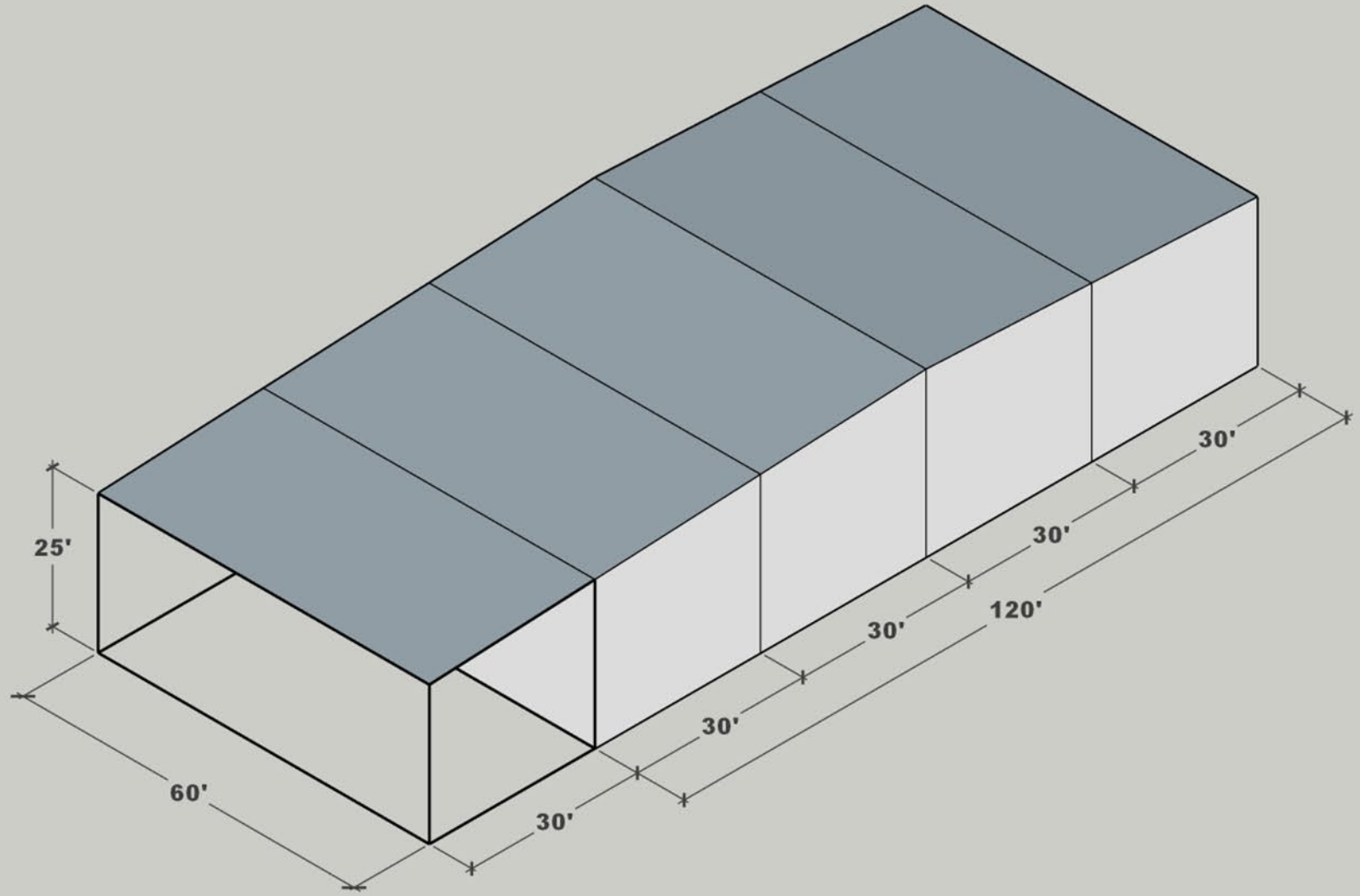
30’ x 60’ Roof Structural Bays

40’ x 60’ Roof Structural Bays

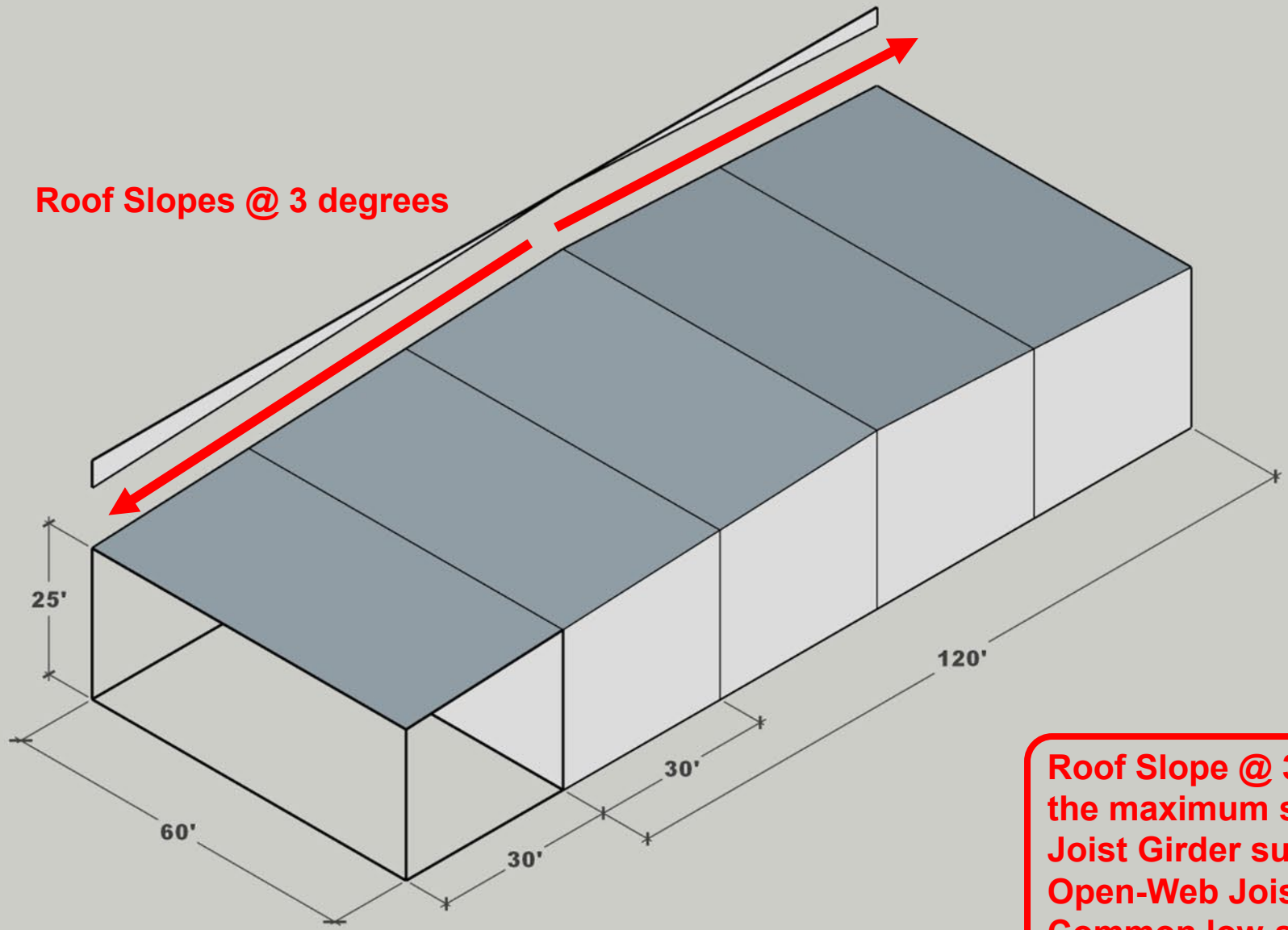
…pick one option for your project



Two Bay Sizes…Ten Roof Configuration Options, each with an open “Porch” at one end

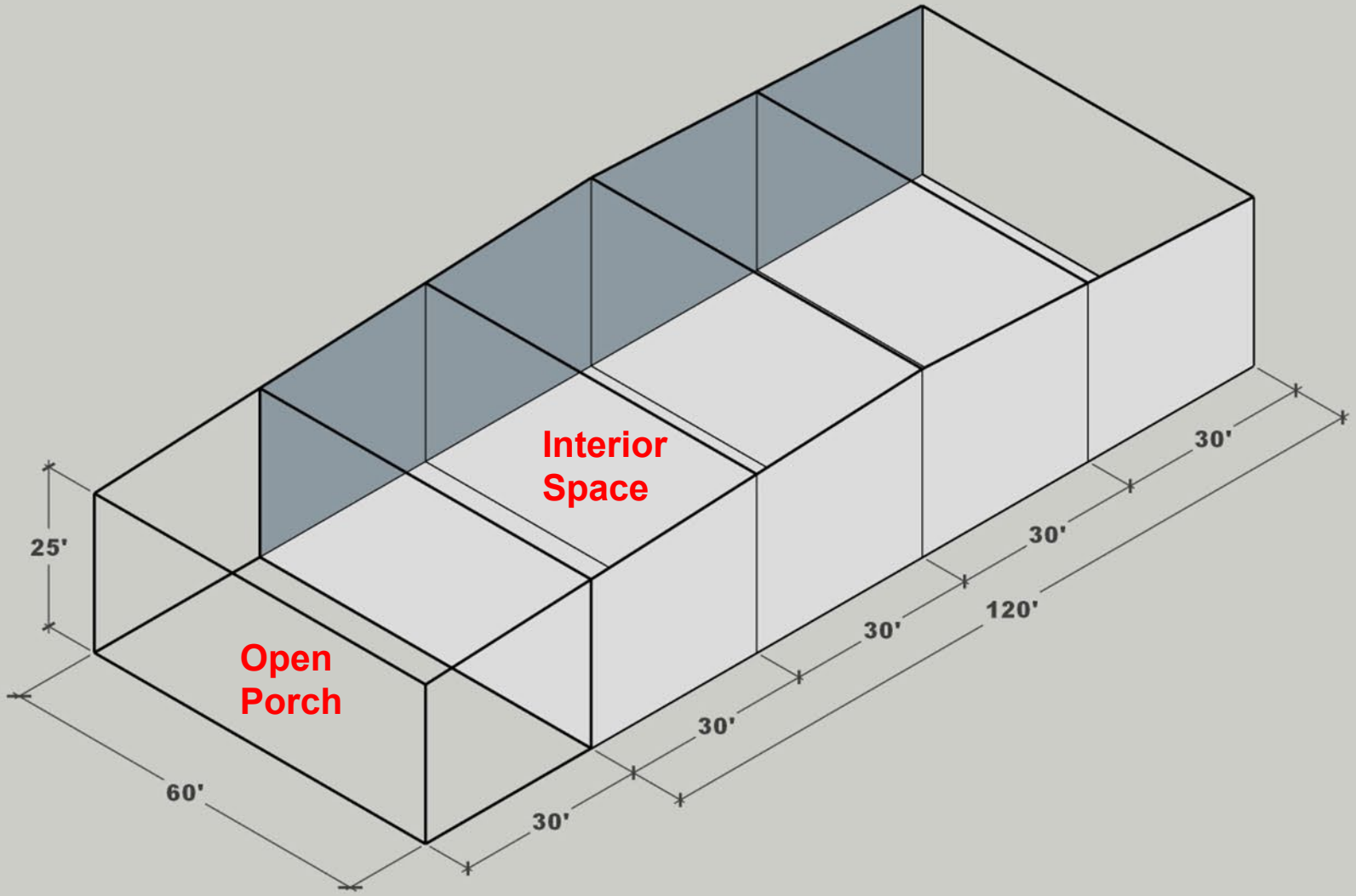


3060 A (30’ x 60’ structural roof bay)



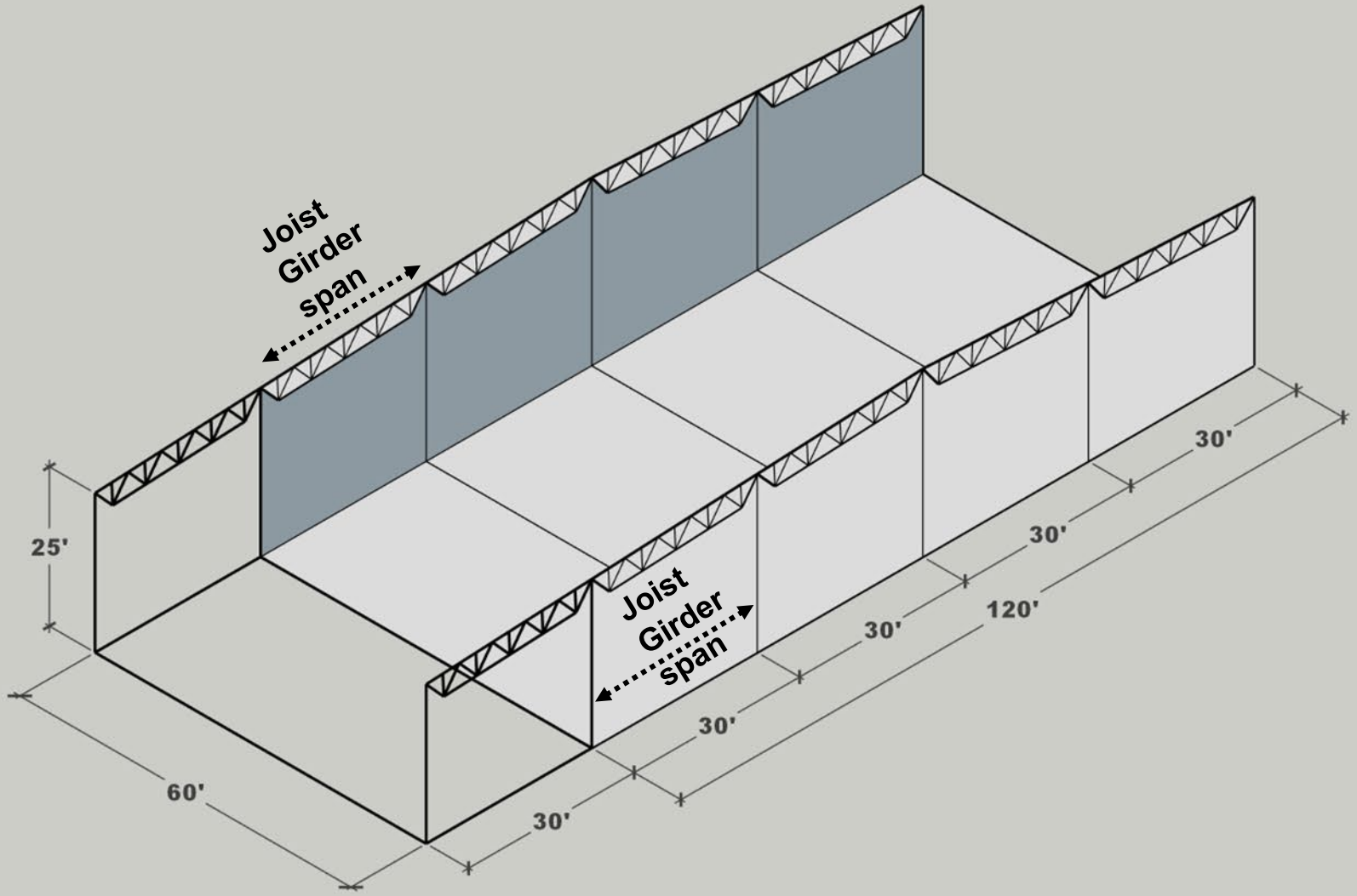
Roof Slope @ 3 degrees is the maximum slope for a Joist Girder supporting Open-Web Joists. Common low-slope for metal roofing or membrane roofing is 3 degrees .

3060 A Roof Slope



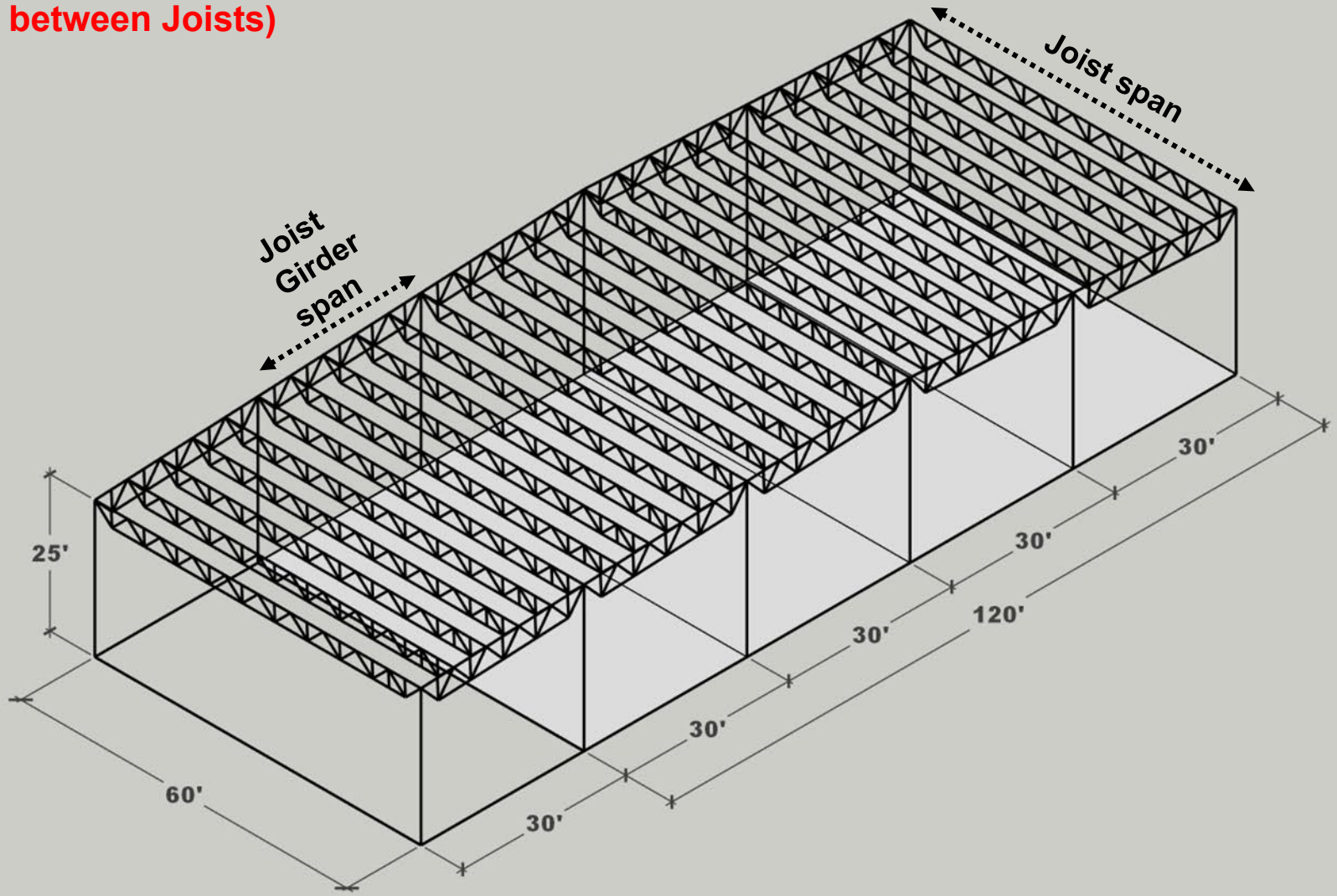
3060 A (Open-Web Joists and Joist Girders can work two ways)

Option 1: Joist Girders span 30’ to Columns



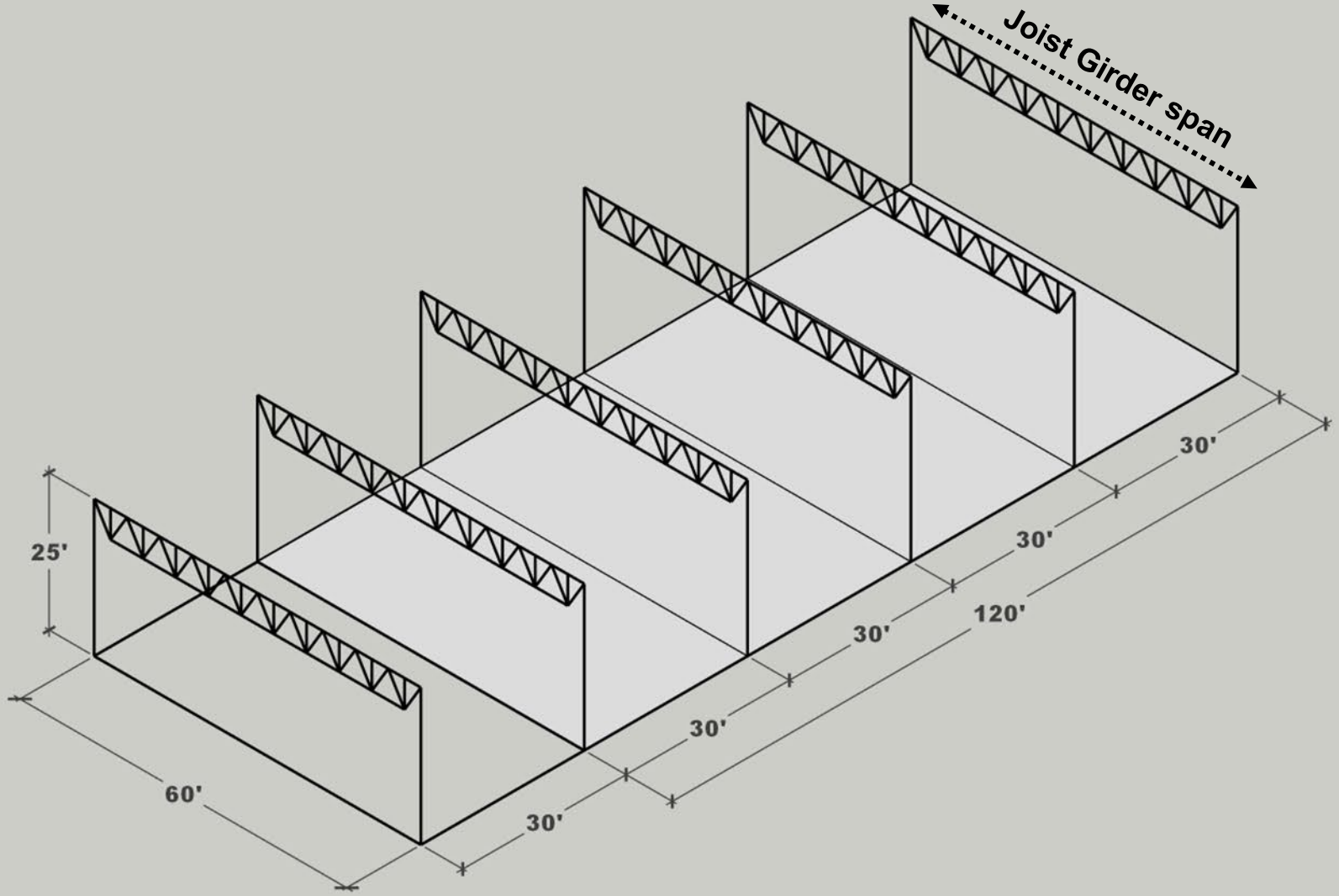
3060 A

Option 1: **Joists span 60’ to Joist Girders**
(Deck spans between Joists)



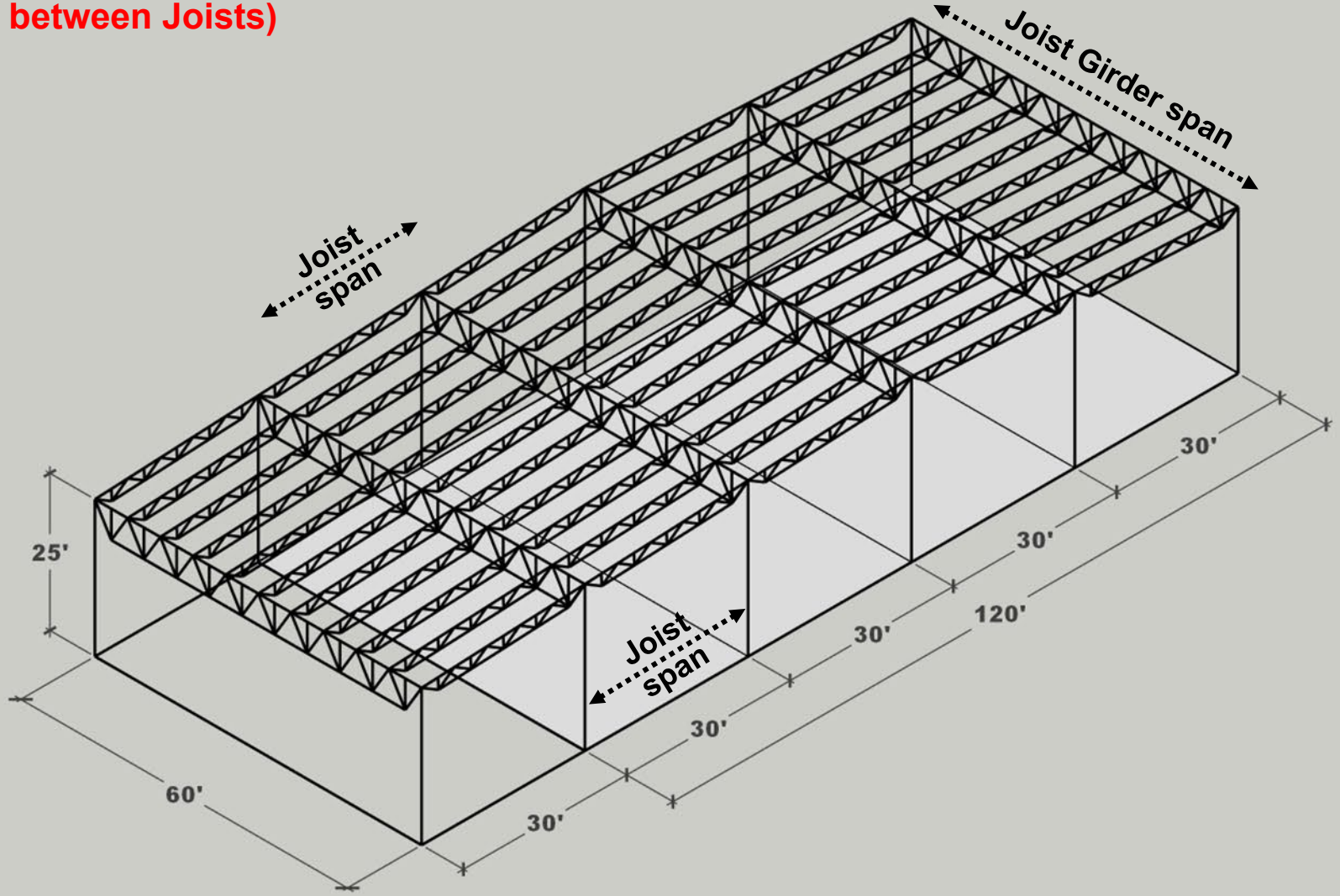
3060 A

Option 2: Joist Girders span 60’ to Columns

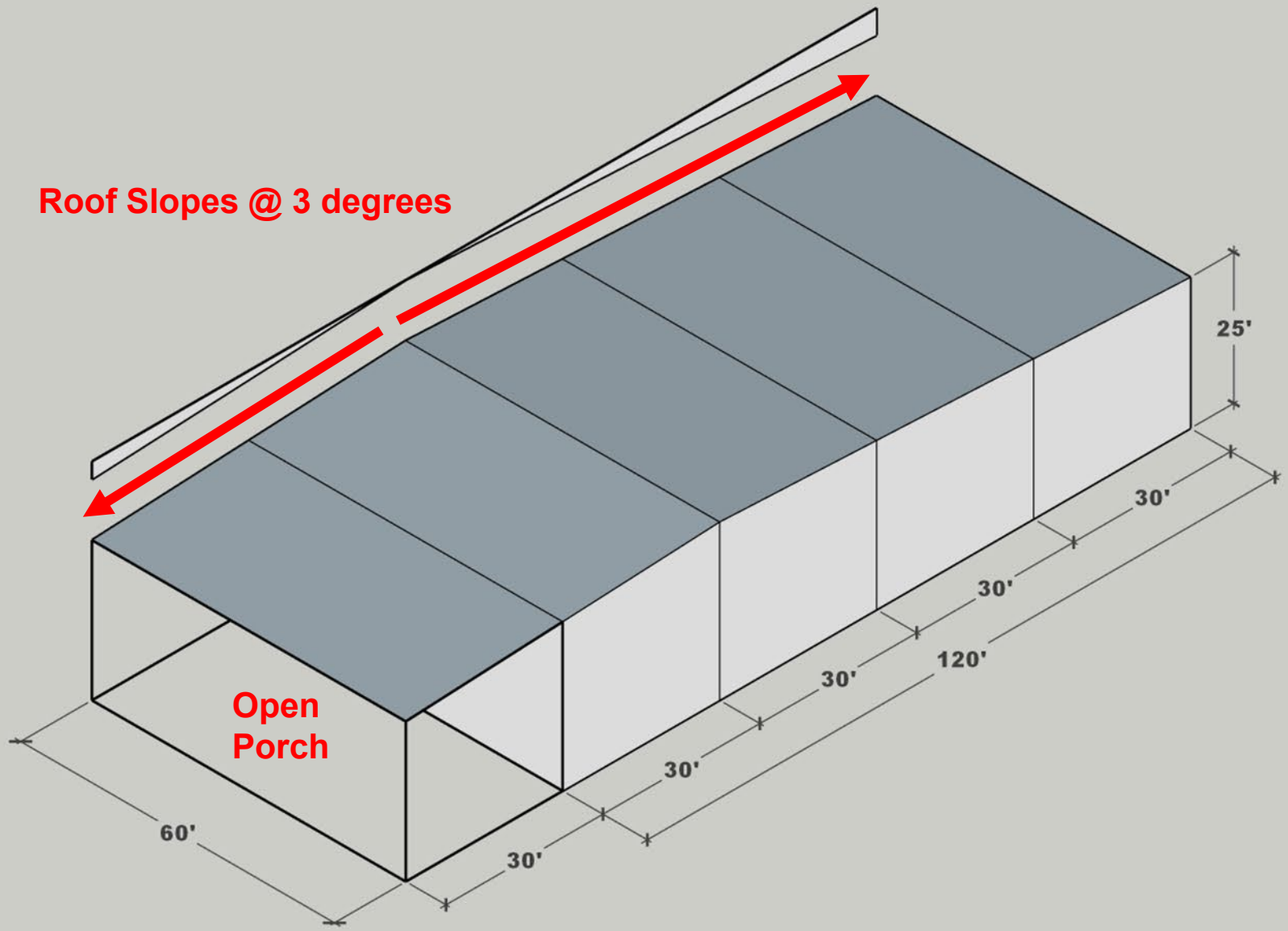


3060 A

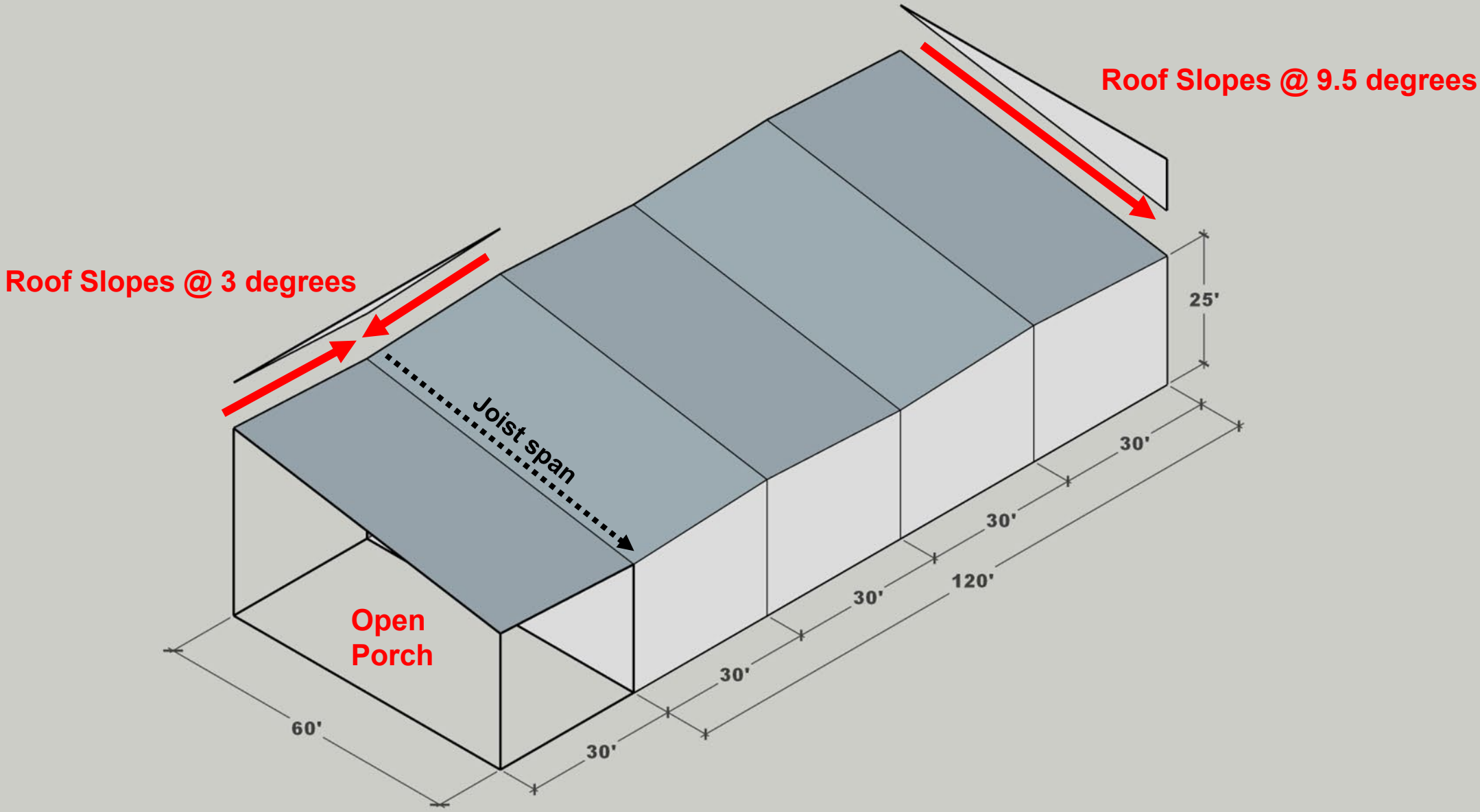
Option 2: **Joists span 30’ to Joist Girders**
(Deck spans between Joists)



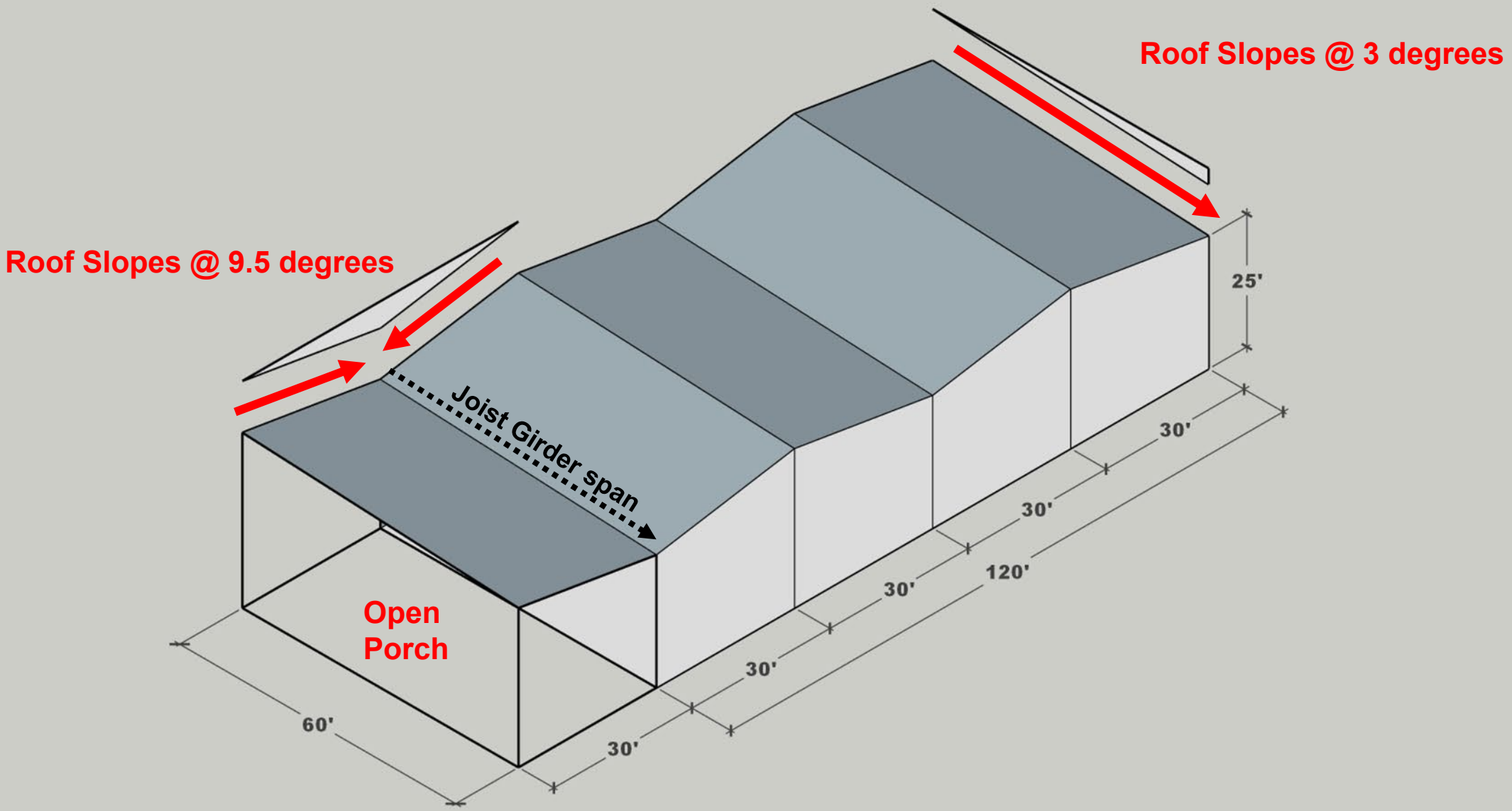
3060 A



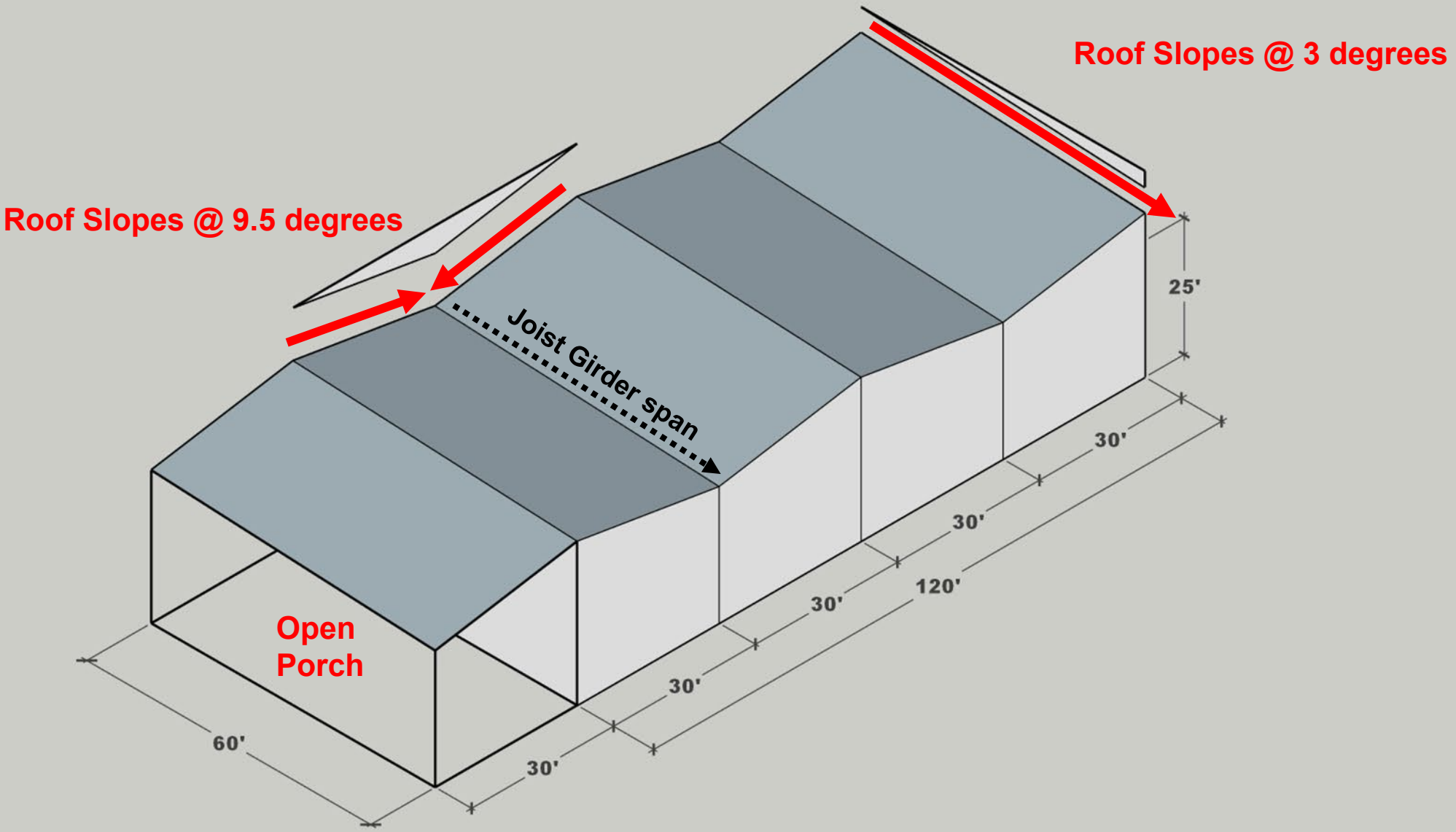
3060 B (30’ x 60’ structural roof bay) **(Joists and Joist Girders have 2 options** like 3060 A)



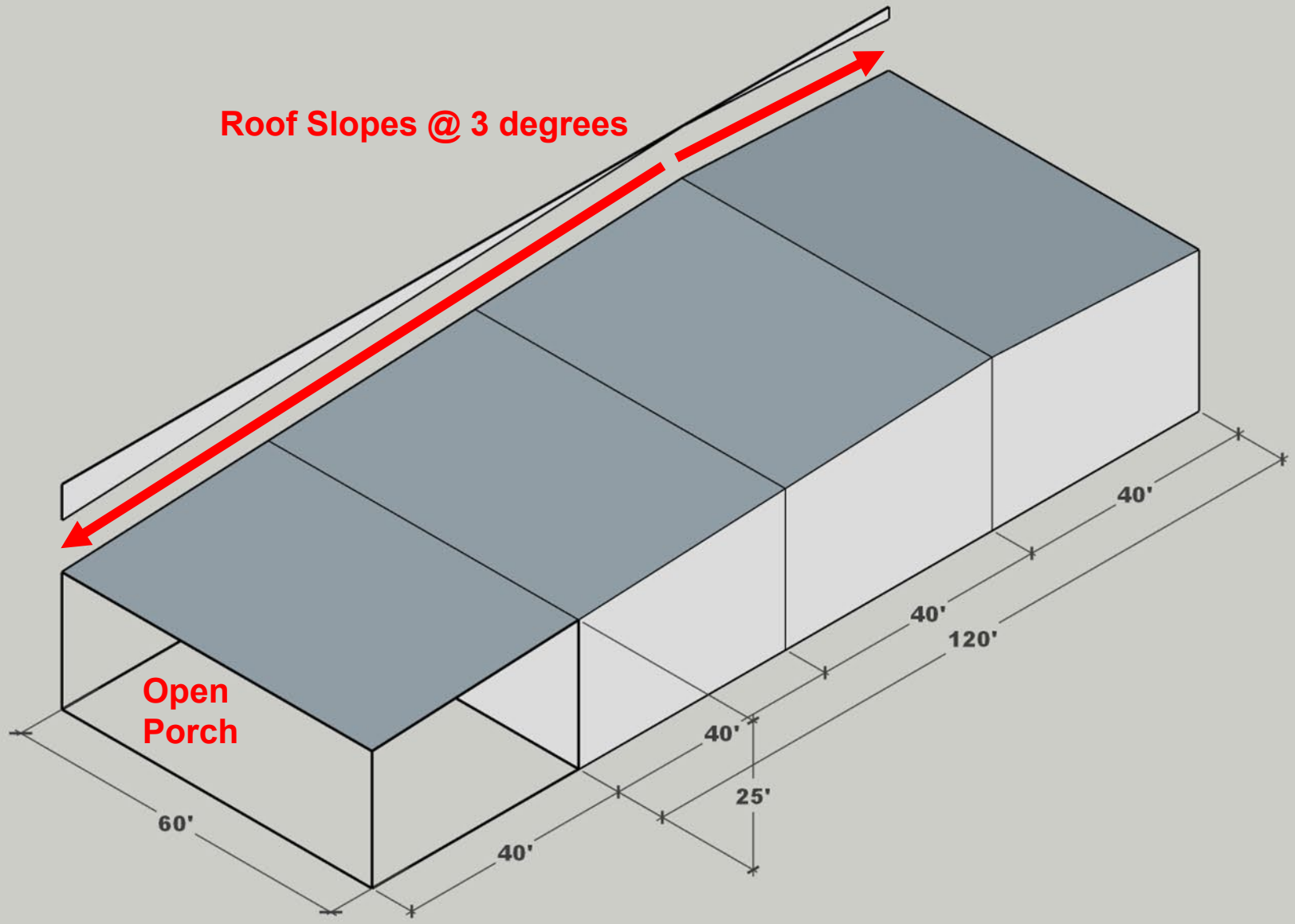
3060 C (30’ x 60’ structural roof bay) **(Joists <u>must</u> span 60’)**



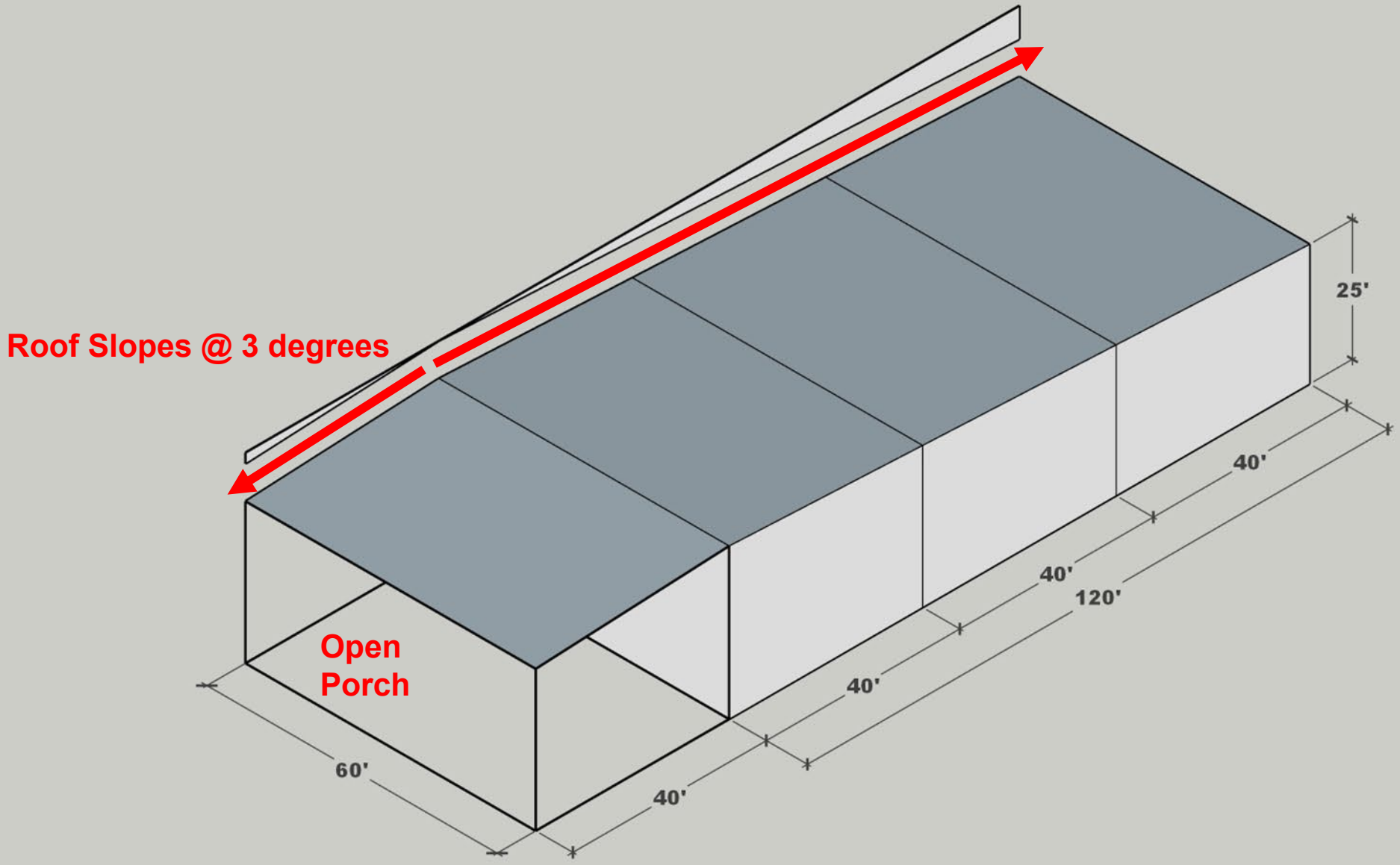
3060 D (30’ x 60’ structural roof bay) **(Joist Girders must span 60’)**



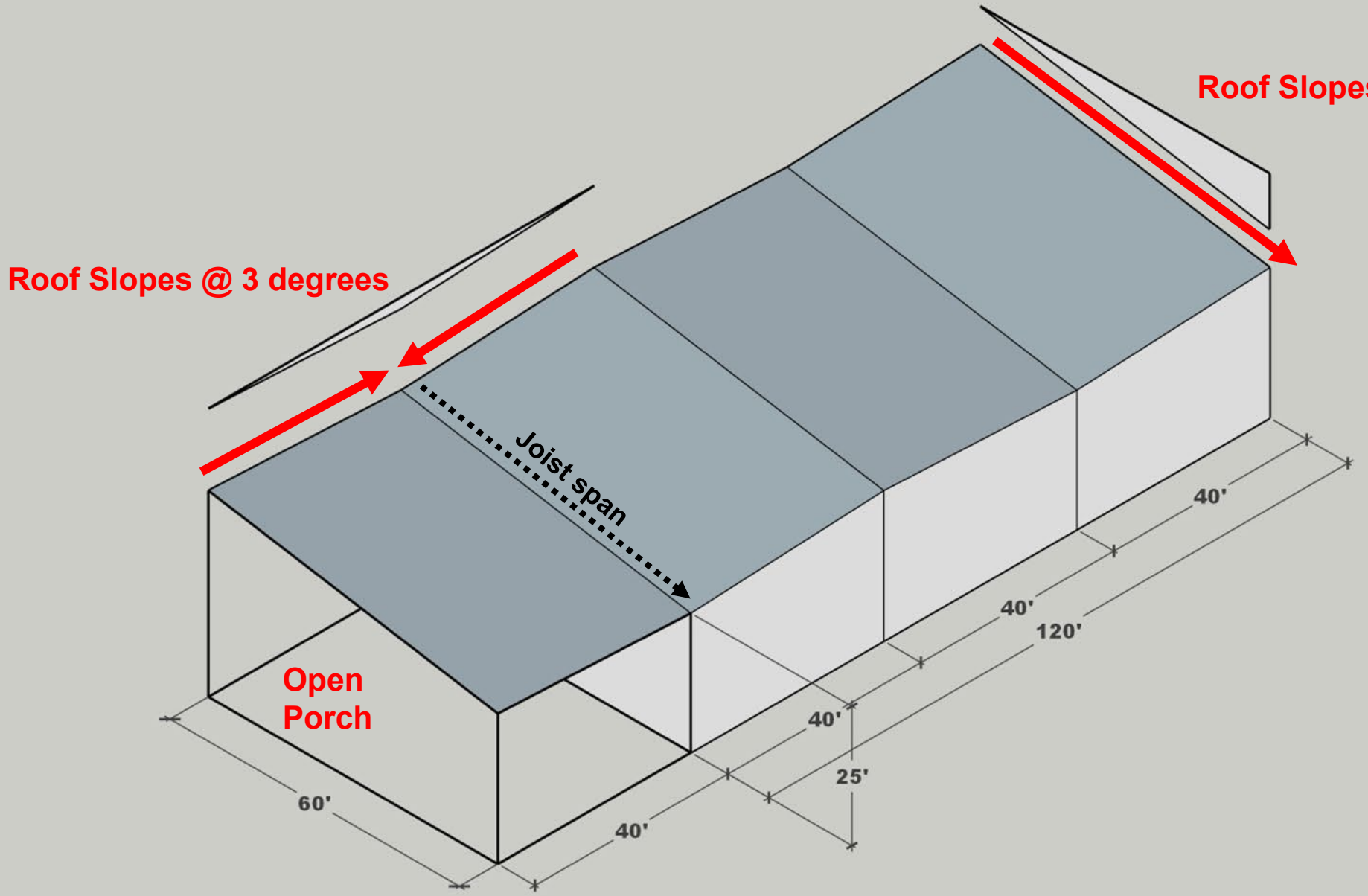
3060 E (30’ x 60’ structural roof bay) **(Joist Girders must span 60’)**



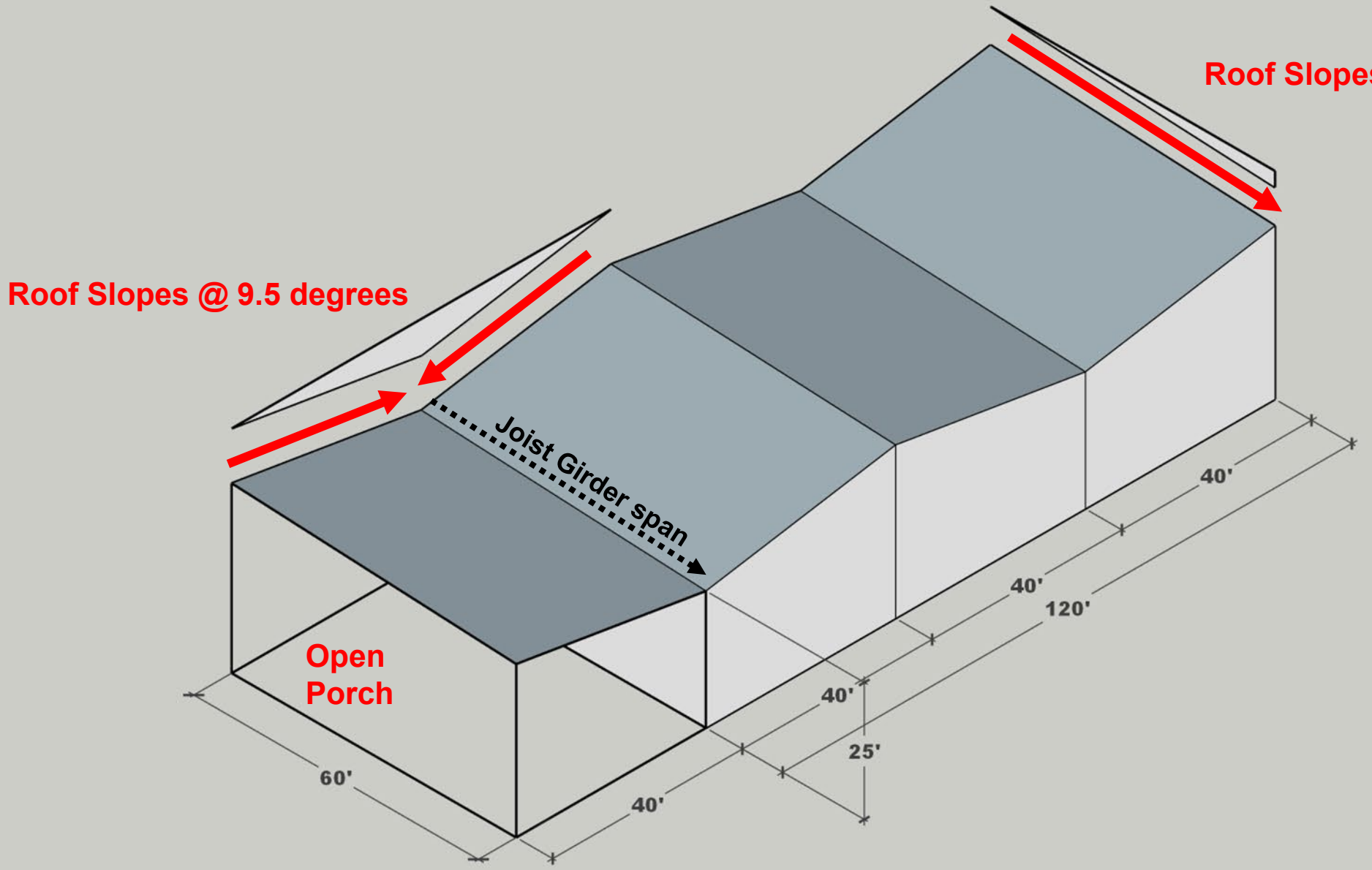
4060 A (40’ x 60’ Bays) **(Open-Web Joists and Joist Girders can work 2 ways)**



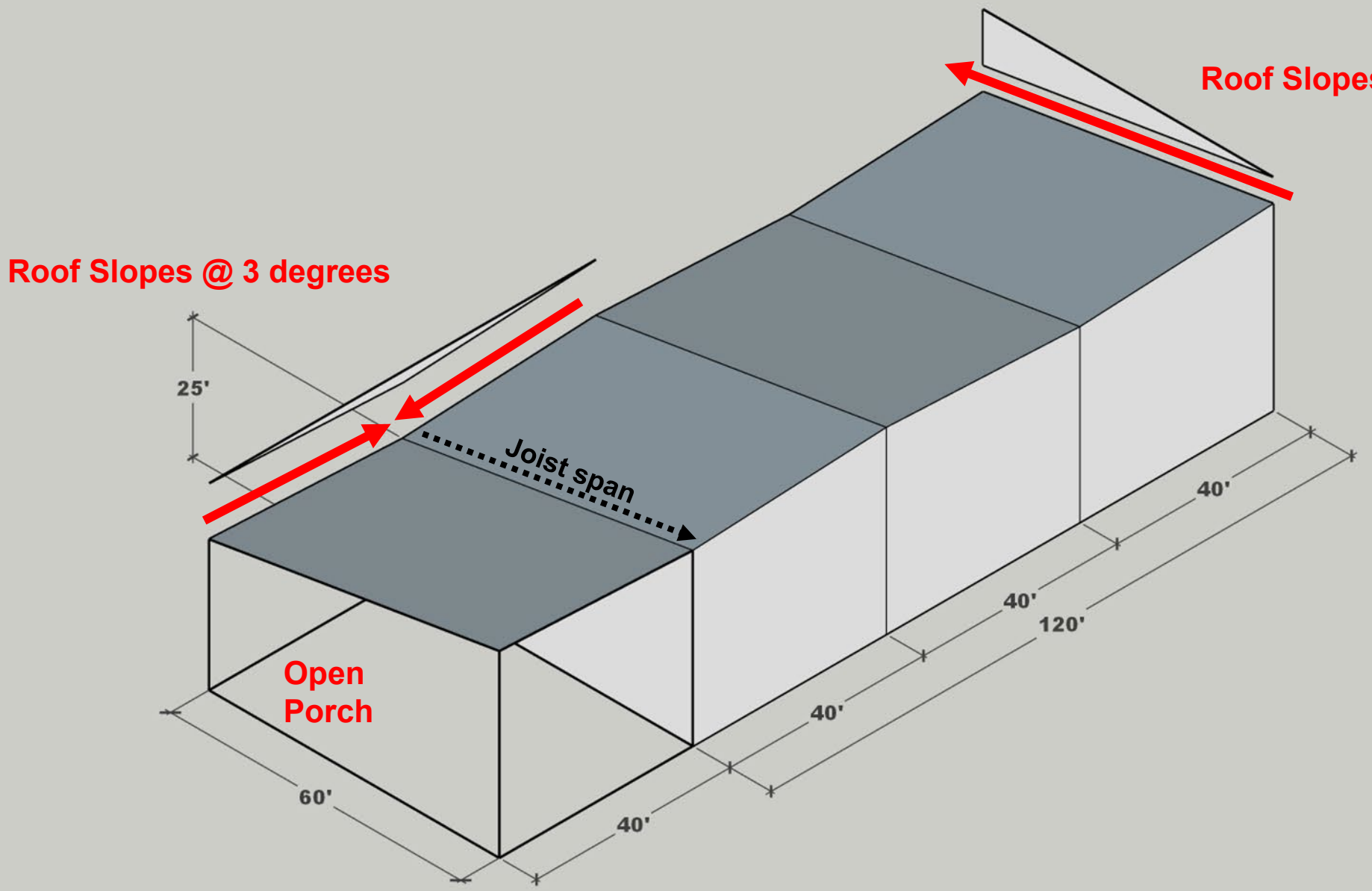
4060 B (40’ x 60’ Bays) **(Open-Web Joists and Joist Girders can work 2 ways)**



4060 C (40’ x 60’ Bays) **(Joists must span 60’)**



4060 D (40’ x 60’ Bays) **(Joist Girders must span 60’)**



4060 E (40’ x 60’ Bays) **(Joists <u>must</u> span 60’)**

STEEL BUILDING project

…what is the process for sizing roof deck, joists, and joist girders?

STEEL BUILDING project

WEEK ONE (Oct 6-12)
…configure roof structure

Step 1: select a 3060 or 4060 layout and roof form, and decide on joist span direction
Step 2: define load on roof deck and roof deck span (related to joist girder panels)
Step 3: identify joist-space options for joist girder (based on joist girder tables)
Step 4: define open-web joists by load in plf from deck (based on joist tables)
Step 5: define primary joist girders (based on joist girder tables)
Step 6: 3d model overall configuration using Revit family (available on Canvas)
(Step 7: define horizontal bracing of open-web joists)

LIBRARY as a STEEL BUILDING

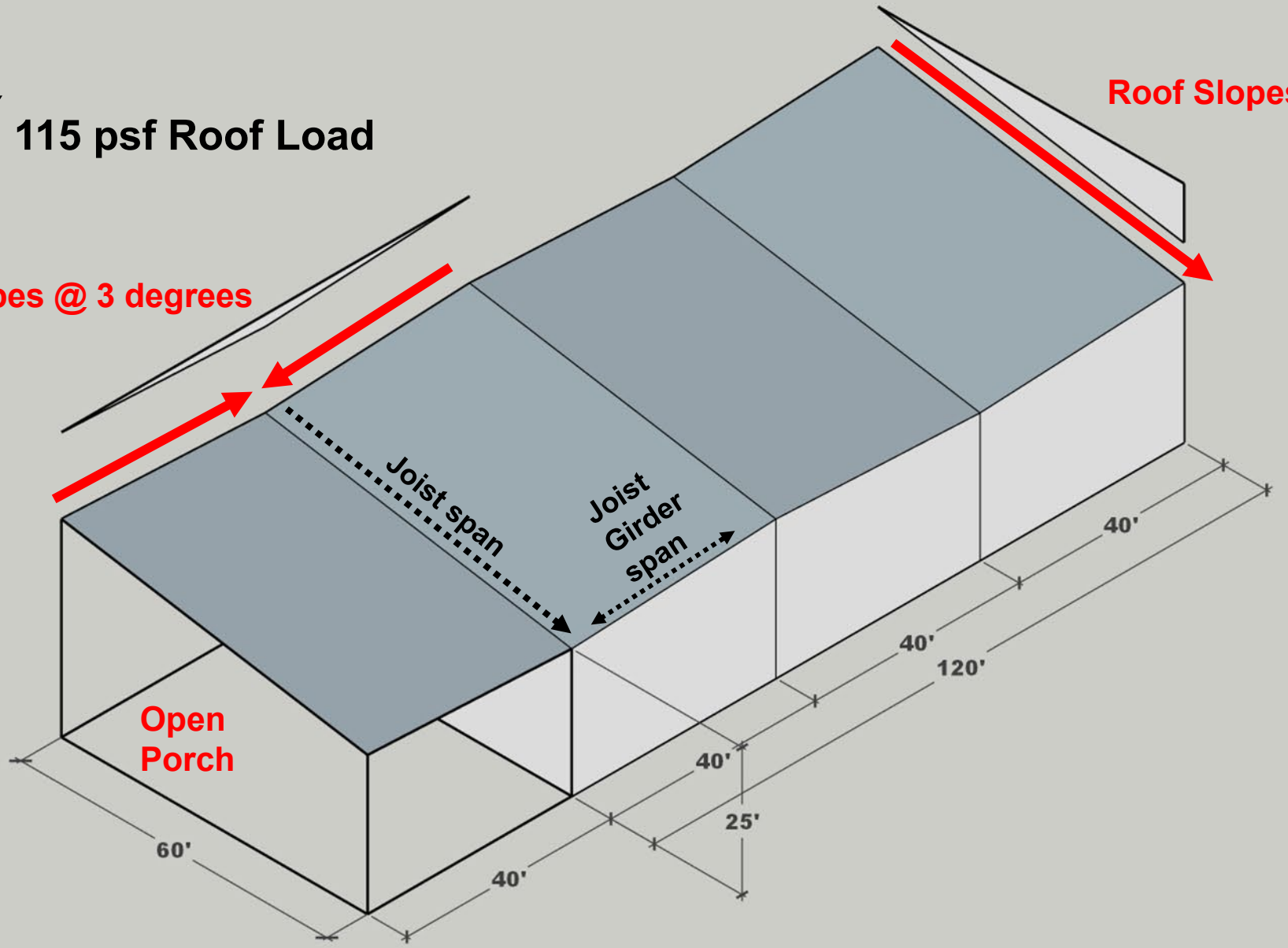
LIBRARY as a STEEL BUILDING

PROJECT PARAMETERS

PUBLIC LIBRARY: Houghton, Michigan (60’ x 120’ nominal footprint, with one open-porch bay)

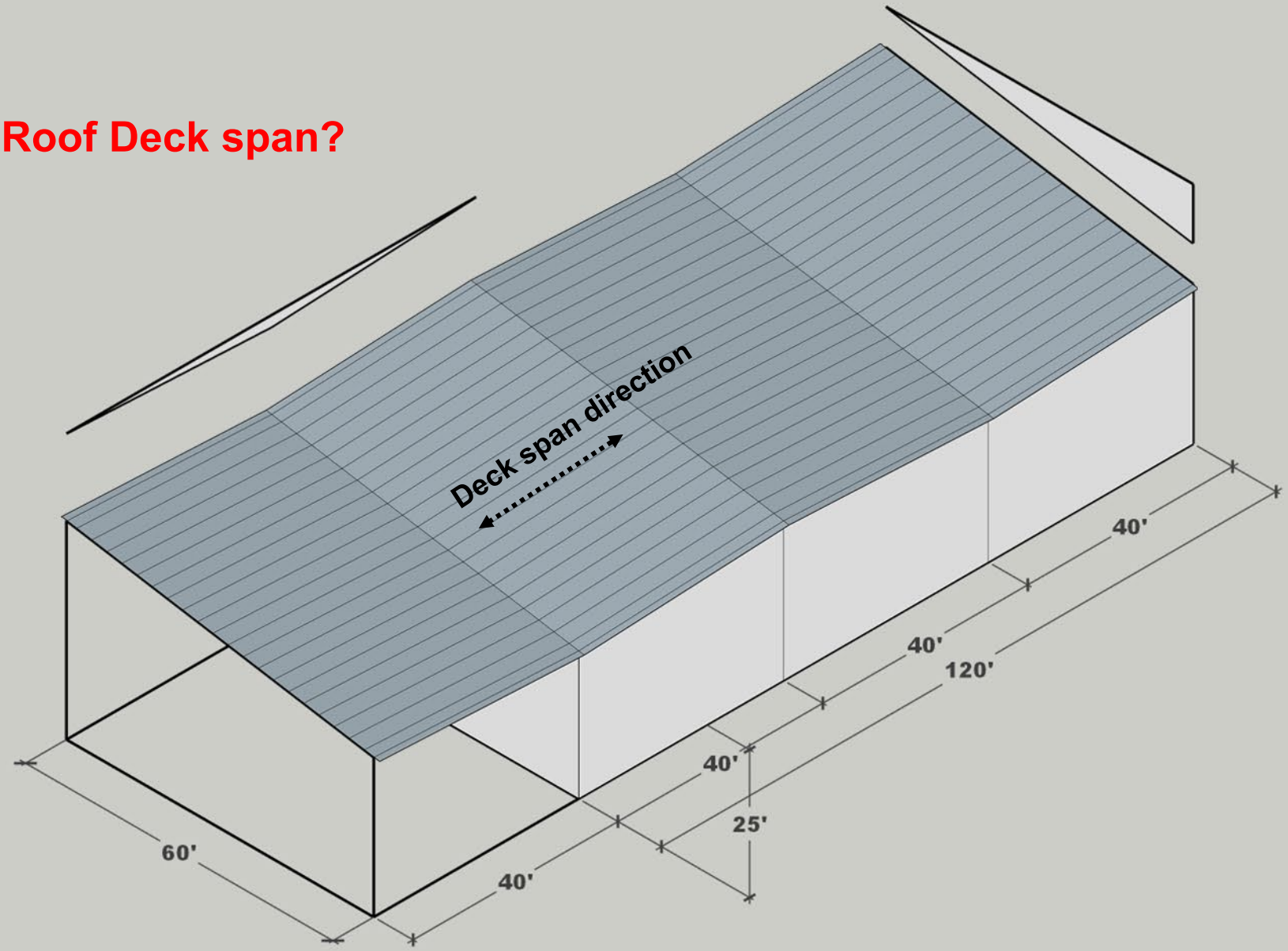
Snow Load: 100 psf Roofing DL: 15 psf **115 psf = Snow Load + Roofing Dead Load**

LIBRARY 115 psf Roof Load



4060 C (40’ x 60’ Bays) **(Joists <u>must</u> span 60’)**

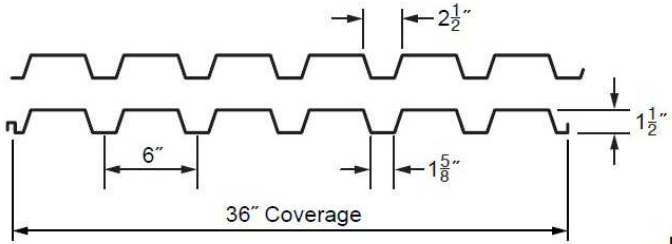
How far can the Roof Deck span?



4060 C Type-B Roof Deck (1.5” x 36” wide panels)

PROPERTIES

Gage	Thickness (in)	Cover (in)	Weight (psf)
22	0.0295	36	1.63
20	0.0358		1.98
18	0.0474		2.62
16	0.0598		3.30



ICC-ES APPROVED
REFER TO ESR-3818

ASD

SECTION PROPERTIES / DESIGN STRENGTHS

Gage	F_y (ksi)	I_D 1 Span (in^4/ft)	I_D 2+ Span (in^4/ft)	I_p (in^4/ft)	I_n (in^4/ft)	S_p (in^3/ft)	S_n (in^3/ft)	Mn,p/Ω (in-lb/ft)	Mn,n/Ω (in-lb/ft)	Vn/Ω (lb/ft)	*Rbe/Ω (lb/ft)	*Rbi/Ω (lb/ft)
22	40	0.163	0.175	0.157	0.175	0.176	0.183	4215	4391	2009	654	1181
20	40	0.205	0.213	0.201	0.213	0.224	0.235	5375	5624	2428	932	1696
18	40	0.280	0.281	0.279	0.281	0.304	0.315	7291	7553	3192	1557	2858
16	40	0.355	0.355	0.355	0.355	0.390	0.395	9332	9454	3996	2381	4396

Notes:
1. Section properties are calculated in accordance with the AISI S100-16.
2. Web crippling design strengths* are based on minimum bearing lengths of 1 1/2" for end bearing and 3" for interior bearing.



MAXIMUM CANTILEVER SPANS

Gage	F_y (ksi)	Back-Span Condition		
		Single	Double	Triple
22	40	1 - 8	1 - 8	1 - 8
20	40	2 - 0	1 - 11	1 - 11
18	40	2 - 6	2 - 3	2 - 3
16	40	3 - 1	2 - 7	2 - 7

ALLOWABLE UNIFORM LOADS AND MAXIMUM CONSTRUCTION SPANS

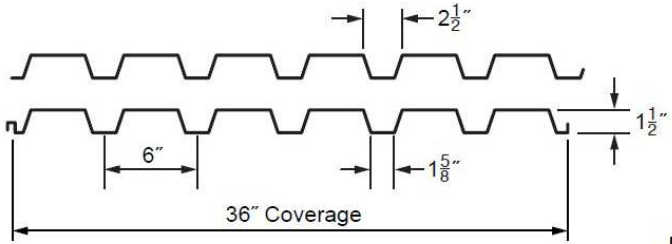
Allowable Load in psf (L/240)

Span Condition	Gage	Allowable Uniform Total Load (psf) / Load that Produces L/240 Deflection (psf) — Center to Center Span (ft - in)										SDI Max. Constr. Span (ctr / ctr)
		5 - 0	5 - 6	6 - 0	6 - 6	7 - 0	8 - 0	9 - 0	10 - 0	11 - 0	12 - 0	
Single	22	112 / 86	93 / 64	78 / 50	67 / 39	57 / 31	-	-	-	-	-	6 - 10
	20	143 / 107	118 / 81	100 / 62	85 / 49	73 / 39	56 / 26	44 / 18	-	-	-	8 - 7
	18	194 / 147	161 / 110	135 / 85	115 / 67	99 / 54	76 / 36	60 / 25	49 / 18	40 / 14	34 / 11	11 - 3
	16	249 / 186	206 / 140	173 / 108	147 / 85	127 / 68	97 / 45	77 / 32	62 / 23	51 / 17	43 / 13	13 - 11
Double	22	115 / 221	95 / 166	80 / 128	69 / 101	59 / 81	45 / 54	36 / 38	-	-	-	8 - 4
	20	147 / 269	122 / 202	103 / 156	88 / 122	76 / 98	58 / 66	46 / 46	37 / 34	31 / 25	-	10 - 6
	18	198 / 356	164 / 267	138 / 206	118 / 162	102 / 130	78 / 87	62 / 61	50 / 44	41 / 33	35 / 26	13 - 9
	16	247 / 448	205 / 337	173 / 260	147 / 204	127 / 163	98 / 109	77 / 77	63 / 56	52 / 42	44 / 32	16 - 11
Triple	22	143 / 173	119 / 130	100 / 100	85 / 79	74 / 63	57 / 42	45 / 30	-	-	-	8 - 6
	20	183 / 211	152 / 158	128 / 122	109 / 96	94 / 77	72 / 51	57 / 36	47 / 26	39 / 20	-	10 - 8
	18	245 / 278	203 / 209	172 / 161	147 / 127	127 / 101	97 / 68	77 / 48	63 / 35	52 / 26	43 / 20	13 - 11
	16	307 / 351	255 / 264	215 / 203	183 / 160	159 / 128	122 / 86	96 / 60	78 / 44	65 / 33	54 / 25	17 - 1

Steel Corrugated Deck B-Type: SPAN & LOAD TABLE

PROPERTIES

Gage	Thickness (in)	Cover (in)	Weight (psf)
22	0.0295	36	1.63
20	0.0358		1.98
18	0.0474		2.62
16	0.0598		3.30



ICC-ES APPROVED
REFER TO ESR-3818

ASD

SECTION PROPERTIES / DESIGN STRENGTHS

Gage	F_y (ksi)	I_D 1 Span (in^4/ft)	I_D 2+ Span (in^4/ft)	I_p (in^4/ft)	I_n (in^4/ft)	S_p (in^3/ft)	S_n (in^3/ft)	Mn,p/Ω (in-lb/ft)	Mn,n/Ω (in-lb/ft)	Vn/Ω (lb/ft)	*Rbe/Ω (lb/ft)	*Rbi/Ω (lb/ft)
22	40	0.163	0.175	0.157	0.175	0.176	0.183	4215	4391	2009	654	1181
20	40	0.205	0.213	0.201	0.213	0.224	0.235	5375	5624	2428	932	1696
18	40	0.280	0.281	0.279	0.281	0.304	0.315	7291	7553	3192	1557	2858
16	40	0.355	0.355	0.355	0.355	0.390	0.395	9332	9454	3996	2381	4396

Notes:
1. Section properties are calculated in accordance with the AISI S100-16.
2. Web crippling design strengths* are based on minimum bearing lengths of 1 1/2" for end bearing and 3" for interior bearing.



MAXIMUM CANTILEVER SPANS

Gage	F_y (ksi)	Back-Span Condition		
		Single	Double	Triple
22	40	1 - 8	1 - 8	1 - 8
20	40	2 - 0	1 - 11	1 - 11
18	40	2 - 6	2 - 3	2 - 3
16	40	3 - 1	2 - 7	2 - 7

ALLOWABLE UNIFORM LOADS AND MAXIMUM CONSTRUCTION SPANS

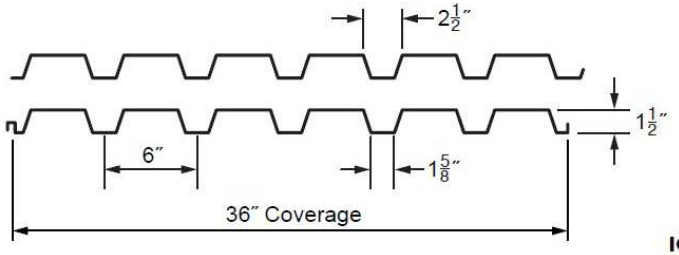
Allowable Load in psf (L/240)

Span Condition	Gage	Allowable Uniform Total Load (psf) / Load that Produces L/240 Deflection (psf) Center to Center Span (ft - in)										SDI Max. Constr. Span (ctr / ctr)
		5 - 0	5 - 6	6 - 0	6 - 6	7 - 0	8 - 0	9 - 0	10 - 0	11 - 0	12 - 0	
Single	22	112 / 86	93 / 64	78 / 50	67 / 39	57 / 31	-	-	-	-	-	6 - 10
	20	143 / 107	118 / 81	100 / 62	85 / 49	73 / 39	56 / 26	44 / 18	-	-	-	8 - 7
	18	194 / 147	161 / 110	135 / 85	115 / 67	99 / 54	76 / 36	60 / 25	49 / 18	40 / 14	34 / 11	11 - 3
	16	249 / 186	206 / 140	173 / 108	147 / 85	127 / 68	97 / 45	77 / 32	62 / 23	51 / 17	43 / 13	13 - 11
Double	22	115 / 221	95 / 166	80 / 128	69 / 101	59 / 81	45 / 54	36 / 38	-	-	-	8 - 4
	20	147 / 269	122 / 202	103 / 156	88 / 122	76 / 98	58 / 66	46 / 46	37 / 34	31 / 25	-	10 - 6
	18	198 / 356	164 / 267	138 / 206	118 / 162	102 / 130	78 / 87	62 / 61	50 / 44	41 / 33	35 / 26	13 - 9
	16	247 / 448	205 / 337	173 / 260	147 / 204	127 / 163	98 / 109	77 / 77	63 / 56	52 / 42	44 / 32	16 - 11
Triple	22	143 / 173	119 / 130	100 / 100	85 / 79	74 / 63	57 / 42	45 / 30	-	-	-	8 - 6
	20	183 / 211	152 / 158	128 / 122	109 / 96	94 / 77	72 / 51	57 / 36	47 / 26	39 / 20	-	10 - 8
	18	245 / 278	203 / 209	172 / 161	147 / 127	127 / 101	97 / 68	77 / 48	63 / 35	52 / 26	43 / 20	13 - 11
	16	307 / 351	255 / 264	215 / 203	183 / 160	159 / 128	122 / 86	96 / 60	78 / 44	65 / 33	54 / 25	17 - 1

Steel Corrugated Deck B-Type (1.5” corrugated) **(20 Gage & triple-span most optimized)**

PROPERTIES

Gage	Thickness (in)	Cover (in)	Weight (psf)
22	0.0295	36	1.63
20	0.0358		1.98
18	0.0474		2.62
16	0.0598		3.30



ICC-ES APPROVED
REFER TO ESR-3818

ASD

SECTION PROPERTIES / DESIGN STRENGTHS

Gage	F_y (ksi)	I_D 1 Span (in^4/ft)	I_D 2+ Span (in^4/ft)	I_p (in^4/ft)	I_n (in^4/ft)	S_p (in^3/ft)	S_n (in^3/ft)	Mn,p/Ω (in-lb/ft)	Mn,n/Ω (in-lb/ft)	Vn/Ω (lb/ft)	*Rbe/Ω (lb/ft)	*Rbi/Ω (lb/ft)
22	40	0.163	0.175	0.157	0.175	0.176	0.183	4215	4391	2009	654	1181
20	40	0.205	0.213	0.201	0.213	0.224	0.235	5375	5624	2428	932	1696
18	40	0.280	0.281	0.279	0.281	0.304	0.315	7291	7553	3192	1557	2858
16	40	0.355	0.355	0.355	0.355	0.390	0.395	9332	9454	3996	2381	4396

Notes:
1. Section properties are calculated in accordance with the AISI S100-16.
2. Web crippling design strengths* are based on minimum bearing lengths of 1 1/2" for end bearing and 3" for interior bearing.



MAXIMUM CANTILEVER SPANS

Gage	F_y (ksi)	Back-Span Condition		
		Single	Double	Triple
22	40	1 - 8	1 - 8	1 - 8
20	40	2 - 0	1 - 11	1 - 11
18	40	2 - 6	2 - 3	2 - 3
16	40	3 - 1	2 - 7	2 - 7

ALLOWABLE UNIFORM LOADS AND MAXIMUM CONSTRUCTION SPANS

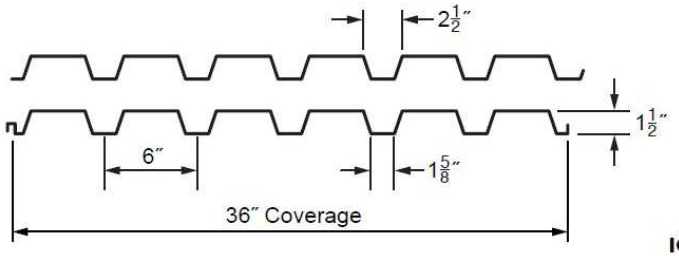
Allowable Load in psf

Span Condition	Gage	Allowable Uniform Total Load (psf) / Load that Produces L/240 Deflection (psf) Center to Center Span (ft - in)										SDI Max. Constr. Span (ctr / ctr)
		5 - 0	5 - 6	6 - 0	6 - 6	7 - 0	8 - 0	9 - 0	10 - 0	11 - 0	12 - 0	
Single	22	112 / 86	93 / 64	78 / 50	67 / 39	57 / 31	-	-	-	-	-	6 - 10
	20	143 / 107	118 / 81	100 / 62	85 / 49	73 / 39	56 / 26	44 / 18	-	-	-	8 - 7
	18	194 / 147	161 / 110	135 / 85	115 / 67	99 / 54	76 / 36	60 / 25	49 / 18	40 / 14	34 / 11	11 - 3
	16	249 / 186	206 / 140	173 / 108	147 / 85	127 / 68	97 / 45	77 / 32	62 / 23	51 / 17	43 / 13	13 - 11
Double	22	115 / 221	95 / 166	80 / 128	69 / 101	59 / 81	45 / 54	36 / 38	-	-	-	8 - 4
	20	147 / 269	122 / 202	103 / 156	88 / 122	76 / 98	58 / 66	46 / 46	37 / 34	31 / 25	-	10 - 6
	18	198 / 356	164 / 267	138 / 206	118 / 162	102 / 130	78 / 87	62 / 61	50 / 44	41 / 33	35 / 26	13 - 9
	16	247 / 448	205 / 337	173 / 260	147 / 204	127 / 163	98 / 109	77 / 77	63 / 56	52 / 42	44 / 32	16 - 11
Triple	22	143 / 173	119 / 130	100 / 100	85 / 79	74 / 63	57 / 42	45 / 30	-	-	-	8 - 6
	20	183 / 211	152 / 158	128 / 122	109 / 96	94 / 77	72 / 51	57 / 36	47 / 26	39 / 20	-	10 - 8
	18	245 / 278	203 / 209	172 / 161	147 / 127	127 / 101	97 / 68	77 / 48	63 / 35	52 / 26	43 / 20	13 - 11
	16	307 / 351	255 / 264	215 / 203	183 / 160	159 / 128	122 / 86	96 / 60	78 / 44	65 / 33	54 / 25	17 - 1

Steel Corrugated Deck B-Type: Spans 6'-0" maximum with 115 psf load (115 < 128)

PROPERTIES

Gage	Thickness (in)	Cover (in)	Weight (psf)
22	0.0295	36	1.63
20	0.0358		1.98
18	0.0474		2.62
16	0.0598		3.30



ICC-ES APPROVED
REFER TO ESR-3818

ASD

SECTION PROPERTIES / DESIGN STRENGTHS

Gage	F_y (ksi)	I_D 1 Span (in^4/ft)	I_D 2+ Span (in^4/ft)	I_p (in^4/ft)	I_n (in^4/ft)	S_p (in^3/ft)	S_n (in^3/ft)	Mn,p/Ω (in-lb/ft)	Mn,n/Ω (in-lb/ft)	Vn/Ω (lb/ft)	*Rbe/Ω (lb/ft)	*Rbi/Ω (lb/ft)
22	40	0.163	0.175	0.157	0.175	0.176	0.183	4215	4391	2009	654	1181
20	40	0.205	0.213	0.201	0.213	0.224	0.235	5375	5624	2428	932	1696
18	40	0.280	0.281	0.279	0.281	0.304	0.315	7291	7553	3192	1557	2858
16	40	0.355	0.355	0.355	0.355	0.390	0.395	9332	9454	3996	2381	4396

Notes:
1. Section properties are calculated in accordance with the AISI S100-16.
2. Web crippling design strengths* are based on minimum bearing lengths of 1 1/2" for end bearing and 3" for interior bearing.



MAXIMUM CANTILEVER SPANS

Gage	F_y (ksi)	Back-Span Condition		
		Single	Double	Triple
22	40	1 - 8	1 - 8	1 - 8
20	40	2 - 0	1 - 11	1 - 11
18	40	2 - 6	2 - 3	2 - 3
16	40	3 - 1	2 - 7	2 - 7

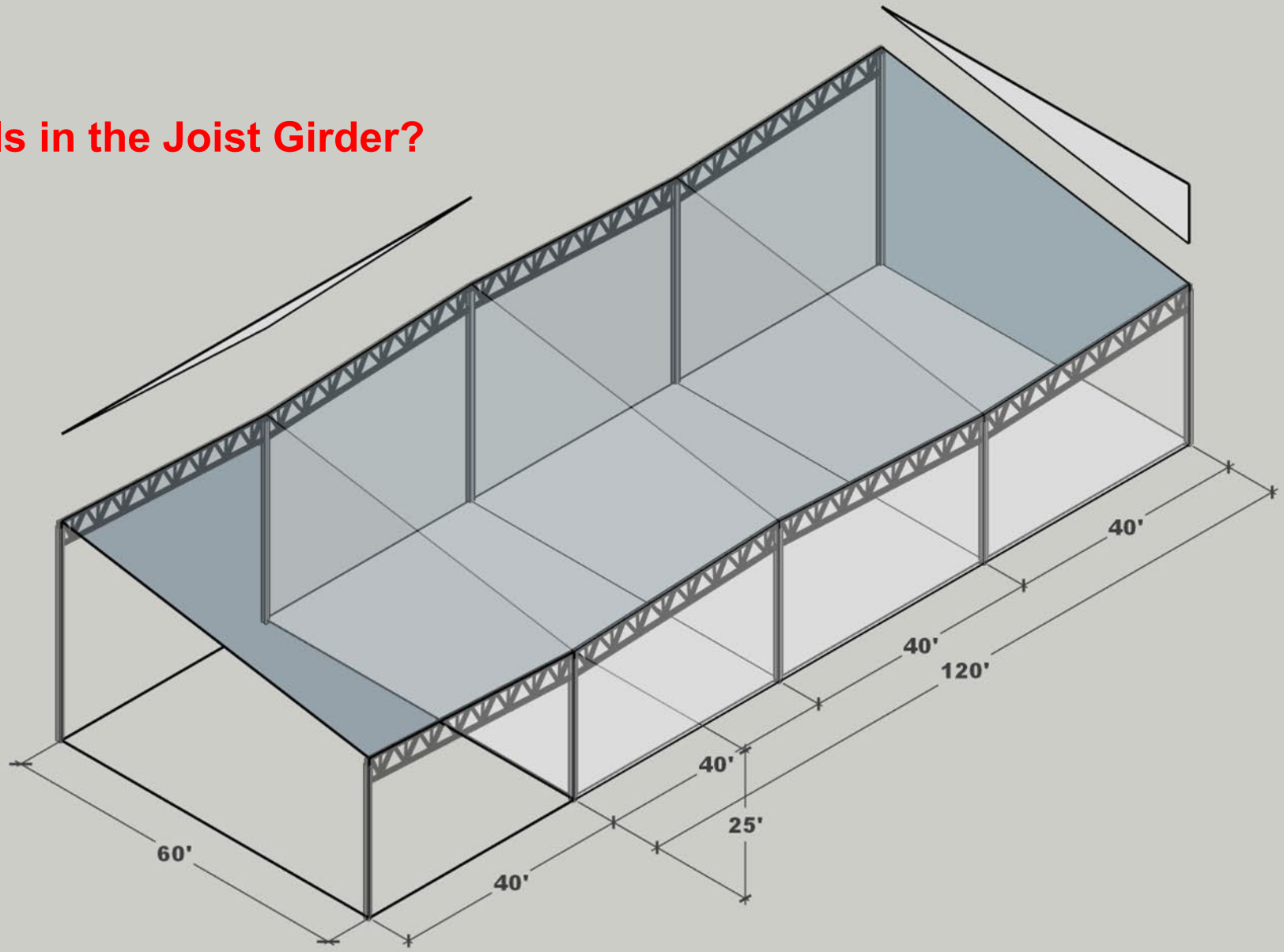
ALLOWABLE UNIFORM LOADS AND MAXIMUM CONSTRUCTION SPANS

Allowable Load in psf

Span Condition	Gage	Allowable Uniform Total Load (psf) / Load that Produces L/240 Deflection (psf) — Center to Center Span (ft - in)										SDI Max. Constr. Span (ctr / ctr)
		5 - 0	5 - 6	6 - 0	6 - 6	7 - 0	8 - 0	9 - 0	10 - 0	11 - 0	12 - 0	
Single	22	112 / 86	93 / 64	78 / 50	67 / 39	57 / 31	-	-	-	-	-	6 - 10
	20	143 / 107	118 / 81	100 / 62	85 / 49	73 / 39	56 / 26	44 / 18	-	-	-	8 - 7
	18	194 / 147	161 / 110	135 / 85	115 / 67	99 / 54	76 / 36	60 / 25	49 / 18	40 / 14	34 / 11	11 - 3
	16	249 / 186	206 / 140	173 / 108	147 / 85	127 / 68	97 / 45	77 / 32	62 / 23	51 / 17	43 / 13	13 - 11
Double	22	115 / 221	95 / 166	80 / 128	69 / 101	59 / 81	45 / 54	36 / 38	-	-	-	8 - 4
	20	147 / 269	122 / 202	103 / 156	88 / 122	76 / 98	58 / 66	46 / 46	37 / 34	31 / 25	-	10 - 6
	18	198 / 356	164 / 267	138 / 206	118 / 162	102 / 130	78 / 87	62 / 61	50 / 44	41 / 33	35 / 26	13 - 9
	16	247 / 448	205 / 337	173 / 260	147 / 204	127 / 163	98 / 109	77 / 77	63 / 56	52 / 42	44 / 32	16 - 11
Triple	22	143 / 173	119 / 130	100 / 100	85 / 79	74 / 63	57 / 42	45 / 30	-	-	-	8 - 6
	20	183 / 211	152 / 158	128 / 122	109 / 96	94 / 77	72 / 51	57 / 36	47 / 26	39 / 20	-	10 - 8
	18	245 / 278	203 / 209	172 / 161	147 / 127	127 / 101	97 / 68	77 / 48	63 / 35	52 / 26	43 / 20	13 - 11
	16	307 / 351	255 / 264	215 / 203	183 / 160	159 / 128	122 / 86	96 / 60	78 / 44	65 / 33	54 / 25	17 - 1

Steel Corrugated Deck B-Type: Cantilever 1’-8” maximum

How many panels in the Joist Girder?



4060 C Joist Girders

Standard Weight Tables for Joist Girders

JOIST GIRDER STANDARD WEIGHTS

PANEL POINT LOAD (kips)		ASD	4	6	8	10	12	14	16	18	20	24	28	32	36	40	44	48	52	56
		LRFD	6	9	12	15	18	21	24	27	30	36	42	48	54	60	66	72	78	84
GIRDER SPAN (ft)	JOIST SPACES (ft)	GIRDER DEPTH (in)	JOIST GIRDER WEIGHT - POUNDS PER LINEAR FOOT (plf)																	
40	4 spaces @ 10.00'	32	17	20	23	29	37	40	47	50	56	64	73	86	103	114	126	128	149	151
		36	17	19	22	29	31	37	40	44	51	57	65	74	87	103	104	125	127	128
		40	17	18	22	25	29	33	37	40	47	52	62	73	77	87	96	104	117	127
		44	16	17	20	24	29	31	36	38	41	49	59	66	74	78	84	96	106	106
		48	17	17	20	24	25	30	32	37	39	48	53	59	67	78	78	85	99	106
	5 spaces @ 8.00'	32	15	21	26	32	38	43	52	55	62	73	86	101	109	124	134			
		36	16	20	24	30	34	39	45	53	55	66	74	88	102	102	112	128	138	
		40	16	20	24	27	32	37	41	46	51	62	68	77	90	100	105	115	130	142
		44	17	20	23	29	32	37	41	49	50	58	70	82	84	99	116	118	130	141
		48	17	20	23	26	31	34	40	41	50	57	68	75	85	95	100	119	120	132
	6 spaces @ 6.67'	32	16	24	30	38	44	52	58	65	72	93	100	115	133					
		36	17	22	27	34	39	47	53	60	67	79	97	102	117	137	141			
		40	16	21	26	30	36	43	48	54	62	71	82	99	103	114	130	142		
		44	17	21	24	28	36	40	47	51	55	66	78	91	102	107	116	134	142	146
		48	17	21	24	31	36	42	46	53	57	69	79	86	100	109	132	133	135	164
	7 spaces @ 5.71'	32	18	26	33	43	52	58	66	74	86	101	115	135						
		36	17	24	31	39	47	53	61	67	75	97	103	117	136					
		40	17	24	29	35	43	49	55	62	69	82	99	105	119	140				
		44	20	22	28	33	39	48	55	59	64	78	92	102	111	122	143			
		48	20	23	28	36	41	48	54	61	66	80	86	108	122	134	136	164	167	
	8 spaces @ 5.00'	32	21	29	38	48	58	67	78	94	96	115	135							
		36	19	27	36	46	53	60	68	80	88	102	118	137						
		40	19	25	34	39	49	58	65	72	82	99	109	120	141					
		44	21	27	33	39	47	56	63	70	75	93	103	120	136	147				
		48	20	25	32	42	47	55	62	69	80	90	104	122	136	155	170			
	10 spaces @ 4.00'	32	29	39	51	64	79	92	112	123	125	149								
		36	25	36	47	60	69	81	94	103	125	150								
		40	24	36	45	56	66	75	82	96	115	129	152							
		44	23	32	41	51	60	71	82	84	99	119	143	161						
		48	23	32	41	52	58	68	76	85	94	121	134	152						

JOIST GIRDER SPAN & LOAD TABLE

Standard Weight Tables for Joist Girders

Joist Spaces is the same as the number of panels in a Truss…

JOIST GIRDER STANDARD WEIGHTS

PANEL POINT LOAD (kips)		ASD	4	6	8	10	12	14	16	18	20	24	28	32	36	40	44	48	52	56
		LRFD	6	9	12	15	18	21	24	27	30	36	42	48	54	60	66	72	78	84
GIRDER SPAN (ft)	JOIST SPACES (ft)	GIRDER DEPTH (in)	JOIST GIRDER WEIGHT - POUNDS PER LINEAR FOOT (plf)																	
40	4 spaces @ 10.00'	32	17	20	23	29	37	40	47	50	56	64	73	86	103	114	126	128	149	151
		36	17	19	22	29	31	37	40	44	51	57	65	74	87	103	104	125	127	128
		40	17	18	22	25	29	33	37	40	47	52	62	73	77	87	96	104	117	127
		44	16	17	20	24	29	31	36	38	41	49	59	66	74	78	84	96	106	106
		48	17	17	20	24	25	30	32	37	39	48	53	59	67	78	78	85	99	106
	5 spaces @ 8.00'	32	15	21	26	32	38	43	52	55	62	73	86	101	109	124	134			
		36	16	20	24	30	34	39	45	53	55	66	74	88	102	102	112	128	138	
		40	16	20	24	27	32	37	41	46	51	62	68	77	90	100	105	115	130	142
		44	17	20	23	29	32	37	41	49	50	58	70	82	84	99	116	118	130	141
		48	17	20	23	26	31	34	40	41	50	57	68	75	85	95	100	119	120	132
	6 spaces @ 6.67'	32	16	24	30	38	44	52	58	65	72	93	100	115	133					
		36	17	22	27	34	39	47	53	60	67	79	97	102	117	137	141			
		40	16	21	26	30	36	43	48	54	62	71	82	99	103	114	130	142		
		44	17	21	24	28	36	40	47	51	55	66	78	91	102	107	116	134	142	146
		48	17	21	24	31	36	42	46	53	57	69	79	86	100	109	132	133	135	164
	7 spaces @ 5.71'	32	18	26	33	43	52	58	66	74	86	101	115	135						
		36	17	24	31	39	47	53	61	67	75	97	103	117	136					
		40	17	24	29	35	43	49	55	62	69	82	99	105	119	140				
		44	20	22	28	33	39	48	55	59	64	78	92	102	111	122	143			
		48	20	23	28	36	41	48	54	61	66	80	86	108	122	134	136	164	167	
	8 spaces @ 5.00'	32	21	29	38	48	58	67	78	94	96	115	135							
		36	19	27	36	46	53	60	68	80	88	102	118	137						
		40	19	25	34	39	49	58	65	72	82	99	109	120	141					
		44	21	27	33	39	47	56	63	70	75	93	103	120	136	147				
		48	20	25	32	42	47	55	62	69	80	90	104	122	136	155	170			
	10 spaces @ 4.00'	32	29	39	51	64	79	92	112	123	125	149								
		36	25	36	47	60	69	81	94	103	125	150								
		40	24	36	45	56	66	75	82	96	115	129	152							
		44	23	32	41	51	60	71	82	84	99	119	143	161						
		48	23	32	41	52	58	68	76	85	94	121	134	152						

JOIST GIRDER SPAN & LOAD TABLE

Standard Weight Tables for Joist Girders

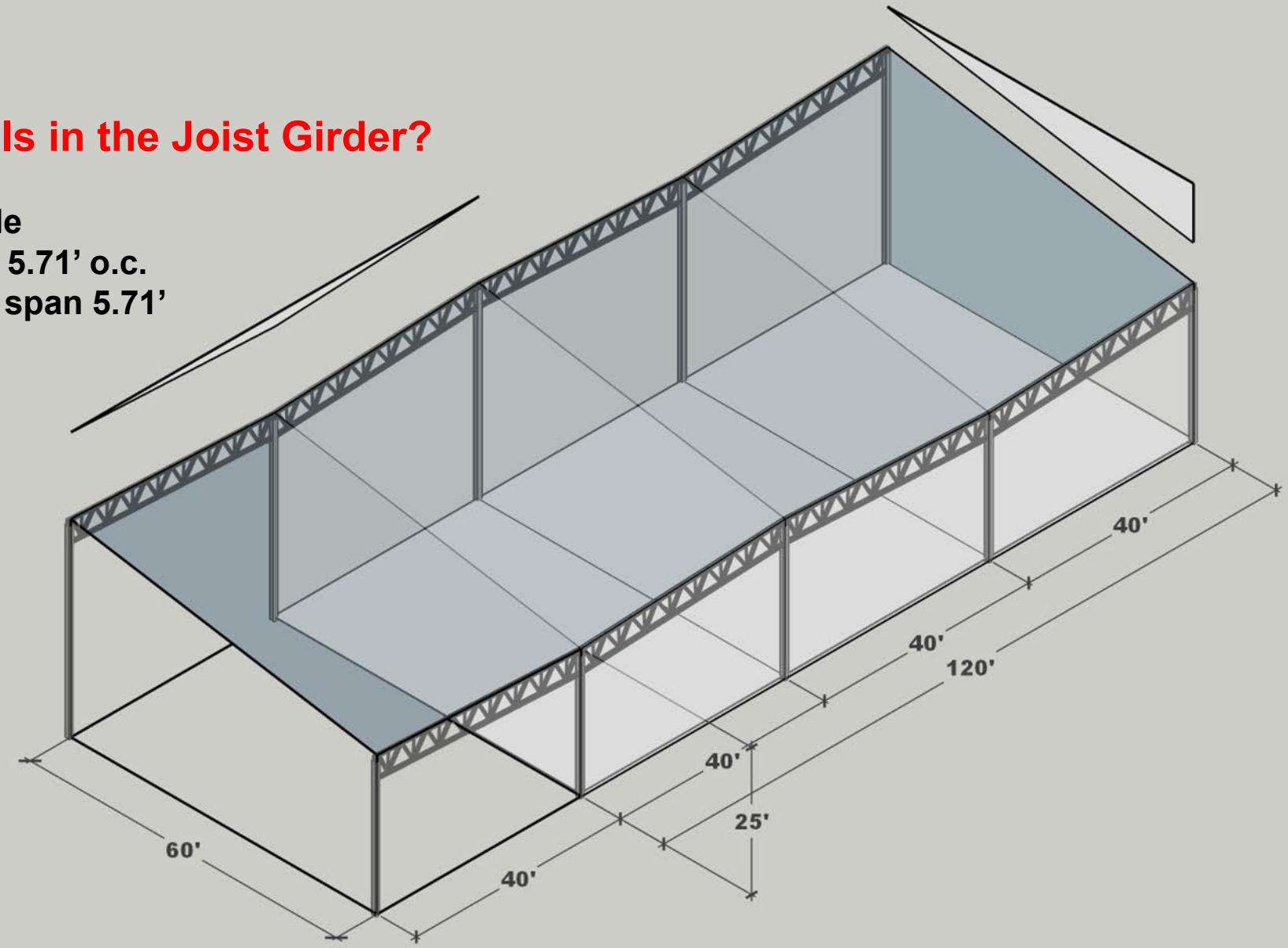
JOIST GIRDER STANDARD WEIGHTS

PANEL POINT LOAD (kips)		ASD	4	6	8	10	12	14	16	18	20	24	28	32	36	40	44	48	52	56
		LRFD	6	9	12	15	18	21	24	27	30	36	42	48	54	60	66	72	78	84
GIRDER SPAN (ft)	JOIST SPACES (ft)	GIRDER DEPTH (in)	JOIST GIRDER WEIGHT - POUNDS PER LINEAR FOOT (plf)																	
40	4 spaces @ 10.00'	32	17	20	23	29	37	40	47	50	56	64	73	86	103	114	126	128	149	151
		36	17	19	22	29	31	37	40	44	51	57	65	74	87	103	104	125	127	128
		40	17	18	22	25	29	33	37	40	47	52	62	73	77	87	96	104	117	127
		44	16	17	20	24	29	31	36	38	41	49	59	66	74	78	84	96	106	106
		48	17	17	20	24	25	30	32	37	39	48	53	59	67	78	78	85	99	106
	5 spaces @ 8.00'	32	15	21	26	32	38	43	52	55	62	73	86	101	109	124	134			
		36	16	20	24	30	34	39	45	53	55	66	74	88	102	102	112	128	138	
		40	16	20	24	27	32	37	41	46	51	62	68	77	90	100	105	115	130	142
		44	17	20	23	29	32	37	41	49	50	58	70	82	84	99	116	118	130	141
		48	17	20	23	26	31	34	40	41	50	57	68	75	85	95	100	119	120	132
	6 spaces @ 6.67'	32	16	24	30	38	44	52	58	65	72	93	100	115	133					
		36	17	22	27	34	39	47	53	60	67	79	97	102	117	137	141			
		40	16	21	26	30	36	43	48	54	62	71	82	99	103	114	130	142		
		44	17	21	24	28	36	40	47	51	55	66	78	91	102	107	116	134	142	146
		48	17	21	24	31	36	42	46	53	57	69	79	86	100	109	132	133	135	164
	7 spaces @ 5.71'	32	18	26	33	43	52	58	66	74	86	101	115	135						
		36	17	24	31	39	47	53	61	67	75	97	103	117	136					
		40	17	24	29	35	43	49	55	62	69	82	99	105	119	140				
		44	20	22	28	33	39	48	55	59	64	78	92	102	111	122	143			
		48	20	23	28	36	41	48	54	61	66	80	86	108	122	134	136	164	167	
	8 spaces @ 5.00'	32	21	29	38	48	58	67	78	94	96	115	135							
		36	19	27	36	46	53	60	68	80	88	102	118	137						
		40	19	25	34	39	49	58	65	72	82	99	109	120	141					
		44	21	27	33	39	47	56	63	70	75	93	103	120	136	147				
		48	20	25	32	42	47	55	62	69	80	90	104	122	136	155	170			
	10 spaces @ 4.00'	32	29	39	51	64	79	92	112	123	125	149								
		36	25	36	47	60	69	81	94	103	125	150								
		40	24	36	45	56	66	75	82	96	115	129	152							
		44	23	32	41	51	60	71	82	84	99	119	143	161						
		48	23	32	41	52	58	68	76	85	94	121	134	152						

JOIST GIRDER SPAN & LOAD TABLE: 40’ span with 7 Joist Spaces at 5.71’

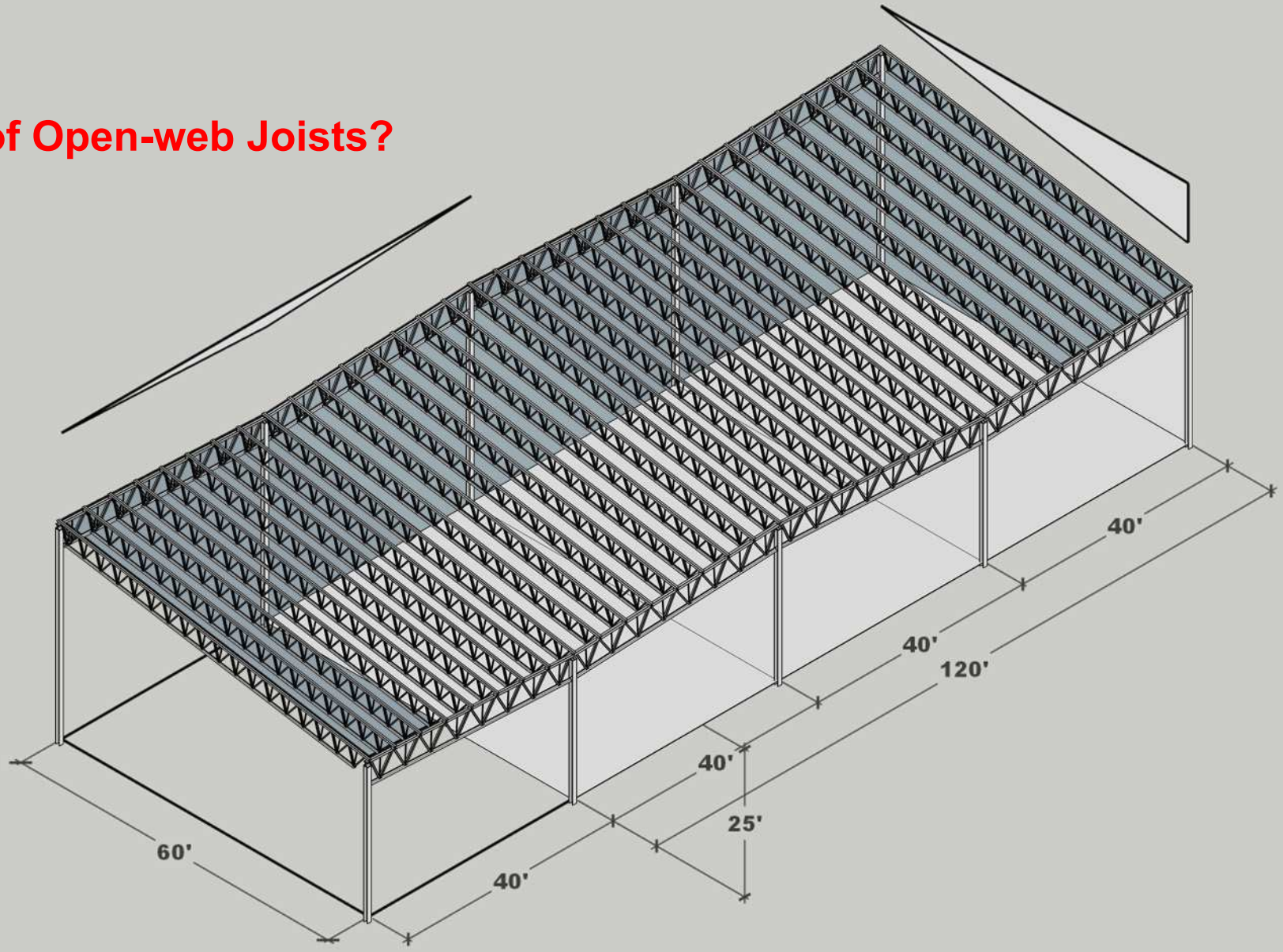
How many panels in the Joist Girder?

7 panels at 5.71’ wide
Joist will be spaced 5.71’ o.c.
Roof Deck can then span 5.71’
(max span 6’)



4060 C **Joist Girders** (7 Panels @ 5.71’ wide = 40’ span)

Size and depth of Open-web Joists?



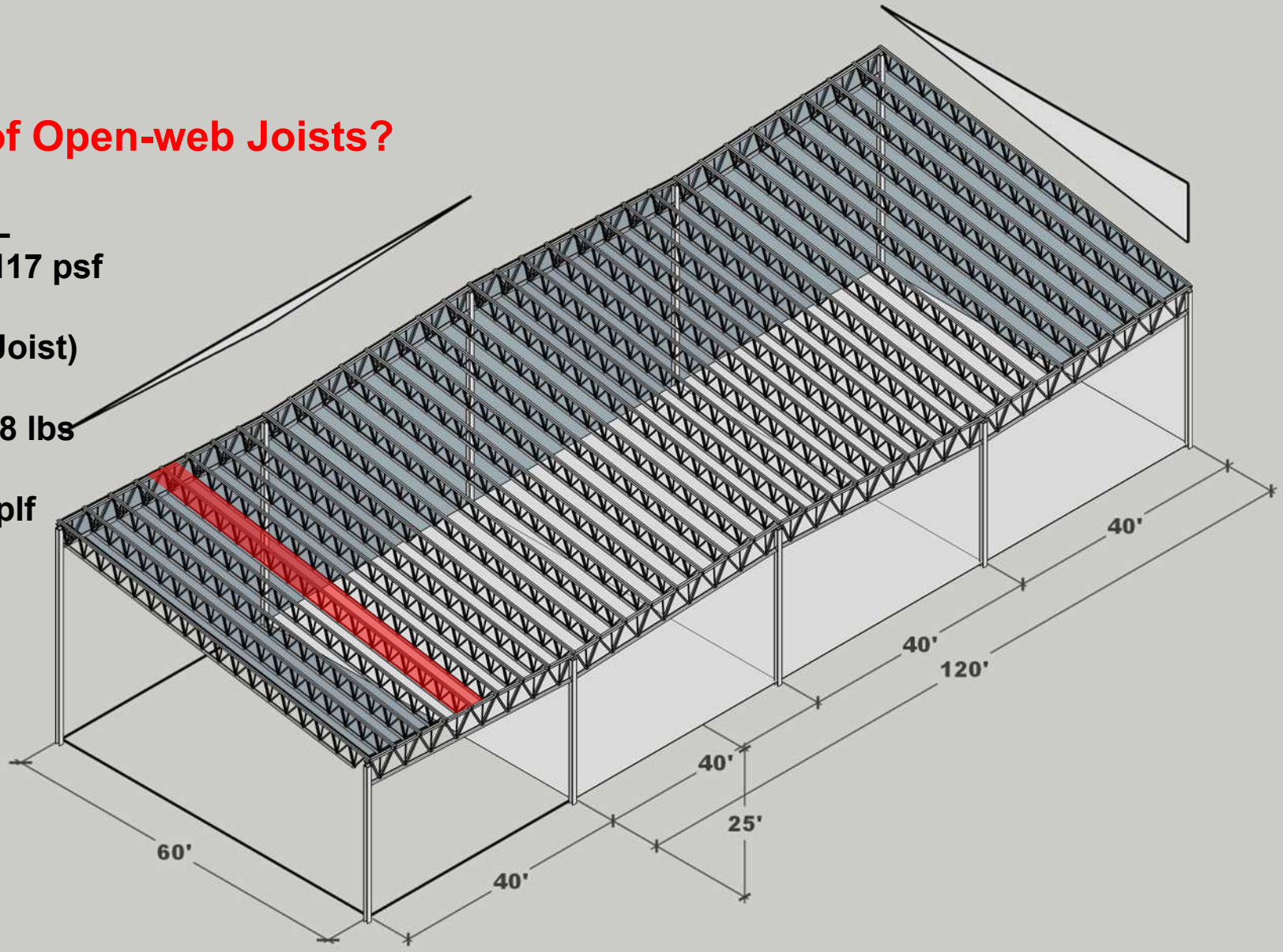
4060 C **Joists** (60’ span, Joists spaced at 5.71’ o.c. to support Roof Deck)

Size and depth of Open-web Joists?

Roof Load + Deck DL
115 psf + 1.98 psf = 117 psf

Tributary Area (one Joist)
5.71’ x 1’ = 5.71 sf
5.71 sf (117 psf) = 668 lbs

Load on Joist = 668 plf



4060 C Joists (60’ span, Joists spaced at 5.71’ o.c. to support Roof Deck)

Economical Load Tables

Joist Span (ft)	Total Load (plf) Factored LRFD	Total Load (plf) Service ASD	Live Load (plf) 1/240	Live Load (plf) 1/360	Joist Designation	Joist Weight (plf)
60 (cont.)	F 301	201	144	96	30K10	11.5
	F 346	231	163	109	30K11	13.0
	F 363	242	187	125	32LH06	13.4
	F 393	262	186	124	30K12	14.7
	F 411	274	240	160	36LH07	14.4
	F 453	302	264	176	36LH08	15.3
	F 568	379	379	297	44LH09	16.9
	F 582	388	388	261	40LH09	17.8
	F 601	401	401	362	48LH10	17.8
	F 639	426	372	248	36LH10	20.6
	F 649	433	433	390	48LH11	17.9
	F 670	447	432	288	40LH10	20.8
	F 705	470	468	312	40LH11	21.2
	F 721	481	481	354	44LH11	21.3
	F 852	568	568	437	44LH12	24.4
	F 856	571	570	380	40LH12	25.8
	F 1017	678	667	445	40LH13	29.9
	F 1161	774	766	511	40LH14	34.1
	F 1174	783	783	593	44LH14	33.3
	F 1192	795	795	690	48LH14	32.5
	F 1366	911	911	790	48LH15	36.0
	F 1588	1059	1059	792	44LH16	43.3
	F 1597	1065	1065	910	48LH16	42.8
	F 1771	1181	1181	847	44LH17	48.3
	F 1816	1211	1155	770	40LH17	50.6

Joist Span (ft)	Total Load (plf) Factored LRFD	Total Load (plf) Service ASD	Live Load (plf) 1/240	Live Load (plf) 1/360	Joist Designation	Joist Weight (plf)
61 (cont.)	F 1332	888	888	751	48LH15	35.6
	F 1545	1030	1030	754	44LH16	43.2
	F 1555	1037	1037	866	48LH16	42.7
	F 1723	1149	1149	806	44LH17	48.4
	F 1746	1164	1164	968	48LH17	47.6
	F 1756	1171	1098	732	40LH17	50.8
	F 2229	1486	1486	1230	48LH18	59.8
	F 2455	1637	1637	1140	44LH19	64.5
	F 2589	1726	1726	1369	48LH19	65.9
	F 3091	2061	2061	1425	44LH20	85.1
	F 3261	2174	2174	1715	48LH20	85.9
	F 3405	2270	2270	1580	44LH21	92.7
	F 3591	2394	2394	1901	48LH21	93.4
	F 4240	2827	2827	2161	48LH22	105.9
	F 4458	2972	2972	2365	48LH23	112.6
	F 340	227	171	114	32LH06	13.5
	F 387	258	219	146	36LH07	14.4
	F 426	284	240	160	36LH08	15.2
	F 541	361	361	269	44LH09	16.7
	F 544	363	306	204	36LH09	18.8
	F 552	368	355	237	40LH09	18.2
	F 573	382	382	327	48LH10	17.6
	F 597	398	398	297	44LH10	18.5
	F 601	401	337	225	36LH10	20.6
	F 619	413	413	353	48LH11	18.5

JOIST SPAN & LOAD TABLE

Economical Load Tables

Joist Span (ft)	Total Load (plf) Factored LRFD	Total Load (plf) Service ASD	Live Load (plf) 1/240	Live Load (plf) 1/360	Joist Designation	Joist Weight (plf)
60 (cont.)	F 301	201	144	96	30K10	11.5
	F 346	231	163	109	30K11	13.0
	F 363	242	187	125	32LH06	13.4
	F 393	262	186	124	30K12	14.7
	F 411	274	240	160	36LH07	14.4
	F 453	302	264	176	36LH08	15.3
	F 568	379	379	297	44LH09	16.9
	F 582	388	388	261	40LH09	17.8
	F 601	401	401	362	48LH10	17.8
	F 639	426	372	248	36LH10	20.6
	F 649	433	433	390	48LH11	17.9
	F 670	447	432	288	40LH10	20.8
	F 705	470	468	312	40LH11	21.2
	F 721	481	481	354	44LH11	21.3
	F 852	568	568	437	44LH12	24.4
	F 856	571	570	380	40LH12	25.8
	F 1017	678	667	445	40LH13	29.9
	F 1161	774	766	511	40LH14	34.1
	F 1174	783	783	593	44LH14	33.3
	F 1192	795	795	690	48LH14	32.5
	F 1366	911	911	790	48LH15	36.0
	F 1588	1059	1059	792	44LH16	43.3
	F 1597	1065	1065	910	48LH16	42.8
	F 1771	1181	1181	847	44LH17	48.3
	F 1816	1211	1155	770	40LH17	50.6

Joist Span (ft)	Total Load (plf) Factored LRFD	Total Load (plf) Service ASD	Live Load (plf) 1/240	Live Load (plf) 1/360	Joist Designation	Joist Weight (plf)
61 (cont.)	F 1332	888	888	751	48LH15	35.6
	F 1545	1030	1030	754	44LH16	43.2
	F 1555	1037	1037	866	48LH16	42.7
	F 1723	1149	1149	806	44LH17	48.4
	F 1746	1164	1164	968	48LH17	47.6
	F 1756	1171	1098	732	40LH17	50.8
	F 2229	1486	1486	1230	48LH18	59.8
	F 2455	1637	1637	1140	44LH19	64.5
	F 2589	1726	1726	1369	48LH19	65.9
	F 3091	2061	2061	1425	44LH20	85.1
	F 3261	2174	2174	1715	48LH20	85.9
	F 3405	2270	2270	1580	44LH21	92.7
	F 3591	2394	2394	1901	48LH21	93.4
	F 4240	2827	2827	2161	48LH22	105.9
	F 4458	2972	2972	2365	48LH23	112.6
	F 340	227	171	114	32LH06	13.5
	F 387	258	219	146	36LH07	14.4
	F 426	284	240	160	36LH08	15.2
	F 541	361	361	269	44LH09	16.7
	F 544	363	306	204	36LH09	18.8
	F 552	368	355	237	40LH09	18.2
	F 573	382	382	327	48LH10	17.6
	F 597	398	398	297	44LH10	18.5
	F 601	401	337	225	36LH10	20.6
	F 619	413	413	353	48LH11	18.5

JOIST SPAN & LOAD TABLE: 60’ Span 678 plf = Joist 40LH13 (40” deep, 13 panels)

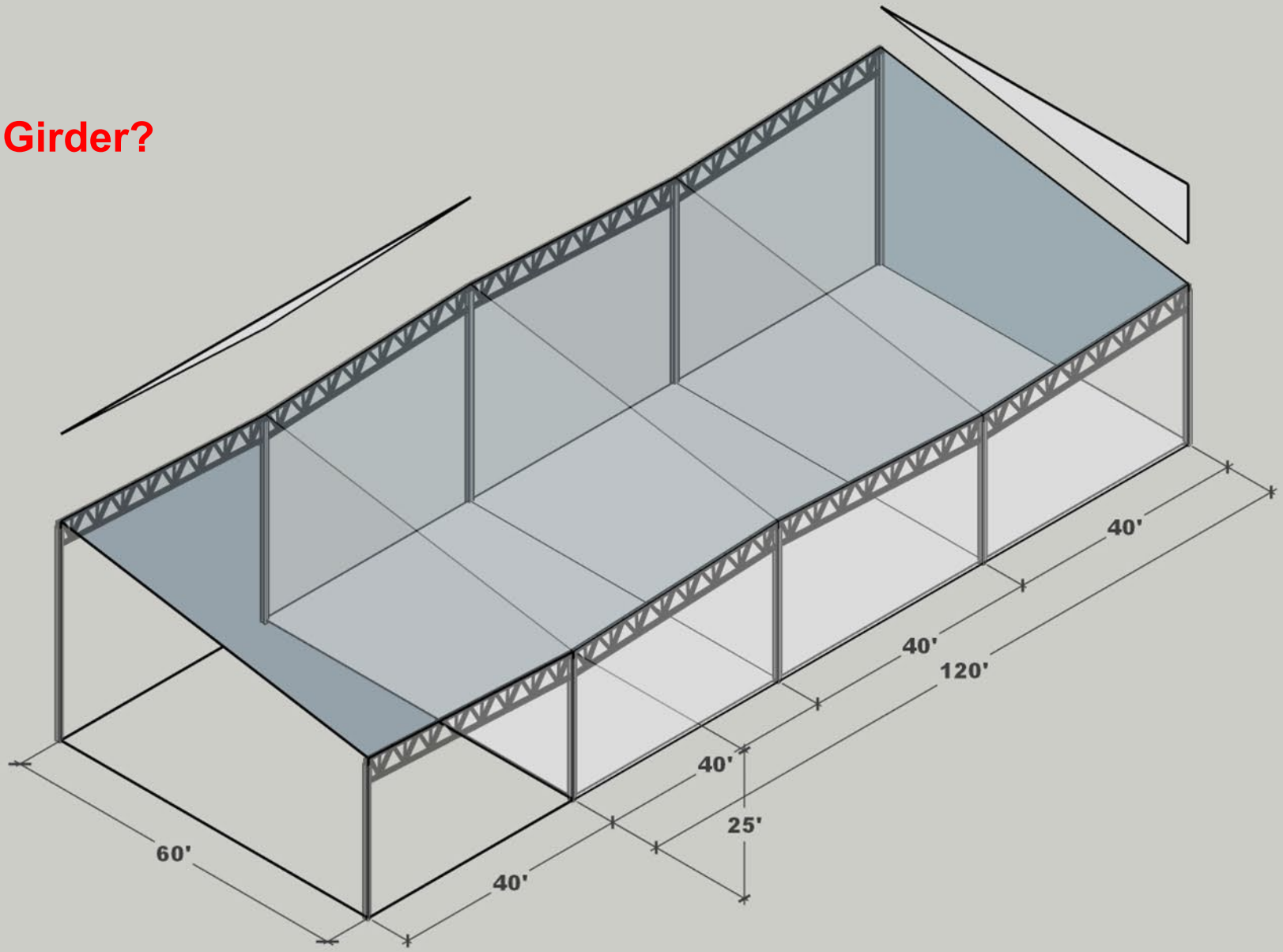
Economical Load Tables

Joist Span (ft)	Total Load (plf) Factored LRFD	Total Load (plf) Service ASD	Live Load (plf) 1/240	Live Load (plf) 1/360	Joist Designation	Joist Weight (plf)
60 (cont.)	F 301	201	144	96	30K10	11.5
	F 346	231	163	109	30K11	13.0
	F 363	242	187	125	32LH06	13.4
	F 393	262	186	124	30K12	14.7
	F 411	274	240	160	36LH07	14.4
	F 453	302	264	176	36LH08	15.3
	F 568	379	379	297	44LH09	16.9
	F 582	388	388	261	40LH09	17.8
	F 601	401	401	362	48LH10	17.8
	F 639	426	372	248	36LH10	20.6
	F 649	433	433	390	48LH11	17.9
	F 670	447	432	288	40LH10	20.8
	F 705	470	468	312	40LH11	21.2
	F 721	481	481	354	44LH11	21.3
	F 852	568	568	437	44LH12	24.4
	F 856	571	570	380	40LH12	25.8
	F 1017	678	667	445	40LH13	29.9
	F 1161	774	766	511	40LH14	34.1
	F 1174	783	783	593	44LH14	33.3
	F 1192	795	795	690	48LH14	32.5
	F 1366	911	911	790	48LH15	36.0
	F 1588	1059	1059	792	44LH16	43.3
	F 1597	1065	1065	910	48LH16	42.8
	F 1771	1181	1181	847	44LH17	48.3
	F 1816	1211	1155	770	40LH17	50.6

Joist Span (ft)	Total Load (plf) Factored LRFD	Total Load (plf) Service ASD	Live Load (plf) 1/240	Live Load (plf) 1/360	Joist Designation	Joist Weight (plf)
61 (cont.)	F 1332	888	888	751	48LH15	35.6
	F 1545	1030	1030	754	44LH16	43.2
	F 1555	1037	1037	866	48LH16	42.7
	F 1723	1149	1149	806	44LH17	48.4
	F 1746	1164	1164	968	48LH17	47.6
	F 1756	1171	1098	732	40LH17	50.8
	F 2229	1486	1486	1230	48LH18	59.8
	F 2455	1637	1637	1140	44LH19	64.5
	F 2589	1726	1726	1369	48LH19	65.9
	F 3091	2061	2061	1425	44LH20	85.1
	F 3261	2174	2174	1715	48LH20	85.9
	F 3405	2270	2270	1580	44LH21	92.7
	F 3591	2394	2394	1901	48LH21	93.4
	F 4240	2827	2827	2161	48LH22	105.9
	F 4458	2972	2972	2365	48LH23	112.6
	F 340	227	171	114	32LH06	13.5
	F 387	258	219	146	36LH07	14.4
	F 426	284	240	160	36LH08	15.2
	F 541	361	361	269	44LH09	16.7
	F 544	363	306	204	36LH09	18.8
	F 552	368	355	237	40LH09	18.2
	F 573	382	382	327	48LH10	17.6
	F 597	398	398	297	44LH10	18.5
	F 601	401	337	225	36LH10	20.6
	F 619	413	413	353	48LH11	18.5

JOIST SPAN & LOAD TABLE: 40LH13 has a self-weight of 29.9 plf (29.9 plf (60') = 1,794 lbs

Size of the Joist Girder?

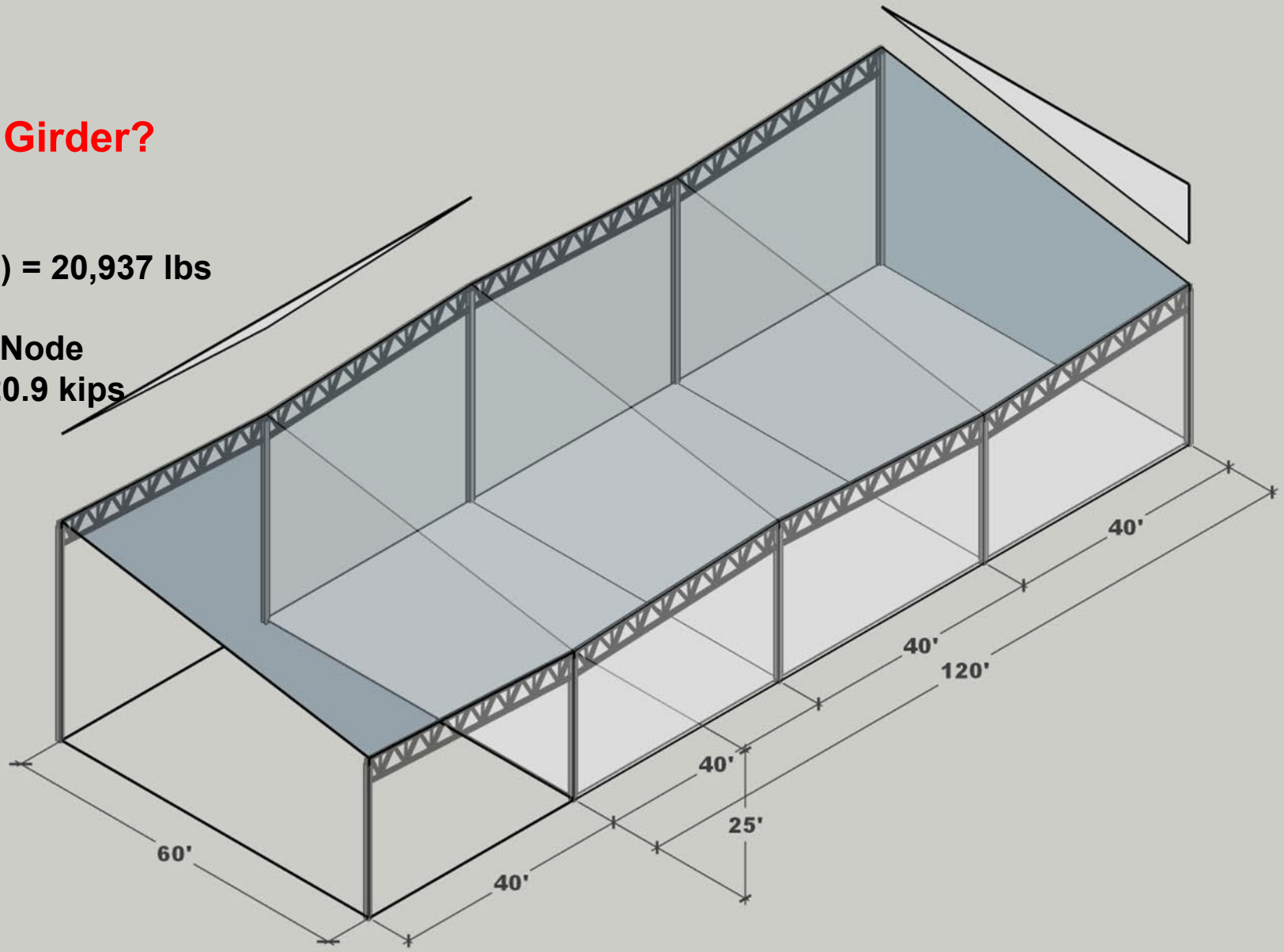


4060 C Joist Girders

Size of the Joist Girder?

Joist Load
668 plf + 29.9 plf (30') = 20,937 lbs

Load at Joist Girder Node
Panel Point Load = 20.9 kips



4060 C Joist Girders

Standard Weight Tables for Joist Girders

JOIST GIRDER STANDARD WEIGHTS

PANEL POINT LOAD (kips)		ASD	4	6	8	10	12	14	16	18	20	24	28	32	36	40	44	48	52	56
		LRFD	6	9	12	15	18	21	24	27	30	36	42	48	54	60	66	72	78	84
GIRDER SPAN (ft)	JOIST SPACES (ft)	GIRDER DEPTH (in)	JOIST GIRDER WEIGHT - POUNDS PER LINEAR FOOT (plf)																	
40	4 spaces @ 10.00'	32	17	20	23	29	37	40	47	50	56	64	73	86	103	114	126	128	149	151
		36	17	19	22	29	31	37	40	44	51	57	65	74	87	103	104	125	127	128
		40	17	18	22	25	29	33	37	40	47	52	62	73	77	87	96	104	117	127
		44	16	17	20	24	29	31	36	38	41	49	59	66	74	78	84	96	106	106
		48	17	17	20	24	25	30	32	37	39	48	53	59	67	78	78	85	99	106
	5 spaces @ 8.00'	32	15	21	26	32	38	43	52	55	62	73	86	101	109	124	134			
		36	16	20	24	30	34	39	45	53	55	66	74	88	102	102	112	128	138	
		40	16	20	24	27	32	37	41	46	51	62	68	77	90	100	105	115	130	142
		44	17	20	23	29	32	37	41	49	50	58	70	82	84	99	116	118	130	141
		48	17	20	23	26	31	34	40	41	50	57	68	75	85	95	100	119	120	132
	6 spaces @ 6.67'	32	16	24	30	38	44	52	58	65	72	93	100	115	133					
		36	17	22	27	34	39	47	53	60	67	79	97	102	117	137	141			
		40	16	21	26	30	36	43	48	54	62	71	82	99	103	114	130	142		
		44	17	21	24	28	36	40	47	51	55	66	78	91	102	107	116	134	142	146
		48	17	21	24	31	36	42	46	53	57	69	79	86	100	109	132	133	135	164
	7 spaces @ 5.71'	32	18	26	33	43	52	58	66	74	86	101	115	135						
		36	17	24	31	39	47	53	61	67	75	97	103	117	136					
		40	17	24	29	35	43	49	55	62	69	82	99	105	119	140				
		44	20	22	28	33	39	48	55	59	64	78	92	102	111	122	143			
		48	20	23	28	36	41	48	54	61	66	80	86	108	122	134	136	164	167	
	8 spaces @ 5.00'	32	21	29	38	48	58	67	78	94	96	115	135							
		36	19	27	36	46	53	60	68	80	88	102	118	137						
		40	19	25	34	39	49	58	65	72	82	99	109	120	141					
		44	21	27	33	39	47	56	63	70	75	93	103	120	136	147				
		48	20	25	32	42	47	55	62	69	80	90	104	122	136	155	170			
	10 spaces @ 4.00'	32	29	39	51	64	79	92	112	123	125	149								
		36	25	36	47	60	69	81	94	103	125	150								
		40	24	36	45	56	66	75	82	96	115	129	152							
		44	23	32	41	51	60	71	82	84	99	119	143	161						
		48	23	32	41	52	58	68	76	85	94	121	134	152						

JOIST GIRDER : 40’ span with 7 Joist Spaces at 5.71’ = 44” depth of Joist Girder (78 plf)

ASD & LRFD two load methods for determining structural elements

ASD Allowable Stress Design (a.k.a. stress design)

Traditional method based on predictable service loads and allowable stresses.
Design method to ensure expected loads on structural elements do not exceed materials allowable strength.
Applies the same safety factor to all types of loads.
Simple method, though often results in material and element over-sizing.

LRFD Load and Resistance Factor Design (a.k.a. strength design)

Contemporary method based on the ultimate strength of material and the probabilistic nature of loads.
Design method to ensure the factored resistance is greater than the factored loads.
Applies different factors to loads and load combinations to account for load variability, emphasizing the uncertainty of live loads and environmental loads.
Distributes the safety factor and margins across both loads and material resistance.
More complex method, though uses materials and elements more efficiently.

LRFD load and resistance factor design

Table A-2.7: Combined load factors[1]

A. Strength Design	
Load Combinations	**Combined Loads and Factors**
Dead load	$1.4D$
Dead, live, and roof or snow	$1.2D + 1.6L + 0.5(L_r \text{ or } S)$
Dead, roof or snow, and live[2] or wind	$1.2D + 1.6(L_r \text{ or } S) + (L \text{ or } 0.5W)$
Dead, wind, live,[2] and roof or snow	$1.2D + 1.0W + L + 0.5(L_r \text{ or } S)$
Dead, earthquake[3], live,[2] and snow	$1.2D + 1.0E + L + 0.2S$
Dead and wind	$0.9D + 1.0W$
Dead and earthquake[3]	$0.9D + 1.0E$

Load Combination Formulas

ASD & LRFD two load methods for determining structural elements

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STEEL BUILDING project

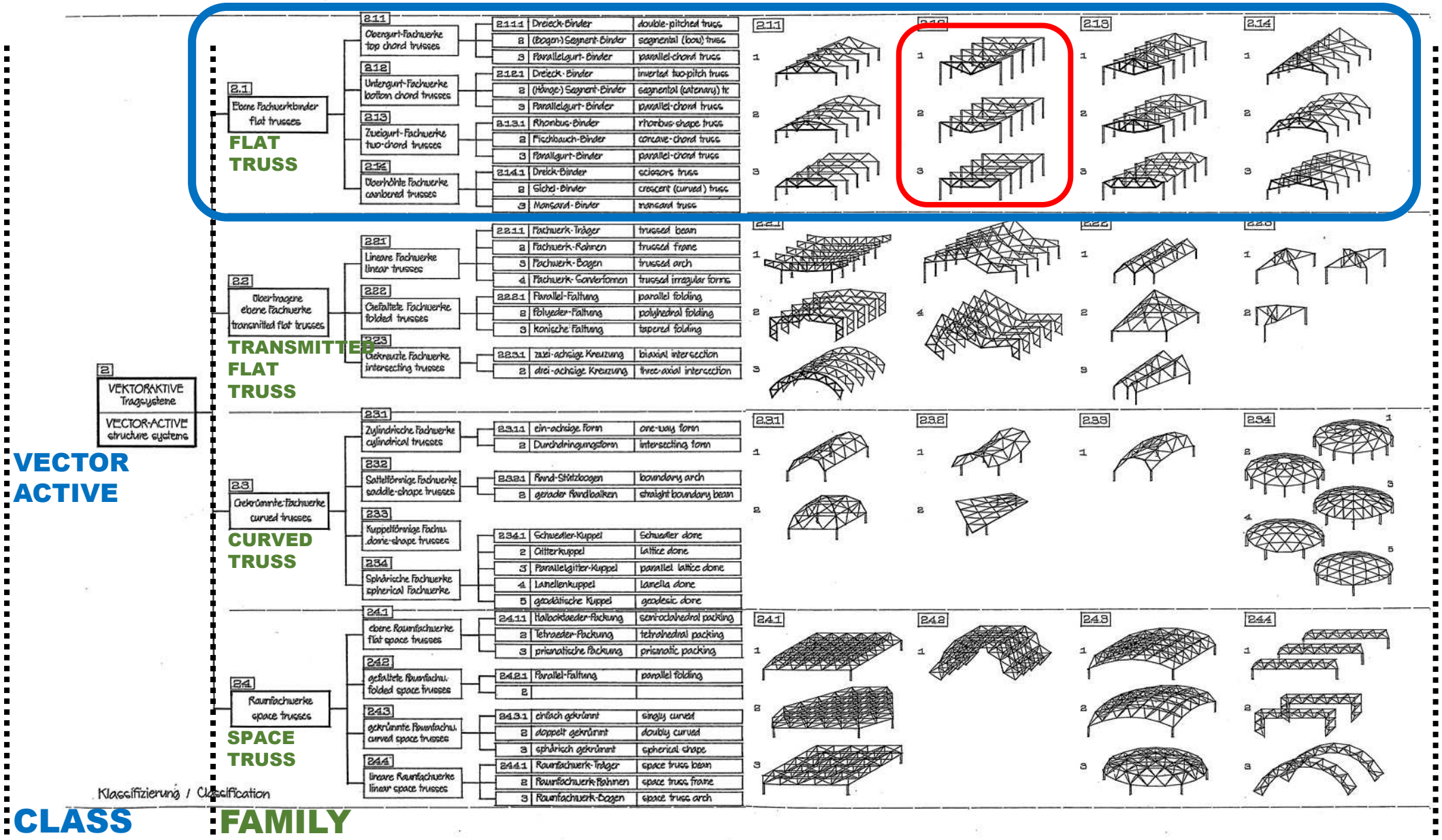
HOMEWORK *(complete by Thursday, October 8, 11am)*

…select a 3060 model or 4060 model option for your project

…review Span & Load Tables on Canvas for Deck, Open-Web Joists, and Joists Girders

…meet with your "Study Group" to work together on general strategy and parameters?

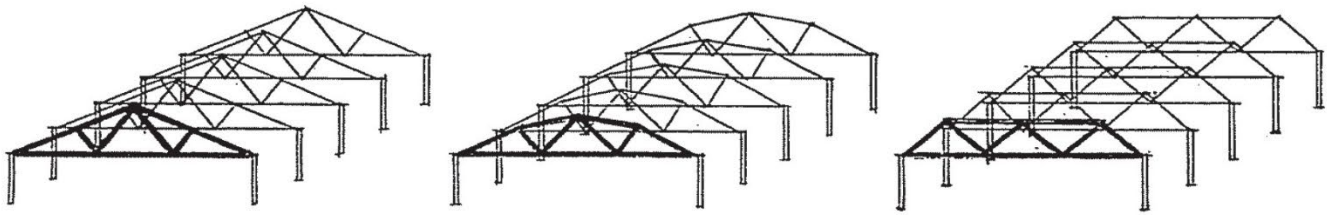
Thursday class will have a brief tutorial, then an in-class working session…



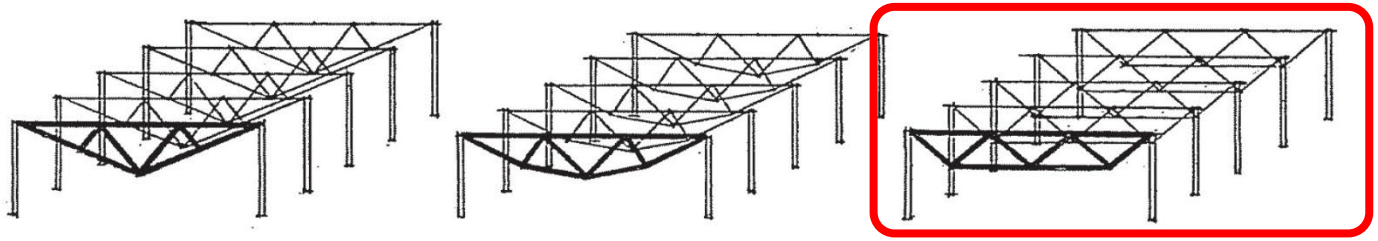
VECTOR-ACTIVE CLASS & FLAT TRUSS FAMILY *…Bottom Chord Branch*

2.1 Ebene Fachwerkbinder / Flat trusses

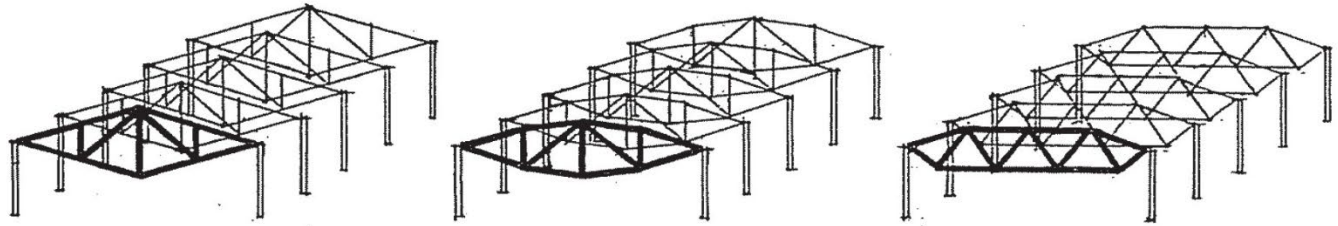
Obergurt-Systeme
Top chord systems



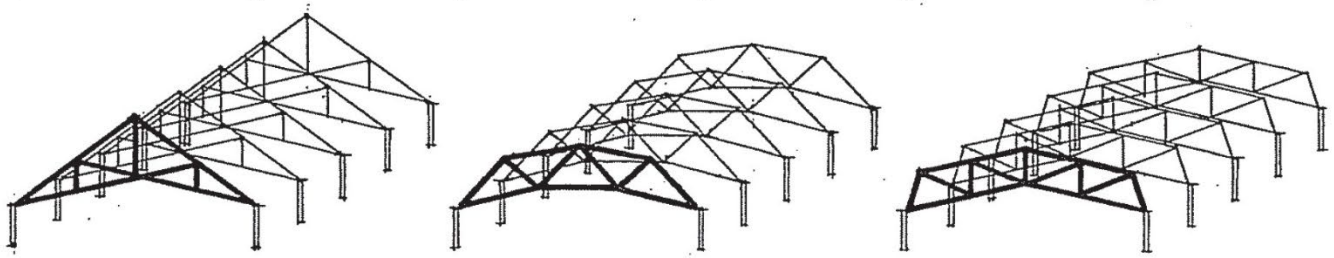
Untergurt-Systeme
Bottom chord systems



Zweigurt-Systeme
Two-chord systems



Überhöhte Systeme
Cambered systems



Bottom Chord Systems (top chord bearing)

…let’s go on a brief FIELD TRIP to the low-bay structural engineering lab